Quantum Mechanics (PHY 203)

Tutorial session

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Q1 – Black-body Radiation & the Ultraviolet Catastrophe

Question 1

What is black-body radiation, and what problem (the *ultraviolet catastrophe*) did the classical Rayleigh–Jeans law create when applied to it?

A **black body** absorbs all incident electromagnetic radiation and re-emits energy with a spectral distribution depending only on its temperature T.

Step 1: Classical (Rayleigh–Jeans) derivation. Treat the radiation field inside a cavity as a set of standing-wave modes. The number of modes per unit volume in the frequency interval $d\nu$ is

$$dn = \frac{8\pi\nu^2}{c^3}\, d\nu.$$

By the equipartition theorem, each mode carries average energy $\langle E\rangle = k_B T$. Hence the spectral energy density is

$$u_{\mathrm{RJ}}(\nu, T) = \frac{8\pi\nu^2}{c^3}\, k_B T.$$

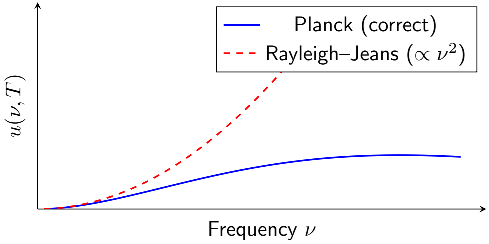
Step 2: The ultraviolet catastrophe. As $\nu \to \infty$, $u_{\mathrm{RJ}} \propto \nu^2 \to \infty$. The total radiated energy

$$\int_0^\infty u_{\mathrm{RJ}}(\nu, T)\, d\nu = \frac{8\pi k_B T}{c^3} \int_0^\infty \nu^2\, d\nu \to \infty$$

diverges. This is unphysical: a body at finite temperature would radiate an *infinite* amount of energy, mostly at ultraviolet and higher frequencies – hence the name **ultraviolet catastrophe**.

Step 3: Resolution (Planck 1900). Planck postulated that energy exchange between matter and radiation is restricted to discrete quanta $E = h\nu$, where $h = 6.626 \times 10^{-34}$ J s. This exponentially suppresses the high-frequency modes and cures the divergence.

Q1 – Black-body Radiation & the Ultraviolet Catastrophe



Question 2

State Planck's radiation law and show how it reduces to the Rayleigh–Jeans law in the low-frequency (classical) limit.

Planck's law:

$$u(\nu, T) = \frac{8\pi h\nu^3}{c^3} \cdot \frac{1}{e^{h\nu/k_BT} - 1}.$$

Step 1: Low-frequency limit $h\nu \ll k_BT$. Expand the exponential in a Taylor series:

$$e^{h\nu/k_BT} = 1 + \frac{h\nu}{k_BT} + \frac{1}{2}\left(\frac{h\nu}{k_BT}\right)^2 + \cdots$$

Keeping only the first two terms:

$$e^{h\nu/k_BT} - 1 \approx \frac{h\nu}{k_BT}.$$

Q2 – Planck's Radiation Law & the Classical Limit

Step 2: Substitute back into Planck's law.

$$u(\nu, T) \approx \frac{8\pi h\nu^3}{c^3} \cdot \frac{k_B T}{h\nu} = \frac{8\pi\nu^2}{c^3} k_B T = u_{\mathrm{RJ}}(\nu, T).$$

Step 3: High-frequency behaviour $h\nu \gg k_B T$. The exponential dominates:

$$e^{h\nu/k_BT} - 1 \approx e^{h\nu/k_BT} \;\Rightarrow\; u(\nu, T) \approx \frac{8\pi h\nu^3}{c^3} e^{-h\nu/k_BT} \xrightarrow{\nu\to\infty} 0.$$

The exponential decay removes the ultraviolet catastrophe. ✓

Conclusion. Planck's formula is the correct generalisation that:

- recovers the Rayleigh–Jeans law at low ν, and
- vanishes exponentially at high ν, giving a finite total energy.

Q3 – Numerical Value of the Planck Spectral Density

Question 3

Compute the spectral energy density $u(\nu, T)$ for $\nu = 3 \times 10^{14}$ Hz at $T = 6000$ K. Give the numerical value in J s m^{-3}.

Step 1: Compute the exponent.

$$\frac{h\nu}{k_B T} = \frac{(6.626 \times 10^{-34})(3 \times 10^{14})}{(1.381 \times 10^{-23})(6000)} \approx 2.399$$

Step 2: Denominator. $e^{2.399} - 1 \approx 10.01$. **Step 3: Prefactor.**

Q3 – Numerical Value of the Planck Spectral Density

$$8\pi h\nu^3 = 8\pi(6.626 \times 10^{-34})(2.7 \times 10^{43}) \approx 4.50 \times 10^{11}, \qquad c^3 = (3 \times 10^8)^3 = 2.7 \times 10^{25}$$

$$\frac{8\pi h\nu^3}{c^3} \approx \frac{4.50 \times 10^{11}}{2.7 \times 10^{25}} \approx 1.67 \times 10^{-14} \text{ J s m}^{-3}$$

Step 4: Combine.

$$\boxed{u \approx \frac{1.67 \times 10^{-14}}{10.01} \approx 1.67 \times 10^{-15} \text{ J s m}^{-3}}$$

Q4 – Temperature Needed to Double the Radiated Power

Question 4

A blackbody at $T_1 = 300$ K radiates total power P_1 per unit area. What temperature T_2 doubles this power? (Stefan–Boltzmann law $j = \sigma T^4$.)

Step 1: Set up the ratio. Since $P \propto T^4$, requiring $P_2 = 2P_1$ gives

$$\left(\frac{T_2}{T_1}\right)^4 = 2.$$

Step 2: Solve. $T_2 = T_1 \cdot 2^{1/4}$, and $2^{1/4} \approx 1.1892$. **Step 3: Evaluate.**

$$\boxed{T_2 = 300 \text{ K} \times 1.1892 \approx 357 \text{ K}}$$

A modest $\approx 19\%$ increase, reflecting the strong T^4 dependence of the radiated power.

Question 5

Starting from Planck's radiation law expressed in terms of *frequency* ν, derive the frequency-space version of Wien's displacement law, $\nu_{\max}/T = \text{const}$, and evaluate the constant.

Step 1: Planck's law in frequency form.

$$u(\nu, T) = \frac{8\pi h\nu^3}{c^3} \cdot \frac{1}{e^{h\nu/k_B T} - 1}.$$

Step 2: Set up the maximisation. Let $x = h\nu/(k_B T)$, so $\nu = x k_B T/h$. Maximising u over ν is equivalent to maximising

$$g(x) = \frac{x^3}{e^x - 1}.$$

Step 3: Differentiate and set to zero.

$$\frac{dg}{dx} = \frac{3x^2(e^x - 1) - x^3 e^x}{(e^x - 1)^2} = 0 \;\Rightarrow\; 3(e^x - 1) = xe^x \;\Rightarrow\; (3 - x)e^x = 3.$$

Step 4: Solve numerically. Iterating (Newton–Raphson) gives the unique positive root

$$x_{\max} \approx 2.8214.$$

(Note this is *different* from the wavelength-space root $x \approx 4.965$ — confirming $\nu_{\max} \neq c/\lambda_{\max}$.)

Step 5: Recover the displacement law. Since $x_{\max} = h\nu_{\max}/(k_B T)$:

$$\boxed{\frac{\nu_{\max}}{T} = \frac{x_{\max} k_B}{h} = \frac{2.8214 \times (1.381 \times 10^{-23})}{6.626 \times 10^{-34}} \approx 5.88 \times 10^{10}\ \text{Hz K}^{-1}.}$$

This is the (lesser-known) frequency-form Wien constant. Naively converting the wavelength-form constant via $\nu = c/\lambda_{\max}$ instead gives $c/b \approx 1.03 \times 10^{11}$ Hz K^{-1} – *larger*

Q5 – Wien's Displacement Law in Frequency Form

than the true value by a factor of ≈ 1.76. So $\nu_{\max} \neq c/\lambda_{\max}$: the actual frequency-space peak sits at roughly $0.57\times$ the naively-converted value, because the two peaks sit at different points of the spectrum (the Jacobian $d\nu/d\lambda$ skews the shape of the distribution).

Q6 – Surface Temperature of a Star from Wien's Law

Question 6

Using Wien's displacement law $\lambda_{\max}T = 2.898 \times 10^{-3}$ m K, find the surface temperature of a star whose blackbody spectrum peaks at $\lambda_{\max} = 145$ nm.

Step 1: Solve Wien's law for T.

$$T = \frac{b}{\lambda_{\max}} = \frac{2.898 \times 10^{-3}\ \text{m K}}{145 \times 10^{-9}\ \text{m}}$$

Step 2: Evaluate.

$$\boxed{T \approx 2.0 \times 10^{4}\ \text{K}}$$

This is a hot B-type star ($T \sim 10{,}000$–$30{,}000$ K); its peak lies in the far UV, consistent with a blue-white appearance. (O-type stars are hotter still, $T \gtrsim 30{,}000$ K.)

Question 7

State Einstein's photoelectric equation and explain the three experimental observations it accounts for that classical wave theory could not.

Einstein's equation (1905):

$$K_{\max} = h\nu - \phi = h\nu - h\nu_0 = eV_0,$$

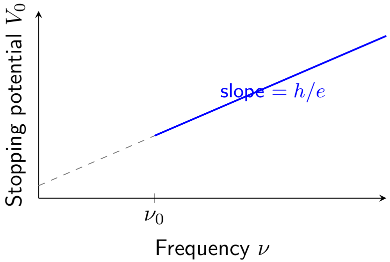
where $\phi = h\nu_0$ is the work function, ν_0 the threshold frequency, and V_0 the stopping potential.

Step 1: Observation 1 – Existence of a threshold frequency. Classically, light of *any* frequency should eject electrons given enough intensity or time. Experimentally, no electrons are emitted for $\nu < \nu_0$, however intense the light. In Einstein's equation, for $\nu < \nu_0$: $K_{\max} = h\nu - h\nu_0 < 0$, which is unphysical – hence no emission, exactly as observed.

Step 2: Observation 2 – Instantaneous emission. Classically, energy would need time to accumulate in the electron from a weak wave. However, photoelectrons are observed to be emitted with *no measurable time delay* ($< 10^{-9}$ s even for very faint light). In the photon picture, a single quantum $h\nu$ is absorbed instantaneously by one electron.

Step 3: Observation 3 – $K_{\max}$ independent of intensity. $K_{\max} = h\nu - \phi$ depends *only* on ν, not on the light intensity I. Increasing I increases the number of photons per second (hence the photocurrent), but not the energy per photon. This predicts a linear V_0 vs. ν graph with slope h/e, independent of intensity – exactly as observed by Millikan.

Q7 – Einstein's Photoelectric Equation



Q8 – Photoelectric Calculation (Copper)

Question 8

The work function of copper is $\phi = 4.5\,\text{eV}$. Find (a) the threshold wavelength, and (b) K_{max} for $\lambda = 200\,\text{nm}$. $(hc = 1240\ \text{eV nm})$

Step 1: Part (a): Threshold wavelength. At threshold, the photon energy exactly equals the work function: $hc/\lambda_0 = \phi$.

$$\lambda_0 = \frac{hc}{\phi} = \frac{1240\ \text{eV nm}}{4.5\ \text{eV}} = \mathbf{275.6}\ \text{nm}.$$

Light with $\lambda > 275.6\,\text{nm}$ (near-UV) cannot eject electrons from copper.

Step 2: Part (b): Check whether emission occurs at $\lambda = 200\,\text{nm}$.

$$E_{\text{photon}} = \frac{hc}{\lambda} = \frac{1240}{200} = 6.20\ \text{eV}.$$

Since $E_{\text{photon}} = 6.20\ \text{eV} > \phi = 4.5\ \text{eV}$ (or equivalently $\lambda < \lambda_0$), photoemission does occur.

Q8 – Photoelectric Calculation (Copper)

Step 3: Apply Einstein's equation.

$$K_{\text{max}} = E_{\textsf{photon}} - \phi = 6.20 - 4.5 = \mathbf{1.70}\ \text{eV}.$$

Step 4: Stopping potential. The minimum retarding potential needed to stop the fastest electrons:

$$eV_0 = K_{\text{max}} \Rightarrow V_0 = 1.70\ \text{V}.$$

Q9 – Photoelectric Effect on a Metal Surface

Question 9

UV light of wavelength $\lambda = 200$ nm strikes a metal with work function $\phi = 4.5$ eV. Find (a) the photon energy in eV, (b) KE_{max} of the photoelectrons, and (c) the stopping potential V_0.

Step 1: (a) Photon energy.

$$h\nu = \frac{hc}{\lambda} = \frac{(6.626 \times 10^{-34})(3 \times 10^{8})}{200 \times 10^{-9}} = 9.94 \times 10^{-19} \text{ J} = 6.21 \text{ eV}$$

Step 2: (b) Einstein's equation.

Q9 – Photoelectric Effect on a Metal Surface

$$KE_{\text{max}} = h\nu - \phi = 6.21 - 4.5 = 1.71 \text{ eV}$$

Step 3: (c) Stopping potential. Since $eV_0 = KE_{\text{max}}$,

$$\boxed{V_0 = 1.71 \text{ V}}$$

Q10 – Checking Whether Photoemission Occurs

Question 10

For the metal of Q9 ($\phi = 4.5$ eV), a second light source of $\lambda = 350$ nm is used. Does emission occur? Calculate $KE_{\max}$ if yes, or find the wavelength gap to the threshold if no.

Step 1: Photon energy at 350 nm.

$$h\nu = \frac{hc}{350 \times 10^{-9}} = 5.68 \times 10^{-19}\ \text{J} = 3.55\ \text{eV}$$

Step 2: Compare with ϕ. Since 3.55 eV $< \phi = 4.5$ eV, no emission occurs. **Step 3:**

Q10 – Checking Whether Photoemission Occurs

Threshold wavelength.

$$\lambda_0 = \frac{hc}{\phi} = \frac{1.988 \times 10^{-25}}{4.5 \times 1.6 \times 10^{-19}} \approx 276 \text{ nm}$$

$$\boxed{\text{The source must be blue-shifted by } 350 - 276 = 74 \text{ nm.}}$$

Q11 – Determining h and ϕ from Two Data Points

Question 11

In a photoelectric experiment the stopping potential is $V_{0,1} = 1.0\,\text{V}$ for $\lambda_1 = 300\,\text{nm}$ and $V_{0,2} = 0.2\,\text{V}$ for $\lambda_2 = 400\,\text{nm}$. Determine Planck's constant h and the work function ϕ.

Step 1: Write Einstein's equation for each case.

$$eV_{0,1} = \frac{hc}{\lambda_1} - \phi, \qquad eV_{0,2} = \frac{hc}{\lambda_2} - \phi.$$

Step 2: Subtract to eliminate ϕ.

$$e(V_{0,1} - V_{0,2}) = hc\left(\frac{1}{\lambda_1} - \frac{1}{\lambda_2}\right).$$

Step 3: Compute the numerical values.

$$V_{0,1} - V_{0,2} = 1.0 - 0.2 = 0.8\ \text{V},$$

$$\frac{1}{\lambda_1} - \frac{1}{\lambda_2} = \frac{1}{300} - \frac{1}{400} = \frac{4-3}{1200} = \frac{1}{1200}\ \text{nm}^{-1} = 8.33 \times 10^{-4}\ \text{nm}^{-1}.$$

Q11 – Determining h and ϕ from Two Data Points

Step 4: Solve for h.

$$h = \frac{e(V_{0,1} - V_{0,2})}{c\left(\frac{1}{\lambda_1} - \frac{1}{\lambda_2}\right)} = \frac{(0.8\text{ eV})}{(3 \times 10^{17}\text{ nm/s})(8.33 \times 10^{-4}\text{ nm}^{-1})}.$$

$$h = \frac{0.8}{2.499 \times 10^{14}}\text{ eV s} \approx 3.20 \times 10^{-15}\text{ eV s} = \mathbf{5.13 \times 10^{-34}}\text{ J s}.$$

(Converting: 3.20×10^{-15} eV s $\times\, 1.602 \times 10^{-19}$ J/eV $= 5.13 \times 10^{-34}$ J s. Deviation from the accepted 6.626×10^{-34} J s reflects rounding in the given data.)

Step 5: Solve for ϕ. Using case 1 with the standard value $hc = 1240\,\text{eV nm}$ for consistency:

$$\frac{hc}{\lambda_1} = \frac{1240}{300} = 4.133\text{ eV},$$

$$\phi = \frac{hc}{\lambda_1} - eV_{0,1} = 4.133 - 1.0 = \mathbf{3.13}\text{ eV}.$$

Q11 – Determining h and ϕ from Two Data Points

Step 6: Verification with case 2.

$$\phi = \frac{hc}{\lambda_2} - eV_{0,2} = \frac{1240}{400} - 0.2 = \mathbf{2.9}\text{ eV}.$$

Note on data consistency. The two independent estimates of ϕ (3.13 eV from case 1 and 2.9 eV from case 2) differ by 0.23 eV. This is not an arithmetic error: the given stopping potentials are mutually inconsistent when combined with the exact value of h. Experimentally, measured stopping potentials always carry uncertainty; treating the two ϕ values as independent estimates and averaging gives $\bar{\phi} \approx 3.0$ eV, a reasonable real-world result.

Q12 – Extracting h/e and ϕ from Stopping-Potential Data

Question 12

Two frequencies $\nu_1 = 6 \times 10^{14}$ Hz and $\nu_2 = 9 \times 10^{14}$ Hz give stopping potentials $V_1 = 0.60$ V and $V_2 = 1.80$ V for the same metal. Determine h/e and the work function ϕ from these data alone.

Step 1: Use the linear relation $V_0 = (h/e)\nu - \phi/e$ at both points and subtract:

$$\frac{h}{e} = \frac{V_2 - V_1}{\nu_2 - \nu_1} = \frac{1.20}{3 \times 10^{14}} = 4.0 \times 10^{-15} \text{ V s}$$

Step 2: Back-substitute to find ϕ.

Q12 – Extracting h/e and ϕ from Stopping-Potential Data

$$\frac{\phi}{e} = \frac{h}{e}\nu_1 - V_1 = (4.0 \times 10^{-15})(6 \times 10^{14}) - 0.60 = 1.8 \text{ V}$$

$$\boxed{\phi = 1.8 \text{ eV}}$$

Q13 – Compton Recoil Electron Kinetic Energy

Question 13

A photon of energy E Compton-scatters off a stationary free electron (mass m_e) through angle θ. Using relativistic conservation of energy and momentum directly (without first deriving the wavelength-shift formula), derive an expression for the kinetic energy T of the recoiling electron as a function of E and θ.

Step 1: Set up momentum conservation. Photon: initial momentum E/c along x; final momentum E'/c at angle θ. Electron: initial momentum 0; final momentum p_e at angle ϕ.

$$x: \ \frac{E}{c} = \frac{E'}{c}\cos\theta + p_e\cos\phi, \qquad y: \ 0 = \frac{E'}{c}\sin\theta - p_e\sin\phi.$$

Step 2: Eliminate ϕ. Squaring and adding:

$$p_e^2c^2 = E^2 - 2EE'\cos\theta + E'^2. \tag{1}$$

Q13 – Compton Recoil Electron Kinetic Energy

Step 3: Use energy conservation. $E + m_e c^2 = E' + E_e$, so $E_e = E - E' + m_e c^2$. With $E_e^2 = p_e^2 c^2 + m_e^2 c^4$:

$$(E - E')^2 + 2(E - E')m_e c^2 + m_e^2 c^4 = p_e^2 c^2 + m_e^2 c^4.$$

Step 4: Substitute (1) and simplify.

$$(E - E')^2 + 2(E - E')m_e c^2 = E^2 - 2EE' \cos\theta + E'^2$$

$$\Rightarrow\ 2(E - E')m_e c^2 = 2EE'(1 - \cos\theta)\ \Rightarrow\ (E - E')m_e c^2 = EE'(1 - \cos\theta).$$

Step 5: Solve for $T = E - E'$. Set $E' = E - T$:

$$T\,m_e c^2 = E(E - T)(1 - \cos\theta)\ \Rightarrow\ T\left[m_e c^2 + E(1 - \cos\theta)\right] = E^2(1 - \cos\theta).$$

$$\boxed{T = \frac{E^2(1 - \cos\theta)}{m_e c^2 + E(1 - \cos\theta)}.}$$

Check: $\theta \to 0 \Rightarrow T \to 0$ (grazing collision transfers no energy); $\theta = \pi$ (backscatter) maximises T, as expected.

Q14 – Compton Shift at $\theta = 60°$

Question 14

X-rays of wavelength $\lambda = 0.071$ nm are scattered through $60°$ from free electrons. Calculate (a) the Compton shift, and (b) the kinetic energy of the recoiling electron.

Step 1: Part (a): Compton shift. With $\theta = 60°$, $\cos 60° = 0.500$:

$$\Delta\lambda = \frac{h}{m_e c}(1 - \cos 60°) = \frac{h}{m_e c}(1 - 0.500) = \frac{h}{2m_e c}.$$

Substituting $h/m_e c = 2.426 \times 10^{-12}$ m:

$$\Delta\lambda = \frac{2.426 \times 10^{-12}}{2}\ \text{m} = 1.213 \times 10^{-12}\ \text{m} = \mathbf{0.001213}\ \text{nm}.$$

The scattered photon has wavelength:

$$\lambda' = \lambda + \Delta\lambda = 0.071 + 0.001213 = 0.072213\ \text{nm}.$$

Q14 – Compton Shift at $\theta = 60^\circ$

Step 2: Part (b): Kinetic energy of the recoiling electron. By energy conservation, $K_e = h\nu - h\nu'$, i.e.

$$K_e = hc\left(\frac{1}{\lambda} - \frac{1}{\lambda'}\right).$$

Step 3: Compute each term.

$$\frac{1}{\lambda} = \frac{1}{0.071\ \text{nm}} = 14.085\ \text{nm}^{-1}, \qquad \frac{1}{\lambda'} = \frac{1}{0.072213\ \text{nm}} = 13.848\ \text{nm}^{-1}.$$

$$\frac{1}{\lambda} - \frac{1}{\lambda'} = 14.085 - 13.848 = 0.237\ \text{nm}^{-1}.$$

Step 4: Final result. Using $hc = 1240\,\text{eV}\,\text{nm}$:

$$K_e = 1240\ \text{eV}\,\text{nm} \times 0.237\ \text{nm}^{-1} \approx \mathbf{294}\ \text{eV}.$$

Q15 – Compton Scattering at $\theta = 135°$

Question 15

X-rays of wavelength 0.0711 nm scatter off electrons at $\theta = 135°$. Find λ' and the kinetic energy gained by the electron in keV.

Step 1: Compton shift. $1 - \cos 135° = 1.7071$, so

$$\Delta\lambda = (2.426 \times 10^{-12})(1.7071) = 4.14 \times 10^{-12}\ \text{m} = 0.00414\ \text{nm}$$

Step 2: Scattered wavelength. $\lambda' = 0.0711 + 0.00414 = 0.07524$ nm. **Step 3: Electron**

Q15 – Compton Scattering at $\theta = 135°$

kinetic energy (energy lost by the photon):

$$KE = hc\left(\frac{1}{\lambda} - \frac{1}{\lambda'}\right) = 1.988 \times 10^{-25} \times (1.407 - 1.329) \times 10^{10}$$

$$\boxed{KE \approx 1.55 \times 10^{-16}\ \mathsf{J} \approx 0.97\ \mathsf{keV}}$$

Q16 – Computing the Compton Wavelength from First Principles

Question 16

Compute the Compton wavelength of the electron, $\lambda_C = h/(m_e c)$, from first principles, and explain its physical significance.

Step 1: Denominator.

$$m_e c = (9.11 \times 10^{-31})(3 \times 10^{8}) = 2.733 \times 10^{-22} \text{ kg m s}^{-1}$$

Step 2: Evaluate.

$$\boxed{\lambda_C = \frac{h}{m_e c} = \frac{6.626 \times 10^{-34}}{2.733 \times 10^{-22}} \approx 2.42 \text{ pm}}$$

When an incoming photon's wavelength is comparable to λ_C, its momentum transfer is comparable to $m_e c$, so the electron recoils relativistically and relativistic kinematics become essential.

Question 17

What is wave–particle duality? Briefly describe two experiments that demonstrate the particle nature and two that demonstrate the wave nature of light/matter.

Concept. Every quantum entity (photons, electrons, etc.) exhibits both wave-like behaviour (interference, diffraction) *and* particle-like behaviour (localised, quantised energy/momentum exchange). The manifestation depends on the experiment performed.

Particle-like behaviour:

(i) Photoelectric effect (Q7–Q11). Light knocks out electrons only above a threshold frequency ν_0, and the energy transfer is instantaneous and quantised ($E = h\nu$). This is explainable *only* by treating light as photons – a wave of any frequency should eventually supply sufficient energy.

(ii) Compton effect (Q13–Q14). X-ray photons scatter off electrons exactly like billiard-ball collisions, conserving energy and momentum as discrete particles. The shift $\Delta\lambda = (h/m_e c)(1 - \cos\theta)$ follows from particle kinematics.

Wave-like behaviour:

(i) Young's double-slit experiment. Light (and electrons, atoms) produces an interference fringe pattern that requires superposition of waves from both slits. The pattern disappears if the path is known (Q52–Q54).

(ii) Davisson–Germer electron diffraction (Q23). Electrons accelerated through 54 V diffract off a nickel crystal lattice and produce a Bragg diffraction peak at exactly the angle predicted by $\lambda = h/p$. This is a wave phenomenon that cannot be explained by classical particle mechanics.

Conclusion. Together, these results led to the unified description in which every particle is associated with a wave of wavelength $\lambda = h/p$ (de Broglie), probability amplitudes interfere as waves (Born), but energy and momentum are still exchanged in discrete quanta.

Question 18

Derive the de Broglie relation $p = \hbar k$ by requiring that the phase of a free particle's matter wave be a Lorentz invariant, rather than by analogy with the photon relation $p = h/\lambda$.

Step 1: Postulate a plane matter wave.

$$\Psi(x,t) = Ae^{i(kx-\omega t)},$$

with phase $\Phi(x,t) = kx - \omega t$.

Step 2: Physical requirement: the phase is observer-independent. Wave crests ($\Phi = \text{const}$) are physical events; every observer must agree on the phase at a given crest. So Φ must be a *Lorentz scalar* — the same number in every inertial frame.

Q18 – De Broglie Relation from Lorentz Invariance

Step 3: Construct the unique linear Lorentz scalar. For a free particle, the energy–momentum four-vector is $p^\mu = (E/c, \mathbf{p})$ and the spacetime four-vector is $x^\mu = (ct, \mathbf{x})$. Using the metric signature $(+, -, -, -)$, the only Lorentz-invariant quantity linear in t and x built from these is the contraction

$$p^\mu x_\mu = Et - \mathbf{p} \cdot \mathbf{x}.$$

Dividing by $\hbar$ gives a dimensionless invariant. Because an overall constant sign in the phase is a convention, we adopt

$$\Phi = \frac{\mathbf{p} \cdot \mathbf{x} - Et}{\hbar},$$

which matches the wave ansatz $\Phi = kx - \omega t$ in Step 1 term-by-term with no sign ambiguity.

Step 4: Match to the wave ansatz. Comparing $\Phi = (\mathbf{p}/\hbar) \cdot \mathbf{x} - (E/\hbar)t$ to $\Phi = kx - \omega t$ identifies directly:

$$\omega = \frac{E}{\hbar}, \qquad k = \frac{p}{\hbar}.$$

Step 5: Conclusion.

$$\boxed{p = \hbar k \implies \lambda = \frac{h}{p}.}$$

Because this argument used only Lorentz covariance (not a photon analogy), it applies equally to massive and massless particles, and is the route usually used to justify why $E = \hbar\omega$, $p = \hbar k$ must hold for *any* relativistic particle, not just light.

Question 19

Calculate the de Broglie wavelength of (a) an electron accelerated through 200 V and (b) a 0.20 kg ball moving at 30 m/s. Comment on the results.

Step 1: Part (a): Electron accelerated through $V = 200$ V. Kinetic energy gained:

$$K = eV = (1.602 \times 10^{-19}\ \text{C})(200\ \text{V}) = 3.204 \times 10^{-17}\ \text{J}.$$

Step 2: Compute the momentum.

$$p = \sqrt{2m_eK} = \sqrt{2(9.109 \times 10^{-31})(3.204 \times 10^{-17})} = \sqrt{5.840 \times 10^{-47}} = 7.642{\times}10^{-24}\ \text{kg m/s}.$$

Step 3: Compute the wavelength.

$$\lambda = \frac{h}{p} = \frac{6.626 \times 10^{-34}}{7.642 \times 10^{-24}} = 8.67 \times 10^{-11}\ \text{m} = \mathbf{0.867}\ \text{Å}.$$

This is comparable to atomic/crystal spacings (∼1–5 Å), explaining why electrons of this energy diffract from crystal lattices.

Step 4: Part (b): Macroscopic ball ($m = 0.20$ kg, $v = 30$ m/s).

$$p = mv = 0.20\ \text{kg} \times 30\ \text{m/s} = 6.0\ \text{kg m/s}.$$

$$\lambda = \frac{h}{p} = \frac{6.626 \times 10^{-34}}{6.0} = 1.10 \times 10^{-34}\ \text{m}.$$

Step 5: Comment. $\lambda = 1.10 \times 10^{-34}$ m is vastly smaller than the nucleus ($\sim 10^{-15}$ m) – far beyond any measurement capability. Wave effects for macroscopic objects are utterly unobservable, which is why classical mechanics is an excellent description at everyday scales.

Question 20

A proton and an electron have the same de Broglie wavelength $\lambda = 0.1$ nm. Find (a) the proton's kinetic energy in eV and (b) the ratio KE_e/KE_p.

Step 1: (a) Momentum.

$$p = \frac{h}{\lambda} = \frac{6.626 \times 10^{-34}}{10^{-10}} = 6.626 \times 10^{-24} \text{ kg m s}^{-1}$$

Step 2: Proton kinetic energy.

$$KE_p = \frac{p^2}{2m_p} \approx 1.313 \times 10^{-20} \text{ J} = 0.082 \text{ eV}$$

Step 3: (b) Ratio. Since p is identical, $KE \propto 1/m$:

Q20 – Equal-Wavelength Proton and Electron: Kinetic Energy Ratio

$$\boxed{\frac{KE_e}{KE_p} = \frac{m_p}{m_e} = \frac{1.673 \times 10^{-27}}{9.11 \times 10^{-31}} \approx 1836}$$

Q21 – Wave Packets

Question 21

What is a wave packet? Explain qualitatively how superposing many plane waves of slightly different wavenumber produces a localised packet representing a moving particle.

Step 1: Problem with a single plane wave. A single plane wave $\psi(x,t) = Ae^{i(kx-\omega t)}$ has a perfectly defined wavenumber k (hence a definite momentum $p = \hbar k$) but is spread over *all* of space – it cannot represent a localised particle.

Step 2: Superposition of plane waves. Because the Schrödinger equation is linear, any superposition is also a valid state:

$$\Psi(x,t) = \int_{-\infty}^{\infty} A(k)\, e^{i(kx-\omega(k)t)}\, dk,$$

where $A(k)$ is appreciable only in a narrow range Δk around some central wavenumber k_0.

Q21 – Wave Packets

Step 3: Constructive and destructive interference. At $t = 0$, the components add **constructively** near one point x_0 (where their phases line up) and **destructively** away from x_0 (where phases are randomly distributed). The result is a packet peaked at x_0 and decaying away from it.

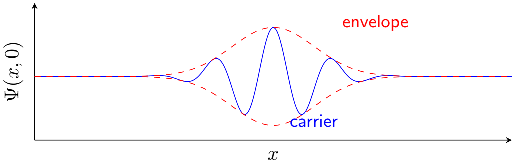
Step 4: Localisation and the uncertainty relation. The narrower the spread Δk in wavenumber, the broader the spatial width Δx of the packet, and vice versa:

$$\Delta x \cdot \Delta k \sim 1 \;\Rightarrow\; \Delta x \cdot \Delta p \sim \hbar.$$

This is the precursor of the Heisenberg uncertainty relation (Q37–Q41).

Step 5: Group vs. phase velocity. The packet's peak moves with the **group velocity** $v_g = d\omega/dk$, which equals the classical particle velocity (Q28–Q29). The individual wave crests inside the packet move with the (different) **phase velocity** $v_p = \omega/k$ (Q27).

Q21 – Wave Packets



Q22 – Superposition Principle

Question 22

State the principle of superposition as applied to quantum wavefunctions, and explain its physical consequence for interference experiments.

Statement. If ψ_1 and ψ_2 are both solutions of the Schrödinger equation, then any linear combination

$$\psi = c_1\psi_1 + c_2\psi_2, \quad c_1, c_2 \in \mathbb{C},$$

is also a physically allowed state. This follows directly from the *linearity* of the Schrödinger equation.

Consequence: interference of probabilities. The probability density of the combined state is

$$|\psi|^2 = |c_1\psi_1 + c_2\psi_2|^2 = |c_1\psi_1|^2 + |c_2\psi_2|^2 + \underbrace{2\,\mathrm{Re}(c_1^* c_2 \psi_1^* \psi_2)}_{\text{interference term}}.$$

Q22 – Superposition Principle

The cross term $2\,\mathrm{Re}(c_1^* c_2 \psi_1^* \psi_2)$ is the **interference term**:

- It is **positive** (constructive interference) when $c_1^* c_2 \psi_1^* \psi_2$ is real and positive – giving $|\psi|^2 > |c_1\psi_1|^2 + |c_2\psi_2|^2$.
- It is **negative** (destructive interference) when the phase difference is π – giving $|\psi|^2 < |c_1\psi_1|^2 + |c_2\psi_2|^2$ (even zero).

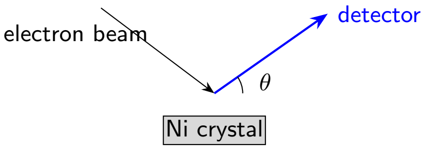
In the double-slit experiment (Q51–Q54), this is why the combined probability pattern from both slits open is *not* the simple sum of the two single-slit patterns, but shows alternating bright and dark fringes – a direct manifestation of the superposition principle.

Q23 – Davisson–Germer Experiment

Question 23

Describe the Davisson–Germer experiment and explain how its results confirmed de Broglie's hypothesis.

Setup. A beam of electrons accelerated through a known potential V is directed at a nickel single crystal. A movable detector records the intensity of scattered electrons as a function of angle θ.



Q23 – Davisson–Germer Experiment

Step 1: Experimental observation. For electrons accelerated through $V = 54\,\text{V}$, a pronounced maximum in scattered intensity was observed at $\theta = 50^\circ$, exactly analogous to a Bragg diffraction peak from X-rays.

Step 2: Bragg's law applied to the nickel lattice. Using the known nickel lattice spacing $d = 0.91\,\text{Å}$ and Bragg's condition $n\lambda = 2d\sin\varphi$ (with φ the glancing angle), the diffraction peak corresponds to:

$$\lambda_{\text{exp}} \approx 1.65\ \text{Å}.$$

Step 3: De Broglie prediction. Independently, from $\lambda = h/\sqrt{2m_e eV}$:

$$\lambda_{dB} = \frac{6.626\times10^{-34}}{\sqrt{2(9.109\times10^{-31})(1.602\times10^{-19})(54)}} = \frac{6.626\times10^{-34}}{3.975\times10^{-24}} \approx 1.67\ \text{Å}.$$

Step 4: Agreement. $\lambda_{\text{exp}} \approx \lambda_{dB}$ to within experimental error. Since electrons produced a diffraction pattern – a *wave* phenomenon – with exactly the wavelength predicted by $\lambda = h/p$, this confirmed that material particles possess wave properties, as de Broglie had hypothesised.

Q24 – Electron de Broglie Wavelength and the Davisson–Germer Angle

Question 24

An electron is accelerated from rest through $V = 54$ V. (a) Find its de Broglie wavelength. (b) Nickel surface rows have spacing $D = 2.15$ Å. At what diffraction angle ϕ is the first-order maximum predicted by $D \sin\phi = \lambda$?

Step 1: (a) de Broglie wavelength using $\lambda = h/\sqrt{2m_e eV}$:

$$\lambda = \frac{6.626 \times 10^{-34}}{\sqrt{2(9.11 \times 10^{-31})(1.6 \times 10^{-19})(54)}} \approx 1.67 \text{ Å}$$

Step 2: (b) Diffraction angle.

$$\sin\phi = \frac{\lambda}{D} = \frac{1.67}{2.15} = 0.777$$

$$\boxed{\phi = \sin^{-1}(0.777) \approx 51.0^\circ}$$

This is in excellent agreement with Davisson and Germer's 1927 observation.

Q25 – Kinetic Energy of a Neutron from Its de Broglie Wavelength

Question 25

A neutron ($m_n = 1.675 \times 10^{-27}$ kg) has a de Broglie wavelength of 1.5 Å. (a) Find its kinetic energy in eV. (b) Is this neutron relativistic?

Step 1: (a) Momentum.

$$p = \frac{h}{\lambda} = \frac{6.626 \times 10^{-34}}{1.5 \times 10^{-10}} = 4.417 \times 10^{-24} \text{ kg m s}^{-1}$$

Step 2: Kinetic energy.

Q25 – Kinetic Energy of a Neutron from Its de Broglie Wavelength

$$KE = \frac{p^2}{2m_n} = \frac{(4.417 \times 10^{-24})^2}{2(1.675 \times 10^{-27})} \approx 5.82 \times 10^{-21} \text{ J}$$

$$\boxed{KE \approx 0.036 \text{ eV}}$$

Step 3: (b) Speed check. $v = p/m_n \approx 2.64 \times 10^3$ m/s, so $v/c \approx 8.8 \times 10^{-6}$: completely non-relativistic. ✓

Q26 – Required Electron Energy for a Bragg Peak

Question 26

What de Broglie wavelength must electrons have for their $n = 1$ Bragg peak off a crystal of lattice spacing $d = 2.0$ Å to appear at $2\theta = 60°$? Find the required electron kinetic energy.

Step 1: Bragg condition ($n = 1$).

$$\lambda = 2d\sin\theta = 2(2.0\ \text{Å})\sin 30° = 2.0\ \text{Å}$$

Step 2: Kinetic energy using $KE = h^2/(2m_e\lambda^2)$:

Q26 – Required Electron Energy for a Bragg Peak

$$KE = \frac{(6.626 \times 10^{-34})^2}{2(9.11 \times 10^{-31})(2.0 \times 10^{-10})^2} \approx 6.02 \times 10^{-18}\ \text{J}$$

$$\boxed{KE \approx 37.6\ \text{eV}}$$

Q27 – Phase Velocity

Question 27

Define phase velocity for a wave and derive its expression in terms of angular frequency ω and wavenumber k.

Definition. The **phase velocity** v_p is the speed at which a point of constant phase (e.g. a crest) of a monochromatic wave travels.

Step 1: Derivation from a plane wave. Consider the plane wave $\psi(x,t) = A\cos(kx - \omega t)$. A point of constant phase satisfies:

$$kx - \omega t = \text{const.}$$

Step 2: Differentiate with respect to time.

$$k\frac{dx}{dt} - \omega = 0 \;\Rightarrow\; \boxed{v_p = \frac{dx}{dt} = \frac{\omega}{k}.}$$

Q27 – Phase Velocity

Step 3: Apply to a non-relativistic matter wave. Using $E = \hbar\omega$ and $p = \hbar k$:

$$v_p = \frac{\omega}{k} = \frac{E/\hbar}{p/\hbar} = \frac{E}{p}.$$

For a free particle with $E = p^2/2m$:

$$v_p = \frac{p^2/2m}{p} = \frac{p}{2m} = \frac{mv}{2m} = \frac{v}{2},$$

where $v = p/m$ is the classical particle speed.

Step 4: Remark. The phase velocity of a matter wave is *half* the particle speed. It is *not* directly observable as the particle's velocity – that role belongs to the group velocity (Q28–Q29).

Q28 – Group Velocity from a Continuous Wave Packet

Question 28

Starting from a continuous superposition (Fourier wave packet) $\Psi(x,t) = \int A(k)\, e^{i(kx-\omega(k)t)}\, dk$ sharply peaked around k_0, derive that the packet's envelope travels at $v_g = d\omega/dk\big|_{k_0}$.

Step 1: Expand $\omega(k)$ about the peak. Since $A(k)$ is sharply peaked at k_0, expand to first order:

$$\omega(k) \approx \omega_0 + v_g(k-k_0), \qquad v_g \equiv \left.\frac{d\omega}{dk}\right|_{k_0}.$$

Step 2: Substitute into the packet integral. Let $s = k - k_0$:

$$\Psi(x,t) = \int A(k_0+s)\, e^{i[(k_0+s)x-(\omega_0+v_g s)t]}\, ds.$$

Q28 – Group Velocity from a Continuous Wave Packet

Step 3: Factor out the carrier wave.

$$\Psi(x,t) = e^{i(k_0 x - \omega_0 t)} \int A(k_0 + s)\, e^{is(x - v_g t)}\, ds \equiv e^{i(k_0 x - \omega_0 t)}\, F(x - v_g t).$$

Step 4: Interpret the result. The envelope F depends on x and t *only through the combination* $x - v_g t$: it is a fixed profile rigidly translating in x at speed v_g, while the rapidly oscillating carrier $e^{i(k_0 x - \omega_0 t)}$ moves at the (generally different) phase velocity ω_0/k_0.

$$\boxed{v_g = \left.\frac{d\omega}{dk}\right|_{k_0}.}$$

Q29 – Phase–Group Velocity Relation for EM Waves in a Plasma

Question 29

An electromagnetic wave propagating in a plasma obeys the dispersion relation $\omega^2 = \omega_p^2 + c^2k^2$, where ω_p (the plasma frequency) is a constant. Derive v_p and v_g for this wave and show that $v_p v_g = c^2$.

Step 1: Phase velocity.

$$v_p = \frac{\omega}{k} = \frac{\sqrt{\omega_p^2 + c^2k^2}}{k}.$$

Note $v_p > c$ always (superluminal phase velocity – but no information travels at v_p).
Step 2: Group velocity. Differentiate $\omega^2 = \omega_p^2 + c^2k^2$ implicitly with respect to k:

$$2\omega\frac{d\omega}{dk} = 2c^2k \;\Rightarrow\; v_g = \frac{d\omega}{dk} = \frac{c^2k}{\omega}.$$

Q29 – Phase–Group Velocity Relation for EM Waves in a Plasma

Step 3: Combine.

$$v_p v_g = \left(\frac{\omega}{k}\right)\left(\frac{c^2 k}{\omega}\right) = c^2.$$

$$\boxed{v_p v_g = c^2.}$$

Step 4: Physical reading. Since $v_p > c$, the relation $v_p v_g = c^2$ forces $v_g < c$ always – the group velocity (which carries energy/information) never exceeds c, even though the phase velocity does. This is the same algebraic identity structure as the relativistic massive-particle case $v_p v_{\text{particle}} = c^2$, but here it arises from electromagnetic dispersion in a medium rather than from $E^2 = p^2c^2 + m^2c^4$.

Question 30

A particle of rest mass m_0 moves relativistically with speed v. Show that the phase velocity of its de Broglie wave is $v_p = c^2/v > c$, while the group velocity equals v.

Step 1: Relativistic energy and momentum.

$$E = \gamma m_0 c^2, \quad p = \gamma m_0 v, \quad \gamma = \frac{1}{\sqrt{1 - v^2/c^2}}.$$

Step 2: Phase velocity. Using $\omega = E/\hbar$ and $k = p/\hbar$:

$$v_p = \frac{\omega}{k} = \frac{E}{p} = \frac{\gamma m_0 c^2}{\gamma m_0 v} = \frac{c^2}{v}.$$

Since $v < c$, this gives $v_p = c^2/v > c$.

Remark: This does *not* violate relativity because v_p is not the speed at which energy or information travels – it is merely the speed of points of constant phase in the (infinite-extent) carrier wave.

Step 3: Relativistic dispersion relation. From $E^2 = p^2c^2 + m_0^2c^4$ with $E = \hbar\omega$ and $p = \hbar k$:

$$\hbar^2\omega^2 = \hbar^2k^2c^2 + m_0^2c^4 \;\Rightarrow\; \omega = c\sqrt{k^2 + \left(\frac{m_0c}{\hbar}\right)^2}.$$

Step 4: Group velocity. Differentiate ω with respect to k:

$$v_g = \frac{d\omega}{dk} = \frac{c \cdot k}{\sqrt{k^2 + (m_0c/\hbar)^2}} = \frac{c^2\hbar k}{\hbar\omega} = \frac{c^2p}{E}.$$

Substituting $p = \gamma m_0 v$ and $E = \gamma m_0 c^2$:

$$v_g = \frac{c^2(\gamma m_0 v)}{\gamma m_0 c^2} = v. \quad \checkmark$$

Step 5: Product check.

$$v_p \cdot v_g = \frac{c^2}{v} \cdot v = c^2,$$

a general result relating phase and group velocity for relativistic matter waves.

Q31 – Photon Number Density of Blackbody Radiation

Question 31

Starting from Planck's law, derive an expression for the number density of photons $n(T)$ (photons per unit volume) in blackbody radiation, and identify the constant in $n(T) = (\text{const})\, T^3$.

Step 1: Photon number per unit volume per unit frequency. The number density of modes per unit frequency is $g(\nu) = 8\pi\nu^2/c^3$, and the average photon *occupation number* per mode is the Bose factor $\bar{n}(\nu) = 1/(e^{h\nu/k_BT} - 1)$ (this is $u(\nu, T)$ divided by the energy per photon $h\nu$). So:

$$n(T) = \int_0^\infty \frac{8\pi\nu^2}{c^3} \cdot \frac{1}{e^{h\nu/k_BT} - 1}\, d\nu.$$

Step 2: Substitute $x = h\nu/k_BT$.

$$n(T) = \frac{8\pi}{c^3} \left(\frac{k_BT}{h}\right)^3 \int_0^\infty \frac{x^2}{e^x - 1}\, dx.$$

Step 3: Evaluate the integral by series expansion. For $x > 0$, $\dfrac{1}{e^x - 1} = \displaystyle\sum_{n=1}^{\infty} e^{-nx}$, so

$$\int_0^\infty \frac{x^2}{e^x - 1}\,dx = \sum_{n=1}^{\infty} \int_0^\infty x^2 e^{-nx}\,dx = \sum_{n=1}^{\infty} \frac{2}{n^3} = 2\zeta(3) \approx 2.4041.$$

(Compare: the energy-density integral, $\int_0^\infty x^3/(e^x - 1)\,dx = 6\zeta(4) = \pi^4/15$, uses $\zeta(4)$ instead – a different zeta value, because the extra power of x shifts which moment of the Bose distribution is being summed.)

Step 4: Assemble the result.

$$\boxed{n(T) = \frac{16\pi\zeta(3)}{(hc)^3}(k_B T)^3 \approx 60.4\left(\frac{k_B T}{hc}\right)^3.}$$

At $T = 2.725$ K (the cosmic microwave background), this gives $n \approx 4 \times 10^8$ photons per cubic metre – the famous CMB photon density.

Q32 – Photon Emission Rate of a Laser Pointer

Question 32

A 5 mW laser pointer emits light at $\lambda = 532$ nm. How many photons does it emit per second?

Step 1: Energy per photon.

$$E = \frac{hc}{\lambda} = \frac{1.988 \times 10^{-25}}{532 \times 10^{-9}} = 3.736 \times 10^{-19}\ \text{J}$$

Step 2: Photon rate.

$$N = \frac{P}{E} = \frac{5 \times 10^{-3}}{3.736 \times 10^{-19}}$$

$$\boxed{N \approx 1.34 \times 10^{16}\ \text{photons/s}}$$

Q33 – Solar Luminosity Estimate

Question 33

The surface temperature of the Sun is about 6000 K and its radius $R_\odot = 6.96 \times 10^8$ m. Treating the Sun as a black body, estimate its total radiated power (luminosity).

Step 1: Power per unit area (Stefan–Boltzmann).

$$E_{\text{total}} = \sigma T^4 = (5.670 \times 10^{-8})(6000)^4 \text{ W/m}^2.$$

Compute T^4:

$$6000^2 = 3.600 \times 10^7, \quad 6000^4 = (3.600 \times 10^7)^2 = 1.296 \times 10^{15}.$$

$$E_{\text{total}} = 5.670 \times 10^{-8} \times 1.296 \times 10^{15} = 7.348 \times 10^7 \text{ W/m}^2.$$

Q33 – Solar Luminosity Estimate

Step 2: Total surface area of the Sun.

$$A = 4\pi R_\odot^2 = 4\pi(6.96\times10^8)^2 = 4\pi\times4.844\times10^{17} = 6.087\times10^{18}\ \mathrm{m}^2.$$

Step 3: Total luminosity.

$$L = E_{\text{total}}\times A = (7.348\times10^7)(6.087\times10^{18}) \approx \mathbf{4.47\times10^{26}}\ \mathrm{W}.$$

Step 4: Comparison with accepted value. The accepted solar luminosity is $L_\odot = 3.846\times10^{26}$ W. Our estimate is ∼16% higher, because we used $T = 6000$ K rather than the more accurate effective temperature of 5778 K, illustrating the strong T^4 sensitivity of the Stefan–Boltzmann law derived entirely from Planck's quantum hypothesis.

Q34 – Solar Surface Temperature from Luminosity

Question 34

Using the Stefan–Boltzmann law, estimate the Sun's surface temperature given total solar luminosity $L_\odot = 3.85 \times 10^{26}$ W and solar radius $R_\odot = 6.96 \times 10^8$ m.

Step 1: Set up the relation. $L_\odot = \sigma T^4 \times 4\pi R_\odot^2$, so

$$T = \left(\frac{L_\odot}{4\pi\sigma R_\odot^2} \right)^{1/4}$$

Step 2: Evaluate the denominator.

Q34 – Solar Surface Temperature from Luminosity

$$4\pi\sigma R_\odot^2 = 4\pi(5.67\times10^{-8})(6.96\times10^{8})^2 \approx 3.453\times10^{11}$$

Step 3: Compute T.

$$\boxed{T = \left(\frac{3.85\times10^{26}}{3.453\times10^{11}}\right)^{1/4} \approx 5775\ \mathrm{K}}$$

Q35 – Bremsstrahlung Short-Wavelength Limit

Question 35

Find the minimum wavelength of X-rays produced when electrons accelerated through $V = 40$ kV strike a target.

Step 1: Physical argument. The entire electron kinetic energy converts to a single photon: $h\nu_{\max} = eV$, so $\lambda_{\min} = hc/(eV)$. **Step 2: Numerator and denominator.**

$$hc = 1.988 \times 10^{-25} \text{ J m}, \qquad eV = (1.6 \times 10^{-19})(40 \times 10^{3}) = 6.4 \times 10^{-15} \text{ J}$$

Step 3: Compute.

Q35 – Bremsstrahlung Short-Wavelength Limit

$$\boxed{\lambda_{\min} = \frac{1.988 \times 10^{-25}}{6.4 \times 10^{-15}} \approx 3.1 \times 10^{-11}\ \text{m} = 0.031\ \text{nm}}$$

This is in the X-ray range, consistent with the electron carrying 40 keV of energy.

Question 36

State the basic axioms (postulates) of quantum mechanics that form the foundation of the theory.

Quantum mechanics rests on six postulates:

Postulate 1 – State. The state of a physical system is completely specified by a wavefunction $\Psi(\mathbf{r}, t)$ (or state vector $|\Psi\rangle$ in Hilbert space), which contains all obtainable information.

Postulate 2 – Observables. Every measurable physical quantity A is represented by a linear, Hermitian operator $\hat{A}$ acting on the space of wavefunctions.

Postulate 3 – Measurement outcomes. The only possible results of a measurement of A are the eigenvalues a_n of $\hat{A}\phi_n = a_n\phi_n$.

Postulate 4 – Probability (Born rule). If the system is in state $\Psi = \sum_n c_n \phi_n$, the probability of obtaining a_n is $|c_n|^2$, where $c_n = \langle \phi_n | \Psi \rangle$.

Postulate 5 – Time evolution. Between measurements the wavefunction evolves according to the time-dependent Schrödinger equation:

$$i\hbar \frac{\partial \Psi}{\partial t} = \hat{H}\Psi.$$

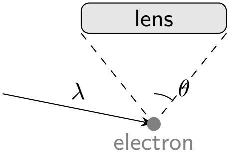
Postulate 6 – Collapse. Immediately after a measurement of A yielding eigenvalue a_n, the state collapses to the corresponding eigenstate ϕ_n.

The correspondence rules (e.g. $\hat{p} = -i\hbar\nabla$) and these six postulates constitute the formal structure of non-relativistic quantum mechanics.

Question 37

Describe Heisenberg's γ-ray microscope thought experiment, and explain how it illustrates $\Delta x \, \Delta p \gtrsim \hbar$.

Setup. Illuminate an electron with a photon of wavelength λ and observe the scattered photon through a microscope lens of half-aperture angle θ.



Q37 – Heisenberg's Gamma-Ray Microscope

Step 1: Position resolution from diffraction. The resolving power of the microscope (Abbe criterion) gives the minimum resolvable distance:

$$\Delta x \sim \frac{\lambda}{\sin\theta}.$$

A shorter wavelength λ gives a smaller Δx (finer resolution).

Step 2: Momentum disturbance from the Compton recoil. For the photon to enter the lens aperture it must scatter within an angle of $\pm\theta$. The photon momentum magnitude is $p_{\mathrm{ph}} = h/\lambda$, so the component transferred to the electron in the x-direction is uncertain by:

$$\Delta p \sim \frac{h}{\lambda}\sin\theta.$$

Step 3: Combine the two effects.

$$\Delta x \cdot \Delta p \sim \frac{\lambda}{\sin\theta} \cdot \frac{h\sin\theta}{\lambda} = h \sim \hbar.$$

Q37 – Heisenberg's Gamma-Ray Microscope

Step 4: Conclusion. Reducing λ to improve position accuracy (Δx smaller) inevitably increases the momentum kick $\Delta p \propto 1/\lambda$. The product $\Delta x\, \Delta p$ is bounded from below by $\hbar$, anticipating the precise result $\Delta x\, \Delta p \geq \hbar/2$ (Q41, Q136).

Q38 – Conjugate Variables

Question 38

What is meant by a pair of conjugate variables in quantum mechanics? Give three standard examples.

Definition. Two dynamical variables A and B are **canonically conjugate** if their quantum operators satisfy a non-zero commutation relation:

$$[\hat{A}, \hat{B}] = \hat{A}\hat{B} - \hat{B}\hat{A} \neq 0.$$

By the generalised uncertainty principle (Q136):

$$\Delta A \, \Delta B \geq \tfrac{1}{2} \left| \langle [\hat{A}, \hat{B}] \rangle \right| .$$

They cannot both be measured with arbitrary precision simultaneously.

Examples:

1. **Position and momentum.** $[\hat{x}, \hat{p}_x] = i\hbar$ (derived in Q76 and Q135), giving:

$$\Delta x\, \Delta p_x \geq \frac{\hbar}{2}.$$

2. Energy and time.

$$\Delta E\,\Delta t \gtrsim \frac{\hbar}{2}.$$

Here Δt is the characteristic timescale over which the system's energy can change (e.g. lifetime of an excited state), not an operator uncertainty. A short-lived state has a large energy uncertainty – the *natural linewidth* (Q60).

3. Azimuthal angle and angular momentum. $[\hat{L}_z, \hat{\varphi}] = i\hbar$, giving:

$$\Delta L_z\,\Delta\varphi \gtrsim \frac{\hbar}{2}.$$

A state of definite L_z (e.g. $e^{im\varphi}$) has a completely undefined azimuthal angle φ, and conversely.

Each pair is connected through a Fourier-transform relationship between their respective wavefunctions (Q39).

Q39 – Position and Momentum as Fourier Transforms

Question 39

Show that the position-space and momentum-space wavefunctions are related by a Fourier transform, and use this to argue why position and momentum are conjugate variables.

Step 1: Expand $\psi(x)$ in momentum eigenfunctions. The momentum eigenfunctions (normalised to a Dirac delta function) are $u_p(x) = \frac{1}{\sqrt{2\pi\hbar}} e^{ipx/\hbar}$. Any wavefunction can be written:

$$\psi(x) = \int_{-\infty}^{\infty} \phi(p) \, \frac{e^{ipx/\hbar}}{\sqrt{2\pi\hbar}} \, dp,$$

where $\phi(p)$ is the **momentum-space wavefunction**.

Step 2: Invert to find $\phi(p)$. By the Fourier inversion theorem:

$$\phi(p) = \int_{-\infty}^{\infty} \psi(x) \, \frac{e^{-ipx/\hbar}}{\sqrt{2\pi\hbar}} \, dx.$$

This is precisely a Fourier transform pair between $\psi(x)$ and $\phi(p)$.

Step 3: Width reciprocity of Fourier transforms. A fundamental mathematical property of the Fourier transform is that if $\psi(x)$ is narrow (small Δx), its transform $\phi(p)$ must be broad (large Δp), and vice versa. Quantitatively, for any Fourier pair:

$$\Delta x\,\Delta k \gtrsim 1 \quad (\text{with } k = p/\hbar),$$

which gives immediately:

$$\Delta x\,\Delta p \gtrsim \hbar.$$

Step 4: Physical interpretation. Because $\psi(x)$ and $\phi(p)$ are not independent but are Fourier transforms of one another, localising the particle in position space (sharp $|\psi(x)|^2$) inevitably spreads out the momentum distribution $|\phi(p)|^2$, and vice versa. This mathematical reciprocity is the underlying origin of the position–momentum uncertainty principle.

Q40 – Gaussian Wave Packet: Minimum Uncertainty State

Question 40

A normalised Gaussian wave packet at $t = 0$ is $\psi(x) = \left(\frac{1}{2\pi\sigma^2}\right)^{1/4} \exp\left(-\frac{x^2}{4\sigma^2}\right)$. Compute Δx and Δp and show $\Delta x\, \Delta p = \hbar/2$.

Step 1: Compute Δx. Since $\psi(x)$ is real and symmetric, $\langle x \rangle = 0$.

$$\langle x^2 \rangle = \left(\frac{1}{2\pi\sigma^2}\right)^{1/2} \int_{-\infty}^{\infty} x^2 \exp\left(-\frac{x^2}{2\sigma^2}\right) dx.$$

Using $\int_{-\infty}^{\infty} x^2 e^{-x^2/2\sigma^2} dx = \sigma^2 \sqrt{2\pi\sigma^2}$:

$$\langle x^2 \rangle = \left(\frac{1}{2\pi\sigma^2}\right)^{1/2} \cdot \sigma^2 \sqrt{2\pi\sigma^2} = \sigma^2.$$

$$\Delta x = \sqrt{\langle x^2 \rangle - \langle x \rangle^2} = \sigma.$$

Step 2: Find the momentum-space wavefunction. The Fourier transform of a Gaussian is again a Gaussian:

$$\phi(p) = \int_{-\infty}^{\infty} \psi(x)\,\frac{e^{-ipx/\hbar}}{\sqrt{2\pi\hbar}}\,dx = \left(\frac{2\sigma^2}{\pi\hbar^2}\right)^{1/4} \exp\left(-\frac{\sigma^2 p^2}{\hbar^2}\right).$$

(The Gaussian integral $\int_{-\infty}^{\infty} e^{-ax^2} e^{ibx} dx = \sqrt{\pi/a}\, e^{-b^2/4a}$ is used here with $a = 1/4\sigma^2$.)

Step 3: Compute Δp. $|\phi(p)|^2 \propto e^{-2\sigma^2 p^2/\hbar^2}$, so this is a Gaussian in p with $\sigma_p^2 = \hbar^2/(2\cdot 2\sigma^2)$...more precisely: writing $|\phi|^2 \propto e^{-p^2/2\sigma_p^2}$ gives $\sigma_p = \hbar/(2\sigma)$. By symmetry $\langle p\rangle = 0$, so

$$\Delta p = \sigma_p = \frac{\hbar}{2\sigma}.$$

Step 4: Product of uncertainties.

$$\Delta x \cdot \Delta p = \sigma \cdot \frac{\hbar}{2\sigma} = \frac{\hbar}{2}.$$

The Gaussian wave packet saturates the minimum-uncertainty bound – it is a **minimum-uncertainty state** (coherent state of the harmonic oscillator).

Q41 – Confinement Energy from the Uncertainty Principle

Question 41

An electron is confined to a region $\Delta x = 5.0 \times 10^{-11}$ m (roughly half an atomic diameter). Use the uncertainty principle to estimate its minimum momentum uncertainty and minimum kinetic energy.

Step 1: Apply Heisenberg's uncertainty relation.

$$\Delta x \, \Delta p \geq \frac{\hbar}{2} \;\Rightarrow\; \Delta p \geq \frac{\hbar}{2\,\Delta x}.$$

Step 2: Minimum momentum uncertainty.

$$\Delta p_{\min} = \frac{\hbar}{2\,\Delta x} = \frac{1.055 \times 10^{-34}}{2 \times 5.0 \times 10^{-11}} = 1.055 \times 10^{-24} \text{ kg m/s}.$$

Q41 – Confinement Energy from the Uncertainty Principle

Step 3: Estimate the kinetic energy. Taking $p \sim \Delta p$ as the momentum scale (the particle must have at least this momentum to be consistent with confinement):

$$K_{\min} \sim \frac{(\Delta p)^2}{2m_e} = \frac{(1.055 \times 10^{-24})^2}{2 \times 9.109 \times 10^{-31}} = \frac{1.113 \times 10^{-48}}{1.822 \times 10^{-30}} = 6.11 \times 10^{-19}\ \text{J}.$$

Step 4: Convert to eV.

$$K_{\min} = \frac{6.11 \times 10^{-19}}{1.602 \times 10^{-19}} \approx \mathbf{3.81}\ \text{eV}.$$

Step 5: Physical significance. This confinement energy is larger than typical atomic binding energies (a few eV), which is expected: tightening the confinement by a factor of 2 relative to an atomic diameter raises the kinetic energy by a factor of 4 (since $K \propto (\Delta x)^{-2}$). Shrinking $\Delta x \to 0$ would require $\Delta p \to \infty$, raising the kinetic energy without bound. The uncertainty principle is therefore partly responsible for the **stability and finite size of atoms**.

Q42 – Minimum Momentum Uncertainty and Kinetic Energy of a Confined Electron

Question 42

An electron's position is known to within $\Delta x = 0.05$ nm. (a) What is the minimum uncertainty in its momentum? (b) Estimate the corresponding minimum kinetic energy in eV.

Step 1: (a) From $\Delta x \, \Delta p \geq \hbar/2$:

$$\Delta p_{\min} = \frac{\hbar}{2\Delta x} = \frac{1.055 \times 10^{-34}}{1.0 \times 10^{-10}} = 1.055 \times 10^{-24} \text{ kg m s}^{-1}$$

Step 2: (b) Treating Δp as the characteristic momentum:

Q42 – Minimum Momentum Uncertainty and Kinetic Energy of a Confined Electron

$$KE_{\min} \sim \frac{(\Delta p)^2}{2m_e} = \frac{(1.055 \times 10^{-24})^2}{2(9.11 \times 10^{-31})} \approx 6.11 \times 10^{-19}\ \text{J}$$

$$\boxed{KE_{\min} \approx 3.82\ \text{eV}}$$

Q43 – Is a γ-Ray Photon Confined Inside the Nucleus?

Question 43

A γ-ray photon has energy $E = 1.0$ MeV and is confined to a nuclear diameter $\Delta x = 10^{-14}$ m. Compute $\Delta p_{\min}$ and compare with p_γ.

Step 1: Minimum momentum uncertainty.

$$\Delta p_{\min} = \frac{\hbar}{2\Delta x} = \frac{1.055 \times 10^{-34}}{2 \times 10^{-14}} = 5.3 \times 10^{-21}\ \text{kg m s}^{-1}$$

Step 2: Photon momentum from $E = pc$:

$$p_\gamma = \frac{1.6 \times 10^{-13}\ \text{J}}{3 \times 10^{8}\ \text{m/s}} = 5.3 \times 10^{-22}\ \text{kg m s}^{-1}$$

Step 3: Compare.

Q43 – Is a γ-Ray Photon Confined Inside the Nucleus?

$$\boxed{\frac{\Delta p_{\min}}{p_\gamma} \approx 10}$$

The required momentum uncertainty is ten times the photon's actual momentum, so a γ-ray photon is far too under-constrained to be *localised* at nuclear scales — it originates from a nuclear transition but is not contained inside the nucleus.

Q44 – Ground-State Kinetic Energy of Hydrogen from the HUP

Question 44

A hydrogen atom is in the $n = 1$ state with Bohr radius $a_0 = 0.529$ Å. Estimate the ground-state kinetic energy using $\Delta x \approx a_0$ and compare with the exact result $KE_1 = 13.6$ eV.

Step 1: Minimum momentum uncertainty.

$$\Delta p \geq \frac{\hbar}{2a_0} = \frac{1.055 \times 10^{-34}}{1.058 \times 10^{-10}} \approx 9.97 \times 10^{-25} \text{ kg m s}^{-1}$$

Step 2: Estimate the kinetic energy.

Q44 – Ground-State Kinetic Energy of Hydrogen from the HUP

$$KE \sim \frac{(\Delta p)^2}{2m_e} \approx 5.46 \times 10^{-19} \text{ J}$$

$$\boxed{KE \approx 3.4 \text{ eV}}$$

This is about a factor of 4 below the exact value of 13.6 eV, due to the factor of 2 in the HUP's denominator. Using $\Delta p \sim \hbar/a_0$ instead gives $KE \sim \hbar^2/(2m_e a_0^2) = 13.6$ eV exactly.

Question 45

Apart from $\Delta x\,\Delta p \geq \hbar/2$, state two other commonly used forms of the uncertainty principle and briefly explain the physical meaning of each.

1. Energy–time uncertainty:

$$\Delta E\,\Delta t \gtrsim \frac{\hbar}{2}.$$

What Δt means. In non-relativistic QM, time is a parameter (not an operator), so Δt is not the standard deviation of a measurement. Instead, Δt is the characteristic time over which the system's properties change appreciably – for example, the **mean lifetime** τ of an excited atomic state.

Consequence. A state with a short lifetime $\Delta t \sim \tau$ has a correspondingly large uncertainty (spread) in its energy:

$$\Delta E \sim \frac{\hbar}{2\tau}.$$

This is the **natural linewidth** of the emitted photon (Q60): shorter lifetimes produce broader spectral lines.

2. Angular-momentum–angle uncertainty:

$$\Delta L_z \, \Delta\varphi \gtrsim \frac{\hbar}{2}.$$

What it means. This relates the z-component of orbital angular momentum L_z to the precision with which the azimuthal angle φ about the z-axis can be specified.

Consequence.

- An eigenstate $e^{im\varphi}$ of $\hat{L}_z$ with eigenvalue $m\hbar$ has completely *undefined* φ (uniform distribution over $[0, 2\pi)$).
- Conversely, a state localised at a definite angle has a large spread in L_z.

Both relations follow the same general pattern as $\Delta x \, \Delta p$: they arise whenever two physical quantities are linked by a Fourier-transform-like relationship or fail to commute as quantum operators (Q136).

Q46 – Electron Diffraction Experiment

Question 46

Describe the basic idea of an electron diffraction experiment and explain what it demonstrates about the nature of matter.

Setup. A monoenergetic beam of electrons (accelerated through voltage V, giving $\lambda = h/\sqrt{2m_e eV}$) is directed at a thin crystalline foil or a double-slit arrangement. The spatial distribution of electrons arriving at a detection screen is recorded.

Observation. Instead of two simple "bright patches" (expected for classical particles), the screen shows a pattern of alternating bright and dark bands – an **interference pattern**, analogous to light through a double slit or X-rays through a crystal.

Key features confirming wave behaviour:

- The fringe spacing matches exactly the prediction using the de Broglie wavelength $\lambda = h/p$ – as if a wave of that wavelength passed through both slits.
- The pattern builds up *one electron at a time*: each electron is detected as a single localised spot (particle-like), yet the statistical distribution of many spots reproduces the wave interference pattern.
- Closing one slit removes the interference fringes – showing that each individual electron's wave genuinely passes through *both* slits simultaneously.

Conclusion. This is direct evidence that electrons exhibit genuine wave interference – confirming wave–particle duality for matter and demonstrating that the probability amplitude (wavefunction) obeys the superposition principle.

Q47 – Born's Probabilistic Interpretation

Question 47

State Born's statistical interpretation of the wavefunction, and explain the physical meaning of $|\Psi(x,t)|^2$.

Born's interpretation (1926). The wavefunction $\Psi(x,t)$ itself has no direct physical meaning, but

$$P(x,t)\,dx = |\Psi(x,t)|^2\,dx$$

gives the **probability of finding the particle in the interval** $[x, x+dx]$ at time t if a position measurement is performed at that instant.
Equivalently, $|\Psi(x,t)|^2$ is a **probability density**.

Consequence 1 – Normalisation. The particle must be found *somewhere*, so the total probability integrates to unity:

$$\int_{-\infty}^{\infty} |\Psi(x,t)|^2\,dx = 1.$$

Q47 – Born's Probabilistic Interpretation

Consequence 2 – Reality and positivity. Because Ψ may be complex while $|\Psi|^2 = \Psi^*\Psi$ is always real and non-negative, $|\Psi|^2$ is a sensible probability density (probabilities cannot be negative or complex).

Consequence 3 – Interference. For a superposition $\Psi = \psi_1 + \psi_2$:

$$|\Psi|^2 = |\psi_1|^2 + |\psi_2|^2 + 2\,\mathrm{Re}(\psi_1^*\psi_2).$$

The interference term $2\,\mathrm{Re}(\psi_1^*\psi_2)$ directly affects the measured probability distribution – this is the mathematical heart of quantum interference: amplitudes interfere; classical probabilities do not.

Departure from classical physics. Deterministic trajectories are replaced by probability distributions. The wavefunction encodes everything that *can* be known; individual outcomes remain fundamentally unpredictable.

Q48 – Normalisation and Born-Rule Probability for a Two-State System

Question 48

A quantum state is written as $\Psi = \frac{3}{5}|\phi_1\rangle + \frac{4}{5}|\phi_2\rangle$, where $|\phi_1\rangle, |\phi_2\rangle$ are orthonormal. Confirm normalisation and find the probability of obtaining the measurement result associated with $|\phi_1\rangle$.

Step 1: Normalisation check.

$$\left(\frac{3}{5}\right)^2 + \left(\frac{4}{5}\right)^2 = \frac{9}{25} + \frac{16}{25} = 1 \checkmark$$

Step 2: Probability. $\langle\phi_1|\Psi\rangle = 3/5$, so

$$\boxed{P_1 = \left|\frac{3}{5}\right|^2 = \frac{9}{25} = 0.36}$$

Q49 – Conditions on a Physically Acceptable Wavefunction

Question 49

List the mathematical conditions a physically acceptable wavefunction $\psi(x)$ must satisfy.

For $\psi(x)$ to represent a physically realisable quantum state it must satisfy:

1. Single-valued. $\psi(x)$ must have one and only one value at each point x, so that the probability density $|\psi(x)|^2$ is unambiguous.

2. Continuous. $\psi(x)$ must be continuous everywhere (no sudden jumps). The Schrödinger equation involves second derivatives, so ψ and (in regions of finite potential) $d\psi/dx$ must be smooth.

3. Finite (bounded). $\psi(x)$ must remain finite everywhere. An infinite ψ at some point would give an infinite, unphysical probability density there.

Q49 – Conditions on a Physically Acceptable Wavefunction

4. Square-integrable (normalisable).

$$\int_{-\infty}^{\infty} |\psi(x)|^2 \, dx \; < \; \infty$$

so that $|\psi|^2$ can be scaled to a genuine probability density summing to unity.

5. Smooth first derivative (except at infinite potentials). $d\psi/dx$ must also be continuous everywhere the potential $V(x)$ is finite (Q56). At points where $V \to \infty$ (e.g. infinite well walls, delta potential), ψ' may be discontinuous even though ψ itself remains continuous.

Summary. A function violating any of these – e.g. diverging at infinity, multi-valued, or discontinuous where the potential is finite – cannot represent an actual quantum state.

Question 50

A particle in the region $0 \le x \le L$ is described by $\psi(x) = A\sin(\pi x/L)$ (and $\psi = 0$ outside). Find the normalisation constant A.

Step 1: Write the normalisation condition.

$$\int_{-\infty}^{\infty} |\psi(x)|^2\,dx = 1 \;\Rightarrow\; A^2 \int_0^L \sin^2\left(\frac{\pi x}{L}\right) dx = 1.$$

Step 2: Evaluate the integral using the half-angle identity.

$$\sin^2\left(\frac{\pi x}{L}\right) = \frac{1 - \cos(2\pi x/L)}{2}.$$

$$\int_0^L \sin^2\left(\frac{\pi x}{L}\right) dx = \int_0^L \frac{1}{2}\left[1 - \cos\left(\frac{2\pi x}{L}\right)\right] dx = \frac{1}{2}\left[x - \frac{L}{2\pi}\sin\left(\frac{2\pi x}{L}\right)\right]_0^L.$$

Q50 – Normalisation of $\psi(x) = A\sin(\pi x/L)$

Step 3: Apply the limits. At $x = L$: $\sin(2\pi) = 0$. At $x = 0$: $\sin(0) = 0$. Therefore:

$$\int_0^L \sin^2\left(\frac{\pi x}{L}\right) dx = \frac{1}{2}[L - 0] - \frac{1}{2}[0 - 0] = \frac{L}{2}.$$

Step 4: Solve for A.

$$A^2 \cdot \frac{L}{2} = 1 \;\Rightarrow\; A^2 = \frac{2}{L} \;\Rightarrow\; \boxed{A = \sqrt{\frac{2}{L}}.}$$

This is exactly the ground-state normalisation constant for the infinite square well (Q87).

Question 51

Explain the classical expectation, the actual quantum result, and the resolution (probability amplitude addition) in Young's double-slit experiment performed with single particles.

Classical (particle) expectation. If each particle behaves as a classical billiard ball going through one slit or the other, the probability of arriving at point x on the screen should be the sum of the individual slit probabilities:

$$P_{12}(x) \stackrel{?}{=} P_1(x) + P_2(x).$$

This would produce two overlapping "humps" on the screen, with no fringes.

Actual quantum result. Experimentally, the pattern with both slits open shows fringes: bright and dark bands that are *not* the simple sum $P_1 + P_2$. At some points $P_{12} < P_1 + P_2$ (destructive interference, even $P_{12} = 0$) and at others $P_{12} > P_1 + P_2$ (constructive).

Q51 – Double-Slit: Amplitudes vs. Probabilities

Resolution: amplitudes, not probabilities, add. Quantum mechanics assigns complex probability amplitudes $\psi_1(x)$ and $\psi_2(x)$ for the particle arriving at x via slits 1 and 2 respectively.

When both slits are open and no path information is recorded, the amplitudes add (superposition principle, Q22):

$$\psi_{12}(x) = \psi_1(x) + \psi_2(x).$$

The probability is the modulus-squared of the *total* amplitude:

$$P_{12}(x) = |\psi_1 + \psi_2|^2 = \underbrace{|\psi_1|^2 + |\psi_2|^2}_{P_1+P_2} + \underbrace{2\,\mathrm{Re}(\psi_1^*\psi_2)}_{\text{interference term}}.$$

This correctly reproduces the observed fringe pattern.

Central lesson: Probability amplitudes (wavefunctions) interfere; classical probabilities themselves do not simply add.

Q52 – Which-Path Detection Destroys Interference

Question 52

What happens to the interference pattern if a detector determines which slit each particle passes through? Explain in terms of the uncertainty principle.

Observation. If any apparatus – however gentle – successfully determines which slit each particle used, the interference fringes *disappear*, leaving the simple sum $P_1(x) + P_2(x)$.

Step 1: Why path determination requires momentum disturbance. To determine which slit (separation d) the particle passed through, one must localise it transversely to within $\Delta y \lesssim d$ at the plane of the slits. By the uncertainty principle:

$$\Delta y \, \Delta p_y \gtrsim \hbar \;\Rightarrow\; \Delta p_y \gtrsim \frac{\hbar}{d}.$$

Step 2: Compare to the fringe-producing momentum scale. The m-th interference maximum satisfies $d \sin \theta_m = m\lambda$, i.e. adjacent fringes differ in transverse momentum by:

$$\delta p_y \sim \frac{h}{d}.$$

Step 3: The disturbance washes out the fringes. The minimum unavoidable kick $\Delta p_y \sim \hbar/d$ is of the same order as the fringe spacing $\delta p_y \sim h/d$. This random, uncontrolled kick randomises the precise phase relationship between the two paths needed for fringes to form.

Step 4: Conclusion – Principle of Complementarity. Any measurement that reveals which-path information necessarily destroys the interference pattern. This is enforced by the uncertainty principle, *not* by any technological limitation: the wave nature (interference) and the particle nature (definite path) are **complementary** – they cannot both be observed in the same experiment.

Q53 – Fringe Visibility and Which-Path Distinguishability

Question 53

In a double-slit experiment, the fringe visibility is measured to be $V = 0.6$. Using Englert's duality relation $V^2 + D^2 \leq 1$, find the which-path distinguishability D.

Step 1: Apply the duality relation at equality.

$$D = \sqrt{1 - V^2} = \sqrt{1 - 0.36} = \sqrt{0.64}$$

$$\boxed{D = 0.8}$$

The path can be determined with 80% reliability, at the cost of reducing fringe visibility to 60%.

Q54 – Double-Slit Maxima: Numerical Example

Question 54

For $I(\theta) = I_0 \cos^2\left(\frac{\pi d \sin\theta}{\lambda}\right)$ with $d = 0.2\,\text{mm}$ and $\lambda = 600\,\text{nm}$, find the angular positions of the first two interference maxima.

Step 1: Condition for maxima. $I(\theta)$ is maximised when $\cos^2(\cdot) = 1$, i.e. when its argument equals $m\pi$:

$$\frac{\pi d \sin\theta}{\lambda} = m\pi \;\Rightarrow\; d\sin\theta = m\lambda, \quad m = 0, 1, 2, \ldots$$

This is the standard double-slit constructive-interference condition.

Step 2: First maximum ($m = 1$).

$$\sin\theta_1 = \frac{\lambda}{d} = \frac{600 \times 10^{-9}}{0.2 \times 10^{-3}} = \frac{6.00 \times 10^{-7}}{2.00 \times 10^{-4}} = 3.00 \times 10^{-3}.$$

Since $\sin\theta_1 \ll 1$: $\theta_1 \approx 3.00 \times 10^{-3}$ rad $= 3.00 \times 10^{-3} \times \frac{180}{\pi} \approx \mathbf{0.172}^\circ$.

Q54 – Double-Slit Maxima: Numerical Example

Step 3: Second maximum ($m = 2$).

$$\sin\theta_2 = \frac{2\lambda}{d} = 2 \times 3.00 \times 10^{-3} = 6.00 \times 10^{-3}.$$

$$\theta_2 \approx 6.00 \times 10^{-3}\ \text{rad} \approx \mathbf{0.344}^\circ.$$

Step 4: Pattern of maxima. Maxima are evenly spaced (for small angles) at angles $\theta_m \approx m\lambda/d$. The angular fringe separation is:

$$\Delta\theta \approx \frac{\lambda}{d} = 3.00 \times 10^{-3}\ \text{rad} = 0.172^\circ.$$

On a screen at distance D from the slits, the fringe spacing on the screen is $\Delta y = D\,\Delta\theta = 3.00 \times 10^{-3} D$.

Q55 – Fringe Spacing in an Electron Double-Slit Experiment

Question 55

In a double-slit experiment with electrons, the slit separation is $d = 50\,\mu$m and the screen is at $D = 2$ m. The electron kinetic energy is $KE = 150$ eV. Find the fringe spacing Δy.

Step 1: de Broglie wavelength. For a non-relativistic electron, $\lambda(\text{Å}) = \sqrt{150/KE(\text{eV})}$, so with $KE = 150$ eV:

$$\lambda = 1.00\ \text{Å} = 1.0 \times 10^{-10}\ \text{m}$$

Step 2: Fringe spacing.

Q55 – Fringe Spacing in an Electron Double-Slit Experiment

$$\Delta y = \frac{\lambda D}{d} = \frac{(1.0 \times 10^{-10})(2)}{50 \times 10^{-6}}$$

$$\boxed{\Delta y = 4.0 \times 10^{-6}\ \text{m} = 4\ \mu\text{m}}$$

Q56 – Continuity Conditions from the TISE

Question 56

Show that continuity of $\psi(x)$ and (where the potential is finite) of $d\psi/dx$ at a point x_0 follow from the time-independent Schrödinger equation.

Step 1: Write the TISE.

$$-\frac{\hbar^2}{2m}\frac{d^2\psi}{dx^2} + V(x)\psi = E\psi \;\Rightarrow\; \frac{d^2\psi}{dx^2} = \frac{2m}{\hbar^2}[V(x) - E]\psi.$$

Step 2: Continuity of ψ. Suppose ψ were discontinuous at x_0. Then $d\psi/dx$ would contain a Dirac delta function at x_0, and $d^2\psi/dx^2$ would contain a derivative of a delta function – infinitely singular.
But the right-hand side $\frac{2m}{\hbar^2}[V(x) - E]\psi$ is *finite* everywhere that $V(x)$ is finite (even if V is discontinuous, it remains a finite number on each side).
A finite right-hand side cannot equal an infinitely singular left-hand side. $\Rightarrow \psi(x)$ must be continuous at x_0.

Step 3: Continuity of $d\psi/dx$. Integrate the TISE across the infinitesimal interval $[x_0 - \varepsilon, x_0 + \varepsilon]$:

$$\left.\frac{d\psi}{dx}\right|_{x_0+\varepsilon} - \left.\frac{d\psi}{dx}\right|_{x_0-\varepsilon} = \frac{2m}{\hbar^2}\int_{x_0-\varepsilon}^{x_0+\varepsilon}[V(x) - E]\psi(x)\,dx.$$

As $\varepsilon \to 0$, if $V(x)$ is finite at x_0 (even if discontinuous), the integrand is bounded, so the integral vanishes:

$$\frac{2m}{\hbar^2}\int_{x_0-\varepsilon}^{x_0+\varepsilon}[\text{bounded}]\,dx \le C \cdot 2\varepsilon \xrightarrow{\varepsilon\to 0} 0.$$

Hence $d\psi/dx|_{x_0^+} = d\psi/dx|_{x_0^-}$.

Step 4: Exception. If $V(x) \to \infty$ at x_0 (e.g. infinite square well wall, or a Dirac delta potential, Q102), the integral on the right need not vanish, and $d\psi/dx$ can be discontinuous there even though ψ itself remains continuous.

Question 57

A particle is in the state $\psi(x) = Ae^{-\alpha|x|}$ for $-\infty < x < \infty$, $\alpha > 0$. Find A and the probability of finding the particle in $[-1/\alpha,\, 1/\alpha]$.

Step 1: Normalisation. By symmetry $|x| = x$ for $x > 0$ and $|x| = -x$ for $x < 0$:

$$\int_{-\infty}^{\infty} |\psi|^2\, dx = A^2 \int_{-\infty}^{\infty} e^{-2\alpha|x|}\, dx = 2A^2 \int_{0}^{\infty} e^{-2\alpha x}\, dx = 2A^2 \cdot \frac{1}{2\alpha} = \frac{A^2}{\alpha}.$$

Setting this equal to 1:

$$\frac{A^2}{\alpha} = 1 \;\Rightarrow\; \boxed{A = \sqrt{\alpha}.}$$

Step 2: Probability in $[-1/\alpha, 1/\alpha]$.

$$P = \int_{-1/\alpha}^{1/\alpha} |\psi|^2 \, dx = \alpha \int_{-1/\alpha}^{1/\alpha} e^{-2\alpha|x|} \, dx = 2\alpha \int_0^{1/\alpha} e^{-2\alpha x} \, dx.$$

Step 3: Evaluate the integral.

$$\int_0^{1/\alpha} e^{-2\alpha x} \, dx = \left[-\frac{1}{2\alpha} e^{-2\alpha x} \right]_0^{1/\alpha} = \frac{1}{2\alpha} \left(1 - e^{-2}\right).$$

Step 4: Final probability.

$$P = 2\alpha \cdot \frac{1}{2\alpha}(1 - e^{-2}) = 1 - e^{-2}.$$

$$P = 1 - \frac{1}{e^2} = 1 - \frac{1}{7.389} = 1 - 0.1353 \approx \mathbf{86.5}\%.$$

Roughly 86.5% of the probability lies within one "decay length" $1/\alpha$ on either side of the origin, consistent with the exponentially localised shape of ψ.

Q58 – Spreading of a Free-Particle Wave Packet

Question 58

A free-particle wave packet $\Psi(x, 0)$ is built from wavenumbers in a range Δk. Using the dispersion relation $\omega = \hbar k^2/2m$, explain qualitatively why the packet broadens with time.

Step 1: The dispersion relation is non-linear in k. For a free particle:

$$\omega(k) = \frac{\hbar k^2}{2m}.$$

Unlike light in vacuum ($\omega = ck$, linear), here $\omega \propto k^2$. This means the medium is **dispersive**: different wavenumbers travel at different speeds.

Step 2: Different wavenumber components have different group velocities.

$$v_g(k) = \frac{d\omega}{dk} = \frac{\hbar k}{m}.$$

Components with larger k (higher momentum) travel faster than those with smaller k:

$$v_g(k_0 + \Delta k) - v_g(k_0 - \Delta k) = \frac{2\hbar\,\Delta k}{m}.$$

Q58 – Spreading of a Free-Particle Wave Packet

Step 3: The packet broadens because components separate. At $t = 0$ the components interfere constructively at one location x_0 (the packet peak). As time progresses, faster components move ahead while slower ones fall behind, and the constructive interference that created the peak degrades – the packet broadens:

$$\Delta x(t) \approx \Delta x(0)\sqrt{1 + \left(\frac{\hbar\,\Delta k\,t}{m\,\Delta x(0)}\right)^2}.$$

Step 4: Physical picture.

- A perfectly monochromatic wave ($\Delta k = 0$, $\Delta p = 0$) would not spread, but it would also be completely delocalised in space – not a particle.
- Any spatially localised state requires $\Delta k > 0$ (uncertainty principle), which inevitably leads to spreading via dispersion.

This spreading is a purely **quantum dispersive effect**, distinct from any classical force.

Question 59

Estimate the time for a free-electron wave packet to double its width, given initial width $\Delta x_0 = 0.2$ nm. Use $\Delta v \approx \hbar/(m_e \Delta x_0)$ and $\Delta v \cdot t \sim \Delta x_0$.

Step 1: Velocity spread.

$$\Delta v \approx \frac{\hbar}{m_e \Delta x_0} = \frac{1.055 \times 10^{-34}}{(9.11 \times 10^{-31})(2.0 \times 10^{-10})} \approx 5.79 \times 10^5 \text{ m/s}$$

Step 2: Doubling time.

Q59 – Time for a Free-Electron Wave Packet to Double in Width

$$t \sim \frac{\Delta x_0}{\Delta v} = \frac{2.0 \times 10^{-10}}{5.79 \times 10^5}$$

$$\boxed{t \approx 3.5 \times 10^{-16}\ \mathsf{s}}$$

Even a 0.2 nm packet spreads in under a femtosecond.

Question 60

An excited atomic state has mean lifetime $\tau = 1.0 \times 10^{-8}$ s. Use the energy–time uncertainty relation to estimate the natural linewidth ΔE and frequency spread $\Delta\nu$.

Step 1: Apply the energy–time uncertainty relation. Identifying $\Delta t \sim \tau$ (the time over which the state "changes" – here, by decaying):

$$\Delta E \sim \frac{\hbar}{2\tau} = \frac{1.055 \times 10^{-34}\ \text{J s}}{2 \times 1.0 \times 10^{-8}\ \text{s}} = 5.28 \times 10^{-27}\ \text{J}.$$

Step 2: Convert to eV.

$$\Delta E = \frac{5.28 \times 10^{-27}}{1.602 \times 10^{-19}}\ \text{eV} = 3.30 \times 10^{-8}\ \text{eV} \approx 33\ \text{neV}.$$

Step 3: Convert to frequency spread. Since $E = h\nu$, $\Delta E = h\,\Delta\nu$:

$$\Delta\nu = \frac{\Delta E}{h} = \frac{5.28 \times 10^{-27}}{6.626 \times 10^{-34}} \approx 7.97 \times 10^{6}\ \text{Hz} \approx \mathbf{8}\ \text{MHz}.$$

Step 4: Physical significance. This is the **natural (intrinsic) linewidth** of the spectral line: even a perfectly isolated, motionless atom emits a photon with a frequency spread of ∼8 MHz purely because the excited state has a finite lifetime.
For an optical transition at $\nu \sim 5 \times 10^{14}$ Hz, the *fractional* linewidth is:

$$\frac{\Delta\nu}{\nu} \approx \frac{8 \times 10^{6}}{5 \times 10^{14}} = 1.6 \times 10^{-8}.$$

A directly verified consequence of the energy–time uncertainty relation. Shorter lifetimes ⇒ broader spectral lines ($\Delta E \propto 1/\tau$).

Q61 – Natural Linewidth from an Excited-State Lifetime

Question 61

Use the energy–time uncertainty relation $\Delta E\,\Delta t \geq \hbar/2$ to estimate the natural linewidth $\Delta\nu$ (in MHz) of a spectral line from an excited state of lifetime $\tau = 2 \times 10^{-9}$ s.

Step 1: Minimum energy uncertainty with $\Delta t = \tau$:

$$\Delta E_{\min} = \frac{\hbar}{2\tau} = \frac{1.055 \times 10^{-34}}{4 \times 10^{-9}} = 2.64 \times 10^{-26}\ \text{J}$$

Step 2: Convert to frequency using $\Delta E = h\,\Delta\nu$:

$$\boxed{\Delta\nu = \frac{2.64 \times 10^{-26}}{6.626 \times 10^{-34}} \approx 39.8\ \text{MHz}}$$

In practice, Doppler broadening ($\sim$GHz) typically dominates at room temperature.

Q62 – Minimum Measurement Time for a Given Energy Uncertainty

Question 62

A particle's energy uncertainty is $\Delta E = 0.01$ eV. What is the minimum time required to measure this energy?

Step 1: Apply $\Delta E\,\Delta t \geq \hbar/2$.

$$\Delta t_{\text{min}} = \frac{\hbar}{2\Delta E}$$

Step 2: Substitute $\Delta E = 1.6 \times 10^{-21}$ J:

$$\boxed{\Delta t_{\text{min}} = \frac{1.055 \times 10^{-34}}{3.2 \times 10^{-21}} \approx 3.3 \times 10^{-14}\ \text{s}}$$

About 33 fs, characteristic of processes resolving meV-scale energy differences.

Question 63

Distinguish clearly between probability amplitude, probability density, and probability in quantum mechanics.

Probability amplitude. The wavefunction $\psi(x,t)$ itself (generally complex-valued) is the probability amplitude. It has no direct physical meaning by itself, but it is the fundamental quantity that is *superposed* (added) when combining different possibilities (Q22). Probabilities are computed from it.

Probability density.

$$\rho(x,t) \equiv |\psi(x,t)|^2 = \psi^*(x,t)\psi(x,t)$$

is real, non-negative, and represents the **probability per unit length** (in 1D) of finding the particle near x at time t.

Probability.

$$P(a \le x \le b,\ t) = \int_a^b |\psi(x,t)|^2 \, dx$$

is the dimensionless actual probability of finding the particle in the interval $[a, b]$.

Relationship summary:

$$\underbrace{\psi(x,t)}_{\text{amplitude (complex)}} \xrightarrow{|\cdot|^2} \underbrace{\rho(x,t)}_{\text{density (real, } \geq 0)} \xrightarrow{\int dx} \underbrace{P}_{\text{probability } (\in[0,1])}$$

Key distinction. Only the amplitude can be negative, complex, or exhibit interference. The density and the probability are always real and non-negative. This is why interference effects appear in $|\psi_1 + \psi_2|^2$ but not when probabilities are added directly.

Q64 – Stationary States

Question 64

Define a stationary state and explain why the probability density $|\Psi(x,t)|^2$ does not depend on time for such a state.

Definition. A stationary state is a solution of the TDSE of the separable form

$$\Psi(x,t) = \psi(x)\, e^{-iEt/\hbar},$$

where $\psi(x)$ satisfies the time-independent Schrödinger equation with definite energy eigenvalue E (Q73).

Step 1: Compute the probability density.

$$|\Psi(x,t)|^2 = \Psi^*(x,t)\,\Psi(x,t) = \left[\psi^*(x)e^{+iEt/\hbar}\right]\left[\psi(x)e^{-iEt/\hbar}\right].$$

Step 2: The time-dependent phase factors cancel.

$$|\Psi(x,t)|^2 = \psi^*(x)\psi(x)\, e^{+iEt/\hbar - iEt/\hbar} = |\psi(x)|^2 \cdot e^0 = |\psi(x)|^2.$$

Step 3: Consequence. The probability density is independent of t – hence the name "stationary". The phase factor $e^{-iEt/\hbar}$ has unit modulus $|e^{-iEt/\hbar}| = 1$ and drops out entirely.

Step 4: What does oscillate. The wavefunction $\Psi(x,t)$ itself oscillates in phase at frequency $\nu = E/h$. This phase oscillation has consequences for:

- Superpositions of stationary states, where the *relative* phases between different $e^{-iE_n t/\hbar}$ factors evolve and produce time-dependent interference ("quantum beats").
- Expectation values of time-dependent operators.

But for any *single* stationary state, all observable probabilities and expectation values of time-independent operators are constant in time.

Q65 – Expectation Value of an Observable

Question 65

Define the expectation value of an observable A in a quantum state $\psi(x)$, and write the general formula.

Definition. The expectation value $\langle A\rangle$ is the average value obtained from a large number of identical measurements of A, each performed on an identically prepared system in state ψ.

General formula.

$$\langle A\rangle = \int_{-\infty}^{\infty} \psi^*(x)\,\hat{A}\,\psi(x)\,dx \;\equiv\; \langle\psi|\hat{A}|\psi\rangle,$$

assuming ψ is normalised ($\int |\psi|^2\,dx = 1$).

Special cases:

- Position: $\langle x\rangle = \int_{-\infty}^{\infty} x\,|\psi(x)|^2\,dx$ (since $\hat{x}$ multiplies by x).
- Momentum: $\langle p\rangle = \int \psi^*(-i\hbar\partial_x)\psi\,dx$.
- Kinetic energy: $\langle K\rangle = -\dfrac{\hbar^2}{2m}\int \psi^*\psi''\,dx$.

Q65 – Expectation Value of an Observable

Origin from Born's rule. If $\Psi = \sum_n c_n \phi_n$ (eigenfunction expansion of $\hat{A}$), then

$$\langle A \rangle = \sum_n |c_n|^2 a_n,$$

the probability-weighted average of all possible eigenvalues – exactly consistent with the general formula above.

Variance and uncertainty. The uncertainty ΔA is defined as the standard deviation:

$$(\Delta A)^2 = \langle A^2 \rangle - \langle A \rangle^2 = \langle (\hat{A} - \langle A \rangle)^2 \rangle.$$

Q66 – Energy Expectation Value and Spread in a Two-State Superposition

Question 66

The wavefunction of a particle is $\Psi = \frac{1}{2}\psi_1 + \frac{\sqrt{3}}{2}\psi_2$, where ψ_1, ψ_2 are orthonormal energy eigenstates with $E_1 = 2$ eV, $E_2 = 6$ eV. Find $\langle E \rangle$ and ΔE.

Step 1: Probabilities. $P_1 = 1/4$, $P_2 = 3/4$ (sum to 1 ✓). **Step 2: Mean energy.**

$$\langle E \rangle = \frac{1}{4}(2) + \frac{3}{4}(6) = 5.0 \text{ eV}$$

Step 3: Spread. $\langle E^2 \rangle = \frac{1}{4}(4) + \frac{3}{4}(36) = 28 \text{ eV}^2$:

$$\boxed{\Delta E = \sqrt{28 - 25} = \sqrt{3} \approx 1.73 \text{ eV}}$$

Question 67

A particle in an infinite square well of width L is in its ground state $\psi(x) = \sqrt{2/L}\,\sin(\pi x/L)$. Calculate $\langle x \rangle$.

Step 1: Set up the integral.

$$\langle x \rangle = \int_0^L x|\psi(x)|^2\,dx = \frac{2}{L}\int_0^L x\sin^2\left(\frac{\pi x}{L}\right)dx.$$

Step 2: Rewrite using the half-angle identity.

$$\sin^2\left(\frac{\pi x}{L}\right) = \frac{1}{2}\left[1 - \cos\left(\frac{2\pi x}{L}\right)\right].$$

$$\langle x \rangle = \frac{1}{L}\int_0^L x\,dx - \frac{1}{L}\int_0^L x\cos\left(\frac{2\pi x}{L}\right)dx.$$

Step 3: First integral.

$$\frac{1}{L}\int_0^L x\,dx = \frac{1}{L}\cdot\frac{L^2}{2} = \frac{L}{2}.$$

Step 4: Second integral – integration by parts. Let $u = x$, $dv = \cos(2\pi x/L)\,dx$, so $du = dx$, $v = \frac{L}{2\pi}\sin(2\pi x/L)$:

$$\int_0^L x\cos\left(\frac{2\pi x}{L}\right)dx = \left[\frac{Lx}{2\pi}\sin\left(\frac{2\pi x}{L}\right)\right]_0^L - \int_0^L \frac{L}{2\pi}\sin\left(\frac{2\pi x}{L}\right)dx.$$

Boundary term: $\sin(2\pi) = 0$ and $\sin(0) = 0$, so it vanishes. Remaining integral:

$$-\frac{L}{2\pi}\left[-\frac{L}{2\pi}\cos\left(\frac{2\pi x}{L}\right)\right]_0^L = \frac{L^2}{4\pi^2}[\cos(2\pi) - \cos(0)] = \frac{L^2}{4\pi^2}(1-1) = 0.$$

Step 5: Combine.

$$\langle x \rangle = \frac{L}{2} - \frac{1}{L}\times 0 = \boxed{\frac{L}{2}.}$$

Q67 – $\langle x \rangle$ for the Infinite Square Well Ground State

By symmetry: the ground-state probability density $|\psi_1|^2 \propto \sin^2(\pi x/L)$ is symmetric about the midpoint $L/2$, so $\langle x \rangle = L/2$. ✓

Question 68

Compute $\langle x \rangle$, $\langle p \rangle$, $\langle x^2 \rangle$, and Δx for $\psi_1(x) = \sqrt{2/L}\sin(\pi x/L)$ in an infinite well of width L.

Step 1: Mean position and momentum. By symmetry $\langle x \rangle = L/2$; since ψ_1 is real, $\langle p \rangle = 0$.
Step 2: Mean square position.

$$\langle x^2 \rangle = L^2\left(\frac{1}{3} - \frac{1}{2\pi^2}\right) \approx 0.2827L^2$$

Step 3: Standard deviation.

Q68 – Position Variance for the Infinite Well Ground State

$$\boxed{\Delta x = \sqrt{0.2827L^2 - \frac{L^2}{4}} = L\sqrt{0.0327} \approx 0.181\, L}$$

Q69 – Eigenfunctions and Eigenvalues

Question 69

Define eigenfunction and eigenvalue of a quantum-mechanical operator, and explain their physical significance.

Definition. Given a linear operator $\hat{A}$, a non-zero function $\phi_n(x)$ is called an **eigenfunction** of $\hat{A}$ with **eigenvalue** a_n if:

$$\hat{A}\,\phi_n(x) = a_n\,\phi_n(x).$$

The operator acting on the eigenfunction simply returns the same function multiplied by the scalar a_n.

Physical significance (Postulate 3, Q36).

- If the system is in state ϕ_n, a measurement of A yields the eigenvalue a_n *with certainty* (probability 1).
- For a general state $\Psi = \sum_n c_n\phi_n$, the measurement yields a_n with probability $|c_n|^2$, and the average over many measurements is $\langle A\rangle = \sum_n |c_n|^2 a_n$ (Q65).

Key properties of eigenfunctions of Hermitian operators.

- **Real eigenvalues** (Q82, Q129): $a_n \in \mathbb{R}$ – a necessary condition for observable quantities.
- **Orthogonality** (Q130): eigenfunctions corresponding to different eigenvalues satisfy $\int \phi_m^* \phi_n \, dx = 0$ for $m \neq n$.
- **Completeness**: the set $\{\phi_n\}$ forms a basis in which any physical wavefunction can be expanded.

Example. The energy eigenfunctions of the infinite square well (Q87) are $\phi_n(x) = \sqrt{2/L} \sin(n\pi x/L)$ with eigenvalues $E_n = n^2\pi^2\hbar^2/2mL^2$.

Question 70

Show that $\cos(kx)$ is *not* an eigenfunction of $\hat{p} = -i\hbar\, d/dx$, but *is* an eigenfunction of $\hat{p}^2 = -\hbar^2\, d^2/dx^2$. Find its eigenvalue and explain the discrepancy.

Step 1: Test $\hat{p}$ on $\cos(kx)$.

$$\hat{p}\cos(kx) = -i\hbar\frac{d}{dx}\cos(kx) = i\hbar k\sin(kx).$$

Since $\sin(kx)$ is not proportional to $\cos(kx)$, this is **not** of the form (const) $\times \cos(kx)$: $\cos(kx)$ is **not** an eigenfunction of $\hat{p}$.

Step 2: Test $\hat{p}^2$ on $\cos(kx)$.

$$\hat{p}^2\cos(kx) = -\hbar^2\frac{d^2}{dx^2}\cos(kx) = -\hbar^2(-k^2\cos(kx)) = \hbar^2k^2\cos(kx).$$

$$\boxed{\hat{p}^2\cos(kx) = (\hbar k)^2\cos(kx) \;\Rightarrow\; \cos(kx) \text{ is an eigenfunction of } \hat{p}^2 \text{ with eigenvalue } \hbar^2k^2.}$$

Step 3: Resolve the discrepancy. Write $\cos(kx) = \frac{1}{2}(e^{ikx} + e^{-ikx})$ – an equal mixture of the $\hat{p}$-eigenstates with eigenvalues $+\hbar k$ and $-\hbar k$. A mixture of two *different* $\hat{p}$ eigenvalues is not itself a $\hat{p}$ eigenstate. But both terms share the same $\hat{p}^2$ eigenvalue $\hbar^2k^2$ (since $(\pm\hbar k)^2$ is the

Q71 – Ehrenfest's Theorem

Question 71

State Ehrenfest's theorem and outline its proof for $\langle x \rangle$ and $\langle p \rangle$.

Statement. Expectation values of position and momentum obey Newton's classical laws:

$$\frac{d\langle x \rangle}{dt} = \frac{\langle p \rangle}{m}, \qquad \frac{d\langle p \rangle}{dt} = -\left\langle \frac{\partial V}{\partial x} \right\rangle = \langle F \rangle.$$

Step 1: General time-derivative formula. For any operator $\hat{A}$ with no explicit time dependence, differentiate $\langle A \rangle = \int \Psi^* \hat{A} \Psi \, dx$ with respect to t and use the TDSE $i\hbar \partial_t \Psi = \hat{H} \Psi$ (and its complex conjugate):

$$\frac{d\langle A \rangle}{dt} = \frac{i}{\hbar} \langle [\hat{H}, \hat{A}] \rangle.$$

Step 2: Apply to $\hat{A} = \hat{x}$. $\hat{H} = \hat{p}^2/2m + V(\hat{x})$. Since $[V(\hat{x}), \hat{x}] = 0$:

$$[\hat{H}, \hat{x}] = \frac{1}{2m}[\hat{p}^2, \hat{x}].$$

Using $[\hat{p}^2, \hat{x}] = \hat{p}[\hat{p}, \hat{x}] + [\hat{p}, \hat{x}]\hat{p}$ and $[\hat{x}, \hat{p}] = i\hbar$ (hence $[\hat{p}, \hat{x}] = -i\hbar$):

$$[\hat{p}^2, \hat{x}] = -i\hbar\hat{p} + (-i\hbar)\hat{p} = -2i\hbar\hat{p}.$$

Therefore $[\hat{H}, \hat{x}] = -\frac{i\hbar}{m}\hat{p}$, and:

$$\frac{d\langle x\rangle}{dt} = \frac{i}{\hbar} \cdot \left(-\frac{i\hbar}{m}\right) \langle \hat{p}\rangle = \frac{\langle p\rangle}{m}. \quad \checkmark$$

Q71 – Ehrenfest's Theorem

Step 3: Apply to $\hat{A} = \hat{p}$. $[\hat{H}, \hat{p}] = [V(\hat{x}), \hat{p}]$ (the kinetic term $[\hat{p}^2/2m, \hat{p}] = 0$). Using $[\hat{p}, f(\hat{x})] = -i\hbar f'(\hat{x})$ (which follows from $[\hat{x}, \hat{p}] = i\hbar$):

$$[V(\hat{x}), \hat{p}] = -[\hat{p}, V(\hat{x})] = i\hbar V'(\hat{x}) = i\hbar \frac{\partial V}{\partial x}.$$

Therefore:

$$\frac{d\langle p \rangle}{dt} = \frac{i}{\hbar} \langle i\hbar \frac{\partial V}{\partial x} \rangle = -\left\langle \frac{\partial V}{\partial x} \right\rangle = \langle F \rangle. \quad \checkmark$$

Conclusion. Ehrenfest's theorem shows how classical mechanics emerges as the average behaviour of quantum expectation values: Newton's second law holds for $\langle x \rangle$ and $\langle p \rangle$ exactly.

Question 72

Postulate $\Psi(x,t) = e^{iS(x,t)/\hbar}$ for some complex function S. By requiring that the equation governing Ψ reduce, as $\hbar \to 0$, to the classical Hamilton–Jacobi equation $\partial S/\partial t + \frac{1}{2m}(\partial S/\partial x)^2 + V(x) = 0$, identify the time-dependent Schrödinger equation.

Step 1: Compute derivatives of $\Psi = e^{iS/\hbar}$.

$$\frac{\partial \Psi}{\partial t} = \frac{i}{\hbar}\frac{\partial S}{\partial t}\Psi, \qquad \frac{\partial \Psi}{\partial x} = \frac{i}{\hbar}\frac{\partial S}{\partial x}\Psi.$$

$$\frac{\partial^2 \Psi}{\partial x^2} = \frac{i}{\hbar}\frac{\partial^2 S}{\partial x^2}\Psi - \frac{1}{\hbar^2}\left(\frac{\partial S}{\partial x}\right)^2\Psi.$$

Step 2: Posit the wave equation and substitute. Try $i\hbar\,\partial\Psi/\partial t = -\dfrac{\hbar^2}{2m}\partial^2\Psi/\partial x^2 + V\Psi$:

$$i\hbar \cdot \frac{i}{\hbar}\frac{\partial S}{\partial t}\Psi = -\frac{\hbar^2}{2m}\left[\frac{i}{\hbar}\frac{\partial^2 S}{\partial x^2} - \frac{1}{\hbar^2}\left(\frac{\partial S}{\partial x}\right)^2\right]\Psi + V\Psi.$$

$$\Rightarrow\ -\frac{\partial S}{\partial t} = \frac{1}{2m}\left(\frac{\partial S}{\partial x}\right)^2 + V - \frac{i\hbar}{2m}\frac{\partial^2 S}{\partial x^2}.$$

Step 3: Take the classical limit $\hbar \to 0$. The last term vanishes, leaving exactly the Hamilton–Jacobi equation:

$$\frac{\partial S}{\partial t} + \frac{1}{2m}\left(\frac{\partial S}{\partial x}\right)^2 + V(x) = 0.$$

Step 4: Conclusion. The *only* linear wave equation for $\Psi = e^{iS/\hbar}$ whose $\hbar \to 0$ limit reproduces classical mechanics (Hamilton–Jacobi) is

$$\boxed{i\hbar\frac{\partial\Psi}{\partial t} = -\frac{\hbar^2}{2m}\frac{\partial^2\Psi}{\partial x^2} + V(x)\Psi.}$$

Q72 – Motivating the TDSE via Hamilton–Jacobi Correspondence

This is the TDSE – motivated here not by matching a single plane wave, but by demanding the correct classical limit, which is the deeper reason this particular equation is the right one.

Q73 – General Time Evolution from a Stationary-State Basis

Question 73

Given a complete set of TISE eigenstates $\{\varphi_n(x)\}$ with $\hat{H}\varphi_n = E_n\varphi_n$, show by direct substitution that $\Psi(x,t) = \sum_n c_n\varphi_n(x)e^{-iE_nt/\hbar}$ solves the TDSE for *any* choice of constants c_n, and derive the formula for c_n in terms of the initial wavefunction $\Psi(x,0)$.

Step 1: Substitute into the TDSE. LHS:

$$i\hbar\frac{\partial\Psi}{\partial t} = i\hbar\sum_n c_n\varphi_n(x)\left(-\frac{iE_n}{\hbar}\right)e^{-iE_nt/\hbar} = \sum_n c_nE_n\varphi_n(x)e^{-iE_nt/\hbar}.$$

RHS (using linearity of $\hat{H}$ and $\hat{H}\varphi_n = E_n\varphi_n$):

$$\hat{H}\Psi = \sum_n c_ne^{-iE_nt/\hbar}\hat{H}\varphi_n(x) = \sum_n c_nE_n\varphi_n(x)e^{-iE_nt/\hbar}.$$

LHS = RHS for *any* c_n, so $\Psi(x,t)$ solves the TDSE term by term.

Q73 – General Time Evolution from a Stationary-State Basis

Step 2: Fix c_n from the initial condition. At $t = 0$: $\Psi(x,0) = \sum_n c_n \varphi_n(x)$.
Step 3: Use orthonormality. Since $\{\varphi_n\}$ is orthonormal, $\int \varphi_m^*(x)\varphi_n(x)\,dx = \delta_{mn}$. Multiply both sides by $\varphi_m^*(x)$ and integrate:

$$\int \varphi_m^*(x)\Psi(x,0)\,dx = \sum_n c_n \int \varphi_m^*(x)\varphi_n(x)\,dx = \sum_n c_n \delta_{mn} = c_m.$$

$$\boxed{c_n = \int \varphi_n^*(x)\,\Psi(x,0)\,dx.}$$

This is how *any* initial wavefunction's subsequent time evolution is computed once the energy eigenbasis is known – the practical payoff of separation of variables that Q73 (original) sets up but doesn't carry through to.

Q74 – Oscillation Period of $\langle x\rangle(t)$ for a Two-State Superposition

Question 74

For $\Psi = \frac{1}{\sqrt{3}}\psi_1 e^{-iE_1 t/\hbar} + \sqrt{\frac{2}{3}}\psi_2 e^{-iE_2 t/\hbar}$ with $E_2 - E_1 = 2$ eV, find the period of oscillation of $\langle x\rangle(t)$.

Step 1: Angular frequency. The cross term oscillates at $\omega = (E_2 - E_1)/\hbar$. **Step 2: Period.**

$$T = \frac{2\pi\hbar}{E_2 - E_1} = \frac{2\pi(1.055 \times 10^{-34})}{2 \times 1.6 \times 10^{-19}}$$

$$\boxed{T \approx 2.07 \times 10^{-15}\ \text{s} \approx 2.07\ \text{fs}}$$

Q75 – Energy Expectation Value and Spread for an Equal Superposition

Question 75

An electron in an infinite well of width $L = 1$ nm is prepared at $t = 0$ in an equal superposition $\Psi(x, 0) = (\psi_1 + \psi_2)/\sqrt{2}$. Find $\langle E \rangle$ and ΔE.

Step 1: Individual energies.

$$E_1 = \frac{\pi^2 \hbar^2}{2m_e L^2} \approx 0.376 \text{ eV}, \qquad E_2 = 4E_1 \approx 1.504 \text{ eV}$$

Step 2: Mean energy. $P_1 = P_2 = 1/2$:

$$\langle E \rangle = \tfrac{1}{2}(0.376 + 1.504) = 0.94 \text{ eV}$$

Step 3: Spread. $\langle E^2 \rangle = \frac{1}{2}(E_1^2 + E_2^2) = 1.202 \text{ eV}^2$:

Q75 – Energy Expectation Value and Spread for an Equal Superposition

$$\boxed{\Delta E = \sqrt{1.202 - 0.940^2} \approx 0.564 \text{ eV}}$$

Question 76

Define the commutator of two operators and evaluate $[\hat{x}, \hat{p}]$ acting on an arbitrary test function $f(x)$.

Definition. For operators $\hat{A}$, $\hat{B}$, the commutator is

$$[\hat{A}, \hat{B}] \equiv \hat{A}\hat{B} - \hat{B}\hat{A},$$

which is itself an operator. $[\hat{A}, \hat{B}] = 0$ means they commute (order does not matter).

Step 1: Apply $\hat{x}\hat{p}$ to $f(x)$. With $\hat{p} = -i\hbar\, d/dx$:

$$\hat{x}\hat{p}\, f(x) = x \cdot \left(-i\hbar \frac{df}{dx}\right) = -i\hbar x f'(x).$$

Step 2: Apply $\hat{p}\hat{x}$ to $f(x)$.

$$\hat{p}\hat{x}\, f(x) = -i\hbar \frac{d}{dx}[x f(x)] = -i\hbar\, [f(x) + x f'(x)]$$

(using the product rule $d(xf)/dx = f + xf'$).

Step 3: Subtract.

$$[\hat{x}, \hat{p}]f(x) = \hat{x}\hat{p}\,f - \hat{p}\hat{x}\,f = -i\hbar x f'(x) - (-i\hbar f(x) - i\hbar x f'(x))\,.$$

$$= -i\hbar x f' + i\hbar f + i\hbar x f' = i\hbar f(x).$$

Step 4: Conclusion. Since this holds for *any* test function $f(x)$:

$$\boxed{[\hat{x}, \hat{p}] = i\hbar.}$$

This fundamental **canonical commutation relation** is the operator-level statement that x and p are conjugate variables (Q38) and is the rigorous origin of the position–momentum uncertainty principle (Q136).

Question 77

Starting from the 3D time-dependent Schrödinger equation, derive the continuity equation $\partial\rho/\partial t + \nabla \cdot \mathbf{j} = 0$, with $\mathbf{j} = \frac{\hbar}{2mi}(\psi^*\nabla\psi - \psi\nabla\psi^*)$.

Step 1: 3D TDSE and its complex conjugate.

$$i\hbar\frac{\partial\psi}{\partial t} = -\frac{\hbar^2}{2m}\nabla^2\psi + V\psi, \qquad -i\hbar\frac{\partial\psi^*}{\partial t} = -\frac{\hbar^2}{2m}\nabla^2\psi^* + V\psi^*$$

(V real, so V is unchanged under conjugation).

Step 2: Multiply and subtract. Multiply the first by ψ^*, the second by ψ, and subtract:

$$i\hbar\left(\psi^*\frac{\partial\psi}{\partial t} + \psi\frac{\partial\psi^*}{\partial t}\right) = -\frac{\hbar^2}{2m}\left(\psi^*\nabla^2\psi - \psi\nabla^2\psi^*\right).$$

(The $V\psi^*\psi$ terms cancel exactly.)

Step 3: Identify the time derivative of ρ. The LHS is $i\hbar\,\partial(\psi^*\psi)/\partial t = i\hbar\,\partial\rho/\partial t$, with $\rho = |\psi|^2$.

Step 4: Use the divergence identity.

$$\psi^*\nabla^2\psi - \psi\nabla^2\psi^* = \nabla\cdot(\psi^*\nabla\psi - \psi\nabla\psi^*),$$

which follows from the product rule $\nabla\cdot(\psi^*\nabla\psi) = \psi^*\nabla^2\psi + \nabla\psi^*\cdot\nabla\psi$ applied to both terms (the $\nabla\psi^*\cdot\nabla\psi$ cross terms cancel between the two).

Step 5: Assemble the continuity equation.

$$i\hbar\frac{\partial\rho}{\partial t} = -\frac{\hbar^2}{2m}\nabla\cdot(\psi^*\nabla\psi - \psi\nabla\psi^*)$$

$$\Rightarrow \boxed{\frac{\partial\rho}{\partial t} + \nabla\cdot\mathbf{j} = 0, \qquad \mathbf{j} = \frac{\hbar}{2mi}(\psi^*\nabla\psi - \psi\nabla\psi^*) = \frac{\hbar}{m}\,\mathrm{Im}(\psi^*\nabla\psi).}$$

Q78 – Probability Current for a Plane Wave

Question 78

Calculate the probability current density $j(x)$ for the plane wave $\Psi = Ae^{i(kx-\omega t)}$.

Step 1: Recall the formula from Q77.

$$j(x,t) = \frac{\hbar}{2mi}\left(\Psi^* \frac{\partial \Psi}{\partial x} - \Psi \frac{\partial \Psi^*}{\partial x}\right).$$

Step 2: Compute the derivatives.

$$\Psi = Ae^{i(kx-\omega t)}, \quad \frac{\partial \Psi}{\partial x} = ikAe^{i(kx-\omega t)} = ik\Psi.$$

$$\Psi^* = A^* e^{-i(kx-\omega t)}, \quad \frac{\partial \Psi^*}{\partial x} = -ikA^* e^{-i(kx-\omega t)} = -ik\Psi^*.$$

Step 3: Substitute into the formula.

$$j = \frac{\hbar}{2mi}[\Psi^*(ik\Psi) - \Psi(-ik\Psi^*)] = \frac{\hbar}{2mi}\left[ik|\Psi|^2 + ik|\Psi|^2\right] = \frac{\hbar}{2mi}(2ik|\Psi|^2).$$

Step 4: Simplify.

$$j = \frac{\hbar \cdot 2ik}{2mi}|A|^2 = \frac{\hbar k}{m}|A|^2.$$

$$\boxed{j = \frac{\hbar k}{m}|A|^2 = v|A|^2.}$$

Step 5: Physical interpretation. Since $p = \hbar k$ and $v = p/m$:
$j =$ (speed) $\times$ (probability density), analogous to a classical flux. This is constant in x and t for a pure plane wave, consistent with $\partial\rho/\partial t = 0$ (since $\rho = |A|^2$ is constant). ✓

Q79 – Stationary vs. Bound States

Question 79

Distinguish between a stationary state and a bound state, giving an example of each that is (a) stationary but not bound, and (b) both stationary and bound.

Stationary state (Q64). Any energy eigenstate $\Psi(x,t) = \psi(x)e^{-iEt/\hbar}$ for which $|\Psi|^2 = |\psi(x)|^2$ is time-independent. This describes the time dependence and applies whether or not the particle is confined.

Bound state. A state in which the particle is confined (localised) to a finite region of space, with $\psi(x) \to 0$ as $|x| \to \infty$, so that ψ is normalisable ($\int |\psi|^2 \, dx$ finite) and the energy spectrum is discrete (quantised).

Key distinction. "Stationary" refers to the time-independence of observable quantities; "bound" refers to spatial confinement. They are logically independent concepts, though they often occur together.

Q79 – Stationary vs. Bound States

(a) Stationary but not bound:

A free-particle plane wave $\psi(x) = Ae^{ikx}$, solving the TISE with $V = 0$ and $E = \hbar^2 k^2/2m$.

- *Stationary*: $|\Psi|^2 = |A|^2$ is time-independent.
- *Not bound*: ψ extends over all space and is not normalisable to 1.

(b) Both stationary and bound:

Any eigenstate of the infinite square well (Q87): $\psi_n(x) = \sqrt{2/L}\sin(n\pi x/L)$ for $0 \le x \le L$.

- *Stationary*: it is an energy eigenstate, so $|\Psi|^2 = |\psi_n(x)|^2$ is time-independent.
- *Bound*: $\psi_n = 0$ outside $[0, L]$ and $\int_0^L |\psi_n|^2 = 1$.

Q80 – $\langle p \rangle = 0$ for the Infinite Well Ground State

Question 80

Calculate $\langle p \rangle$ for the ground state $\psi(x) = \sqrt{2/L}\,\sin(\pi x/L)$ of the infinite square well.

Step 1: Set up the integral.

$$\langle p \rangle = \int_0^L \psi^*(x)\left(-i\hbar \frac{d}{dx}\right)\psi(x)\,dx = -i\hbar\frac{2}{L}\int_0^L \sin\left(\frac{\pi x}{L}\right)\frac{d}{dx}\sin\left(\frac{\pi x}{L}\right)dx.$$

Step 2: Compute the derivative.

$$\frac{d}{dx}\sin\left(\frac{\pi x}{L}\right) = \frac{\pi}{L}\cos\left(\frac{\pi x}{L}\right).$$

$$\langle p \rangle = -i\hbar\frac{2}{L}\cdot\frac{\pi}{L}\int_0^L \sin\left(\frac{\pi x}{L}\right)\cos\left(\frac{\pi x}{L}\right)dx.$$

Step 3: Evaluate the integral using $\sin\theta\cos\theta = \frac{1}{2}\sin 2\theta$.

$$\int_0^L \sin\left(\frac{\pi x}{L}\right)\cos\left(\frac{\pi x}{L}\right)dx = \frac{1}{2}\int_0^L \sin\left(\frac{2\pi x}{L}\right)dx.$$

$$= \frac{1}{2}\left[-\frac{L}{2\pi}\cos\left(\frac{2\pi x}{L}\right)\right]_0^L = -\frac{L}{4\pi}[\cos(2\pi) - \cos(0)] = -\frac{L}{4\pi}(1-1) = 0.$$

Step 4: Conclusion.

$$\langle p \rangle = -i\hbar\frac{2\pi}{L^2} \times 0 = \boxed{0.}$$

Physical interpretation. The ground state $\psi_1 = \sqrt{2/L}\sin(\pi x/L)$ can be written as an equal superposition of right- and left-moving plane waves:

$$\sin\left(\frac{\pi x}{L}\right) = \frac{e^{i\pi x/L} - e^{-i\pi x/L}}{2i}.$$

Right-moving momentum $+\hbar\pi/L$ and left-moving momentum $-\hbar\pi/L$ contribute equally, so $\langle p \rangle = 0$. However $\langle p^2 \rangle = (\hbar\pi/L)^2 \neq 0$, giving non-zero kinetic energy.

Q81 – Why $\hat{p} = -i\hbar\, d/dx$

Question 81

Show from the de Broglie–Einstein relations why it is natural to represent the momentum operator as $\hat{p} = -i\hbar\, d/dx$ in the position representation.

Step 1: Definite-momentum state. A particle with exactly defined momentum p_0 should, by the de Broglie relation $p_0 = \hbar k_0$, be represented by the plane wave:

$$\psi(x) = Ae^{ik_0 x} = Ae^{ip_0 x/\hbar}.$$

Step 2: Require this to be an eigenfunction. By Postulate 3 (Q36), if the system has a definite value p_0 of momentum, then ψ must be an eigenfunction of $\hat{p}$ with eigenvalue p_0:

$$\hat{p}\,\psi(x) = p_0\,\psi(x).$$

Step 3: Find the operator satisfying this. Differentiate $\psi(x) = Ae^{ip_0x/\hbar}$:

$$\frac{d\psi}{dx} = \frac{ip_0}{\hbar}\psi(x) \;\Rightarrow\; -i\hbar\frac{d\psi}{dx} = -i\hbar \cdot \frac{ip_0}{\hbar}\psi(x) = p_0\psi(x).$$

Step 4: Identify the operator. Comparing with $\hat{p}\,\psi = p_0\psi$:

$$\boxed{\hat{p} = -i\hbar\frac{d}{dx}.}$$

Step 5: Generalisation. For a general wavefunction $\psi(x) = \int \phi(p)e^{ipx/\hbar}/\sqrt{2\pi\hbar}\,dp$, applying $-i\hbar\partial_x$ brings down p from the exponent, confirming that this operator returns each Fourier component weighted by its momentum eigenvalue p. This bridges the wave description of matter with the operator formalism of quantum mechanics.

Q82 – Eigenvalues of an Anti-Hermitian Operator are Purely Imaginary

Question 82

An operator $\hat{A}$ is *anti-Hermitian* if $\hat{A}^\dagger = -\hat{A}$. Prove that its eigenvalues are purely imaginary.

Step 1: Set up. Let $\hat{A}\psi = a\psi$ with $\psi \neq 0$, and $\hat{A}^\dagger = -\hat{A}$.

Step 2: Take the inner product and its complex conjugate.

$$\langle\psi|\hat{A}|\psi\rangle = a\langle\psi|\psi\rangle.$$

Taking the complex conjugate of both sides, and using $\big(\langle\psi|\hat{A}|\psi\rangle\big)^* = \langle\psi|\hat{A}^\dagger|\psi\rangle$:

$$\langle\psi|\hat{A}^\dagger|\psi\rangle = a^*\langle\psi|\psi\rangle.$$

Step 3: Substitute $\hat{A}^\dagger = -\hat{A}$.

$$\langle\psi|(-\hat{A})|\psi\rangle = a^*\langle\psi|\psi\rangle \;\Rightarrow\; -a\langle\psi|\psi\rangle = a^*\langle\psi|\psi\rangle.$$

Since $\langle\psi|\psi\rangle \neq 0$:

$$\boxed{a^* = -a \implies \text{Re}(a) = 0,\ \text{i.e. } a \text{ is purely imaginary.}}$$

(Anti-Hermitian operators show up as the generators of unitary time evolution, e.g.

Q83 – Normalisation of $\psi(x) = Nxe^{-x^2/2a^2}$

Question 83

A wavefunction is given by $\psi(x) = Nxe^{-x^2/2a^2}$ for $-\infty < x < \infty$. Determine the normalisation constant N.

Step 1: Write the normalisation condition.

$$\int_{-\infty}^{\infty} |\psi|^2 \, dx = N^2 \int_{-\infty}^{\infty} x^2 e^{-x^2/a^2} \, dx = 1.$$

Step 2: Evaluate the Gaussian integral. Use the standard result with $\beta = 1/a^2$:

$$\int_{-\infty}^{\infty} x^2 e^{-\beta x^2} \, dx = \frac{\sqrt{\pi}}{2\beta^{3/2}}.$$

With $\beta = 1/a^2$, $\beta^{3/2} = 1/a^3$:

$$\int_{-\infty}^{\infty} x^2 e^{-x^2/a^2} \, dx = \frac{\sqrt{\pi}}{2/a^3} = \frac{a^3\sqrt{\pi}}{2}.$$

Step 3: Solve for N.

$$N^2 \cdot \frac{a^3\sqrt{\pi}}{2} = 1 \;\Rightarrow\; N^2 = \frac{2}{a^3\sqrt{\pi}} \;\Rightarrow\; \boxed{N = \sqrt{\frac{2}{a^3\sqrt{\pi}}} = \left(\frac{2}{\sqrt{\pi}\,a^3}\right)^{1/2}.}$$

Remark. This is the form of the first excited state of the quantum harmonic oscillator (Q105–Q106) – an odd function of x vanishing at the origin with exactly one node. Specifically, with $a = \sqrt{\hbar/m\omega}$, N matches the normalisation of $\psi_1(x)$.

Q84 – Conservation of Total Probability

Question 84

Explain the physical meaning of $\partial\rho/\partial t + \partial j/\partial x = 0$ and use it to show that $\int_{-\infty}^{\infty} |\Psi|^2\,dx$ is conserved in time.

Physical meaning. The continuity equation is the **local statement of probability conservation**: probability density $\rho = |\Psi|^2$ behaves like an incompressible "fluid". If ρ increases in some region, it must be because probability current j is flowing *into* that region from its boundaries. Probability is neither created nor destroyed – only transported. (Analogous to mass conservation in fluid dynamics or charge conservation in electromagnetism.)

Step 1: Integrate the continuity equation over all space.

$$\int_{-\infty}^{\infty} \frac{\partial \rho}{\partial t}\,dx = -\int_{-\infty}^{\infty} \frac{\partial j}{\partial x}\,dx.$$

Step 2: Left side – bring derivative outside.

$$\int_{-\infty}^{\infty} \frac{\partial \rho}{\partial t}\,dx = \frac{d}{dt}\int_{-\infty}^{\infty} \rho\,dx = \frac{d}{dt}\int_{-\infty}^{\infty} |\Psi|^2\,dx.$$

Step 3: Right side – fundamental theorem of calculus.

$$-\int_{-\infty}^{\infty} \frac{\partial j}{\partial x}\,dx = -\left[j(x,t)\right]_{x=-\infty}^{x=+\infty}.$$

Step 4: Boundary condition. For any normalisable wavefunction, $\Psi \to 0$ as $|x| \to \infty$. Since $j \propto \Psi^* \partial_x \Psi$, $j \to 0$ at $\pm\infty$ as well.

Step 5: Conclusion.

$$\frac{d}{dt}\int_{-\infty}^{\infty} |\Psi(x,t)|^2\,dx = 0.$$

If Ψ is normalised to 1 at $t = 0$, it remains normalised for all time. The probabilistic interpretation of $|\Psi|^2$ remains self-consistent under Schrödinger evolution.

Q85 – Free Particle: Continuous Spectrum

Question 85

Solve the TISE for a free particle ($V = 0$) and explain why the energy spectrum is continuous and the states are not normalisable in the usual sense.

Step 1: Write the TISE. With $V = 0$:

$$-\frac{\hbar^2}{2m}\frac{d^2\psi}{dx^2} = E\psi \;\Rightarrow\; \frac{d^2\psi}{dx^2} = -k^2\psi, \quad k^2 = \frac{2mE}{\hbar^2} \geq 0.$$

Step 2: General solution. For $E > 0$ (we exclude $E < 0$ since that gives exponentially growing solutions):

$$\psi(x) = Ae^{ikx} + Be^{-ikx},$$

a superposition of right-moving (e^{ikx}) and left-moving (e^{-ikx}) plane waves, valid for any real $k > 0$.

Step 3: Continuous spectrum. k (and hence $E = \hbar^2 k^2/2m$) can take any non-negative real value. There are *no boundary conditions* to restrict k – the particle is unconfined, $x \in (-\infty, \infty)$. Therefore the energy spectrum is **continuous**, in stark contrast to bound systems (Q87, Q102, Q105) where boundary conditions force quantisation.

Step 4: Non-normalisability.

$$\int_{-\infty}^{\infty} |\psi|^2 \, dx = \int_{-\infty}^{\infty} |Ae^{ikx} + Be^{-ikx}|^2 \, dx \to \infty,$$

since $|\psi|^2$ is constant (or slowly oscillating), never decaying.
Step 5: Physical resolution. Free-particle states are normalised to a Dirac delta function in k-space: $\langle k|k'\rangle = \delta(k - k')$. They are physically realised only as part of a localised wave packet (Q21), built from a continuous superposition of many such plane waves – the packet is normalisable even though the individual plane waves are not.

Q86 – Wavenumber in a Classically Allowed Constant-Potential Region

Question 86

An electron in a region of constant potential $V_0 = 3$ eV has total energy $E = 5$ eV. Write the spatial wavefunction form in this region and find the wavenumber k.

Step 1: Kinetic energy. $E - V_0 = 2$ eV $= 3.2 \times 10^{-19}$ J. **Step 2: Wavenumber.**

$$k = \frac{\sqrt{2m(E - V_0)}}{\hbar} = \frac{\sqrt{2(9.11 \times 10^{-31})(3.2 \times 10^{-19})}}{1.055 \times 10^{-34}}$$

$$\boxed{k \approx 7.24 \times 10^9 \text{ m}^{-1}}$$

Spatial form: $\psi(x) = Ce^{ikx} + De^{-ikx}$ (oscillatory, classically allowed).

Q87 – Infinite Square Well: Energy Levels and Eigenfunctions

Question 87

Solve the TISE for the infinite square well ($V = 0$ for $0 < x < L$, $V = \infty$ outside). Find the quantised energies and normalised eigenfunctions.

Step 1: Boundary conditions. Since $V = \infty$ outside, $\psi = 0$ for $x \leq 0$ and $x \geq L$. Continuity requires:

$$\psi(0) = 0, \quad \psi(L) = 0.$$

Step 2: Solve the TISE inside ($0 < x < L$, $V = 0$).

$$\frac{d^2\psi}{dx^2} = -k^2\psi, \quad k^2 = \frac{2mE}{\hbar^2}.$$

General solution:

$$\psi(x) = A\sin(kx) + B\cos(kx).$$

Step 3: Apply boundary condition at $x = 0$.

$$\psi(0) = A\sin(0) + B\cos(0) = B = 0.$$

So $\psi(x) = A\sin(kx)$.

Step 4: Apply boundary condition at $x = L$.

$$\psi(L) = A\sin(kL) = 0.$$

For a non-trivial solution ($A \neq 0$): $\sin(kL) = 0$, so $kL = n\pi$ for $n = 1, 2, 3, \ldots$ (We exclude $n = 0$ since that gives $\psi = 0$ everywhere.)

$$k_n = \frac{n\pi}{L}.$$

Step 5: Quantised energies.

$$\boxed{E_n = \frac{\hbar^2 k_n^2}{2m} = \frac{n^2\pi^2\hbar^2}{2mL^2}, \quad n = 1, 2, 3, \ldots}$$

Step 6: Normalise. From Q50, $\int_0^L \sin^2(n\pi x/L)\,dx = L/2$, so:

$$A^2 \cdot \frac{L}{2} = 1 \;\Rightarrow\; A = \sqrt{\frac{2}{L}}.$$

$$\boxed{\psi_n(x) = \sqrt{\frac{2}{L}}\sin\left(\frac{n\pi x}{L}\right).}$$

Properties:

- $E_n \propto n^2$: levels are not equally spaced.
- ψ_n has $(n-1)$ nodes inside $(0, L)$.
- Ground state $(n = 1)$ has non-zero energy $E_1 = \pi^2\hbar^2/2mL^2 > 0$ (zero-point energy from the uncertainty principle).

Question 88

An electron is confined in an infinite square well of width $L = 0.50\,\text{nm}$. Calculate E_1 (in eV) and E_2.

Step 1: Apply the energy formula from Q87.

$$E_n = \frac{n^2\pi^2\hbar^2}{2m_eL^2}.$$

Constants: $\hbar = 1.055 \times 10^{-34}$ J s, $m_e = 9.109 \times 10^{-31}$ kg, $L = 0.50 \times 10^{-9}$ m.

Step 2: Compute the numerator ($n = 1$).

$$\pi^2\hbar^2 = (9.870)(1.055 \times 10^{-34})^2 = 9.870 \times 1.113 \times 10^{-68} = 1.099 \times 10^{-67}\ \text{J}^2\text{s}^2.$$

Step 3: Compute the denominator.

$$2m_eL^2 = 2\times 9.109\times 10^{-31}\times(0.50\times 10^{-9})^2 = 2\times 9.109\times 10^{-31}\times 2.5\times 10^{-19} = 4.555\times 10^{-49}\ \text{kg m}^2.$$

Step 4: Ground-state energy.

$$E_1 = \frac{1.099 \times 10^{-67}}{4.555 \times 10^{-49}} = 2.412 \times 10^{-19}\ \text{J}.$$

$$E_1 = \frac{2.412 \times 10^{-19}}{1.602 \times 10^{-19}}\ \text{eV} = \mathbf{1.506}\ \text{eV} \approx 1.51\ \text{eV}.$$

Step 5: First excited state. Since $E_n \propto n^2$:

$$E_2 = 4E_1 = 4 \times 1.506 = \mathbf{6.024}\ \text{eV} \approx 6.02\ \text{eV}.$$

Energy level spacing. $E_2 - E_1 = 3E_1 = 4.52\,\text{eV}$ – in the UV range, reflecting the stronger confinement compared to a 1 nm well (where the same spacing is ∼1.13 eV). Halving L raises all energies by a factor of 4, as expected from $E_n \propto L^{-2}$.

Q89 – Probability of Finding a Particle in the Central Half of a Well

Question 89

For $\psi(x) = \sqrt{2/L}\sin(\pi x/L)$ on $[0, L]$, find the probability that the particle is found in $[L/4, 3L/4]$.

Step 1: Integrate.

$$P = \int_{L/4}^{3L/4} \frac{2}{L}\sin^2\left(\frac{\pi x}{L}\right) dx = \frac{1}{L}\left[x - \frac{L}{2\pi}\sin\left(\frac{2\pi x}{L}\right)\right]_{L/4}^{3L/4}$$

Step 2: Evaluate boundary terms. At $3L/4$: $\sin(3\pi/2) = -1$; at $L/4$: $\sin(\pi/2) = +1$.

$$\boxed{P = \frac{1}{2} + \frac{1}{2\pi} \approx 0.659}$$

Question 90

A particle in a 1D box of width L is described at $t = 0$ by $\Psi(x, 0) = \sqrt{30/L^5}\, x(L - x)$. Using $\int_0^L x(L - x)\sin(\pi x/L)dx = 4L^3/\pi^3$, find the probability of measuring E_1.

Step 1: Overlap coefficient.

$$c_1 = \int_0^L \psi_1^* \Psi(x, 0)\, dx = \sqrt{\frac{60}{L^6}} \cdot \frac{4L^3}{\pi^3} = \frac{4\sqrt{60}}{\pi^3}$$

Step 2: Evaluate. $\sqrt{60} \approx 7.746$, $\pi^3 \approx 31.006$:

$$c_1 = \frac{4(7.746)}{31.006} \approx 0.9993$$

$$\boxed{P_1 = |c_1|^2 = \frac{960}{\pi^6} \approx 0.9986}$$

The tent state is an excellent approximation to the $n = 1$ ground state ($> 99.8\%$ overlap).

Q91 – Position Spread of the First Excited State

Question 91

A particle of mass m is in the first excited state $(n = 2)$ of an infinite well of width L. Find $\langle x \rangle$, $\langle x^2 \rangle$, and Δx.

Step 1: Mean position. By symmetry of ψ_2, $\langle x \rangle = L/2$. **Step 2: Mean square position.**

$$\langle x^2 \rangle_2 = L^2 \left(\frac{1}{3} - \frac{1}{8\pi^2} \right) \approx 0.3207 L^2$$

Step 3: Standard deviation.

Q91 – Position Spread of the First Excited State

$$\boxed{\Delta x = L\sqrt{0.3207 - 0.25} = L\sqrt{0.0707} \approx 0.266\, L}$$

Question 92

For an electron in an infinite well of width $L = 0.3$ nm, find the four lowest energy levels in eV and the three photon wavelengths corresponding to transitions $n = 4 \to 3$, $n = 3 \to 2$, and $n = 2 \to 1$.

Step 1: Ground-state energy.

$$E_1 = \frac{\pi^2\hbar^2}{2m_eL^2} \approx 4.18 \text{ eV}$$

Step 2: Excited levels $(E_n = n^2E_1)$.

$$E_2 = 16.73 \text{ eV}, \quad E_3 = 37.64 \text{ eV}, \quad E_4 = 66.91 \text{ eV}$$

Step 3: Transition wavelengths via $\lambda = hc/\Delta E$.

Q92 – Energy Levels and Transition Wavelengths in a Narrow Well

$$n = 4 \to 3: \ \Delta E = 29.28 \text{ eV} \Rightarrow \lambda = 42.4 \text{ nm (XUV)}$$

$$n = 3 \to 2: \ \Delta E = 20.91 \text{ eV} \Rightarrow \lambda = 59.3 \text{ nm (EUV)}$$

$$\boxed{n = 2 \to 1: \ \Delta E = 12.55 \text{ eV} \Rightarrow \lambda = 98.8 \text{ nm (UV)}}$$

Q93 – Step Potential, $E < V_0$: Total Reflection and Phase Shift

Question 93

For the step potential $V(x) = 0$ $(x < 0)$, $V(x) = V_0$ $(x > 0)$, consider a particle with energy $E < V_0$ incident from the left. Derive the reflection coefficient R and the phase shift of the reflected wave.

Step 1: Wavefunctions in each region.

$$\psi_I(x) = Ae^{ikx} + Be^{-ikx} \ (x < 0), \qquad \psi_{II}(x) = Ce^{-\kappa x} \ (x > 0),$$

with $k = \sqrt{2mE}/\hbar$ and $\kappa = \sqrt{2m(V_0 - E)}/\hbar$ (the growing exponential is discarded since ψ must stay finite as $x \to \infty$).

Step 2: Match ψ and ψ' at $x = 0$.

$$A + B = C, \qquad ik(A - B) = -\kappa C.$$

Step 3: Solve for the reflection amplitude B/A. Substituting $C = A + B$ into the derivative condition:

$$ik(A - B) = -\kappa(A + B) \;\Rightarrow\; A(ik + \kappa) = B(ik - \kappa) \;\Rightarrow\; \frac{B}{A} = \frac{\kappa + ik}{ik - \kappa}.$$

Step 4: Compute the reflection coefficient.

$$R = \left|\frac{B}{A}\right|^2 = \frac{\kappa^2 + k^2}{k^2 + \kappa^2} = 1.$$

$R = 1$ – total reflection, as expected since the particle cannot propagate in region II.

Step 5: Extract the phase shift. Write $\kappa + ik = \sqrt{k^2 + \kappa^2}\, e^{i\alpha}$ with $\alpha = \arctan(k/\kappa)$, and $ik - \kappa = -(\kappa - ik) = -\sqrt{k^2 + \kappa^2}\, e^{-i\alpha}$. Then

$$\frac{B}{A} = \frac{e^{i\alpha}}{-e^{-i\alpha}} = -e^{2i\alpha} = e^{i(\pi + 2\alpha)}.$$

Q93 – Step Potential, $E < V_0$: Total Reflection and Phase Shift

$$\boxed{\delta = \pi + 2\arctan(k/\kappa).}$$

Even though $R = 1$ (no energy is reflected back attenuated), the reflected wave is shifted in phase relative to the incident wave – a purely quantum effect with no classical analogue, and the wavefunction is nonzero (evanescent) inside the classically forbidden region $x > 0$.

Q94 – Step Potential ($E > V_0$): Partial Transmission

Question 94

For the step potential $V(x) = 0$ $(x < 0)$, $V(x) = V_0$ $(x > 0)$, consider a particle with energy $E > V_0$ incident from the left. Derive the transmission and reflection coefficients T and R, and verify $R + T = 1$.

Step 1: Wavefunctions in each region. Both regions are classically allowed, so the solutions are oscillatory:

$$\psi_{\rm I} = Ae^{ik_1x} + Be^{-ik_1x}\ (x < 0), \qquad \psi_{\rm II} = Ce^{ik_2x}\ (x > 0),$$

with $k_1 = \sqrt{2mE}/\hbar$ and $k_2 = \sqrt{2m(E - V_0)}/\hbar < k_1$. The left-going wave in Region II is discarded (no source on the right).

Step 2: Boundary conditions at $x = 0$.

$$A + B = C, \qquad ik_1(A - B) = ik_2C.$$

Q94 – Step Potential ($E > V_0$): Partial Transmission

Step 3: Solve for B/A and C/A. From the derivative condition $A - B = (k_2/k_1)C$ and the continuity condition $A + B = C$:

$$2A = C\left(1 + \frac{k_2}{k_1}\right) \;\Rightarrow\; \frac{C}{A} = \frac{2k_1}{k_1 + k_2},$$

$$2B = C\left(1 - \frac{k_2}{k_1}\right) \;\Rightarrow\; \frac{B}{A} = \frac{k_1 - k_2}{k_1 + k_2}.$$

Step 4: Reflection and transmission coefficients. R is the ratio of reflected to incident probability current; T uses the factor k_2/k_1 to account for the change in speed:

$$R = \left|\frac{B}{A}\right|^2 = \left(\frac{k_1 - k_2}{k_1 + k_2}\right)^2, \qquad T = \frac{k_2}{k_1}\left|\frac{C}{A}\right|^2 = \frac{4k_1k_2}{(k_1 + k_2)^2}.$$

Step 5: Verify $R + T = 1$.

$$R + T = \frac{(k_1 - k_2)^2 + 4k_1k_2}{(k_1 + k_2)^2} = \frac{k_1^2 + 2k_1k_2 + k_2^2}{(k_1 + k_2)^2} = \frac{(k_1 + k_2)^2}{(k_1 + k_2)^2} = 1. \checkmark$$

Q94 – Step Potential ($E > V_0$): Partial Transmission

Classical contrast. A classical particle with $E > V_0$ would *always* be transmitted ($R = 0$, $T = 1$). Quantum mechanically, $R > 0$ whenever $k_1 \neq k_2$, i.e. whenever there is any potential step – a purely wave-mechanical effect analogous to partial reflection of light at a refractive-index boundary. As $V_0 \to 0$, $k_2 \to k_1$ and $R \to 0$, recovering the classical result.

Question 95

An electron ($E = 4$ eV) approaches a step of height $V_0 = 2$ eV. Find the reflection coefficient R and verify $R + T = 1$.

Step 1: Wavenumbers (proportional values).

$$k_1 \propto \sqrt{4} = 2, \qquad k_2 \propto \sqrt{4-2} = \sqrt{2}$$

Step 2: Reflection coefficient.

$$R = \left(\frac{k_1 - k_2}{k_1 + k_2}\right)^2 = \left(\frac{0.5858}{3.4142}\right)^2 \approx 0.0295$$

Step 3: Transmission coefficient.

$$T = \frac{4k_1k_2}{(k_1+k_2)^2} \approx 0.9706$$

Step 4: Verify conservation.

$$\boxed{R + T \approx 0.0295 + 0.9706 \approx 1 \checkmark}$$

Q96 – Tunnelling Transmission in the Opaque-Barrier Limit

Question 96

Given the exact rectangular-barrier transmission coefficient $T^{-1} = 1 + \dfrac{V_0^2 \sinh^2(\kappa L)}{4E(V_0 - E)}$, derive the simplified asymptotic formula for T in the opaque-barrier (thick-barrier) limit $\kappa L \gg 1$.

Step 1: Approximate $\sinh(\kappa L)$ for large κL.

$$\sinh(\kappa L) = \frac{e^{\kappa L} - e^{-\kappa L}}{2} \approx \frac{e^{\kappa L}}{2} \quad \text{for } \kappa L \gg 1,$$

so

$$\sinh^2(\kappa L) \approx \frac{e^{2\kappa L}}{4}.$$

Step 2: Substitute into T^{-1}.

$$T^{-1} \approx 1 + \frac{V_0^2}{4E(V_0 - E)} \cdot \frac{e^{2\kappa L}}{4} = 1 + \frac{V_0^2 e^{2\kappa L}}{16E(V_0 - E)}.$$

Step 3: Drop the subleading constant. For $\kappa L \gg 1$, the exponential term dominates the "1", so $T^{-1} \approx V_0^2 e^{2\kappa L}/[16E(V_0 - E)]$. Inverting:

$$\boxed{T \approx \frac{16E(V_0 - E)}{V_0^2}\, e^{-2\kappa L} = 16\frac{E}{V_0}\left(1 - \frac{E}{V_0}\right) e^{-2\kappa L}.}$$

Step 4: Physical reading. The transmission probability is dominated by the exponential decay $e^{-2\kappa L}$, with $\kappa = \sqrt{2m(V_0 - E)}/\hbar$: doubling the barrier width L *squares* the transmission probability's suppression factor, which is why tunneling currents in devices like the STM are exquisitely sensitive to tip-to-sample distance.

Q97 – Decay Constant in a Classically Forbidden Region

Question 97

What happens to the wavefunction in a region where $E < V_0$? Evaluate the decay constant $\kappa = \sqrt{2m(V_0 - E)}/\hbar$ for an electron with $E = 2$ eV and $V_0 = 5$ eV.

Step 1: Energy gap. $V_0 - E = 3 \text{ eV} = 4.8 \times 10^{-19}$ J. **Step 2: Decay constant.**

$$\kappa = \frac{\sqrt{2(9.11 \times 10^{-31})(4.8 \times 10^{-19})}}{1.055 \times 10^{-34}}$$

$$\boxed{\kappa \approx 8.87 \times 10^{9} \text{ m}^{-1}}$$

Question 98

An electron of energy $E = 1.0\,\text{eV}$ encounters a barrier of height $V_0 = 2.0\,\text{eV}$ and width $a = 0.5\,\text{nm}$. Estimate T.

Step 1: Compute κ.

$$V_0 - E = 2.0 - 1.0 = 1.0 \text{ eV} = 1.602 \times 10^{-19} \text{ J}.$$

$$2m_e(V_0 - E) = 2(9.109 \times 10^{-31})(1.602 \times 10^{-19}) = 2.919 \times 10^{-49} \text{ kg}^2\text{m}^2\text{s}^{-2}.$$

$$\kappa = \frac{\sqrt{2m_e(V_0 - E)}}{\hbar} = \frac{\sqrt{2.919 \times 10^{-49}}}{1.055 \times 10^{-34}} = \frac{5.403 \times 10^{-25}}{1.055 \times 10^{-34}} = 5.12 \times 10^9 \text{ m}^{-1}.$$

Step 2: Compute κa.

$$\kappa a = (5.12 \times 10^9)(0.5 \times 10^{-9}) = 2.56.$$

Step 3: Apply the opaque-barrier approximation. Since $\kappa a = 2.56 > 1$, use $T \approx \frac{16E(V_0 - E)}{V_0^2} e^{-2\kappa a}$:

$$\frac{16E(V_0 - E)}{V_0^2} = \frac{16 \times 1.0 \times 1.0}{(2.0)^2} = \frac{16}{4} = 4.$$

$$e^{-2\kappa a} = e^{-2\times 2.56} = e^{-5.12}.$$

$$e^{-5.12} \approx e^{-5} \times e^{-0.12} \approx 6.738 \times 10^{-3} \times 0.887 = 5.98 \times 10^{-3}.$$

Step 4: Final result.

$$T \approx 4 \times 5.98 \times 10^{-3} \approx \mathbf{0.024} = \mathbf{2.4}\%.$$

About 2.4% of incident electrons tunnel through despite having insufficient classical energy. At the atomic scale, tunnelling is a small but physically real and observable effect.

Question 99

A proton ($m_p = 1.673 \times 10^{-27}$ kg) with $E = 0.1$ MeV tunnels through a barrier of height $V_0 = 1$ MeV and width $a = 2$ fm. Estimate the transmission coefficient.

Step 1: Decay constant. $V_0 - E = 0.9$ MeV $= 1.44 \times 10^{-13}$ J:

$$\kappa = \frac{\sqrt{2(1.673 \times 10^{-27})(1.44 \times 10^{-13})}}{1.055 \times 10^{-34}} \approx 2.08 \times 10^{14}\ \text{m}^{-1}$$

Step 2: Exponent.

Q99 – Proton Tunnelling Through a Nuclear-Scale Barrier

$$2\kappa a = 2(2.08 \times 10^{14})(2 \times 10^{-15}) \approx 0.83$$

Step 3: Transmission coefficient.

$$\boxed{T \approx e^{-0.83} \approx 0.44}$$

Since $\kappa a \approx 0.83 < 1$, the barrier is not very "opaque," so this thin-barrier estimate should really use the full rectangular-barrier formula (Q96) rather than the opaque-barrier limit; the crude exponential estimate here already shows tunnelling is far from negligible for a proton at nuclear-scale barriers.

Question 100

A scanning tunnelling microscope (STM) tip–sample gap changes by $\delta a = 0.05$ nm. If the barrier height above E is $V_0 - E = 4$ eV, by what factor does the tunnelling current change? (Current $\propto T \propto e^{-2\kappa a}$.)

Step 1: Decay constant.

$$\kappa = \frac{\sqrt{2(9.11\times10^{-31})(4\times1.6\times10^{-19})}}{1.055\times10^{-34}} \approx 1.023\times10^{10}\ \text{m}^{-1}$$

Step 2: Change in exponent.

$$2\kappa\,\delta a = 2(1.023\times10^{10})(0.05\times10^{-9}) \approx 1.023$$

Step 3: Factor change in current.

Q100 – Sensitivity of STM Tunnelling Current to Tip–Sample Gap

$$\boxed{e^{-1.023} \approx 0.36}$$

The current drops to $\sim 36\%$ of its original value (a decrease by a factor of ≈ 2.8). This extreme sensitivity gives the STM its sub-atomic resolution.

Question 101

Describe two real-world physical phenomena that are direct manifestations of quantum tunnelling.

1. Alpha (α) decay of radioactive nuclei.

Setup. Inside a heavy nucleus, an α-particle (helium-4 nucleus) is bound by the strong nuclear force but faces a Coulomb potential barrier as it tries to escape. The nuclear-plus-Coulomb potential has the shape of a finite barrier.

Classical expectation. The α-particle's kinetic energy ($\sim$4–9 MeV) is far below the Coulomb barrier height ($\sim$25–30 MeV), so the nucleus should be perfectly stable.

Quantum result. The α-particle's wavefunction has a non-zero amplitude outside the nucleus. It escapes by tunnelling through the Coulomb barrier with probability per unit time $\lambda \propto e^{-2\kappa a}$ (Gamow's theory, 1928).

This exponential dependence on $\kappa \propto \sqrt{V_0 - E}$ and a quantitatively explains the enormous range of observed half-lives of α-emitters (from microseconds to billions of years).

2. Scanning Tunnelling Microscope (STM).

Setup. An atomically sharp conducting tip is brought to within a few Å of a conducting sample surface, with a small bias voltage applied. The vacuum gap acts as a potential barrier.

Classical expectation. Electrons cannot cross the vacuum gap – they lack sufficient energy.

Quantum result. Electrons tunnel across the gap, producing a measurable current:

$$I \propto e^{-2\kappa d}, \quad \kappa \approx \frac{\sqrt{2m_e\phi_{\text{work}}}}{\hbar},$$

where d is the tip–sample distance. Since the current depends *exponentially* on d, a change of 1 Å in gap width changes the current by about one order of magnitude, giving atomic-scale (<0.1 Å) vertical sensitivity.

By scanning the tip and keeping I constant (constant d), one obtains an image of the surface topography with atomic resolution.

Other examples: field emission, tunnel diodes, Josephson junctions, enzymatic proton tunnelling.

Q102 – Delta Potential: Exactly One Bound State

Question 102

Solve the TISE for $V(x) = -\alpha\delta(x)$ $(\alpha > 0)$ and show it supports exactly one bound state.

Step 1: TISE away from $x = 0$. For $x \neq 0$, $V = 0$. For a bound state $E < 0$, write $E = -|E|$:

$$\psi'' = \frac{2m|E|}{\hbar^2}\psi \equiv \beta^2\psi, \quad \beta = \frac{\sqrt{2m|E|}}{\hbar} > 0.$$

Step 2: Select normalisable solutions.

- $x < 0$: $\psi = Ae^{\beta x}$ (grows toward $x = -\infty$ only for the wrong sign; we need decay, so keep $e^{+\beta x}$). ✓
- $x > 0$: $\psi = Be^{-\beta x}$. ✓

Step 3: Impose continuity of ψ at $x = 0$. $\psi(0^-) = \psi(0^+)$: $A = B \equiv C$.

$$\psi(x) = Ce^{-\beta|x|} \quad \text{(even/symmetric function).}$$

Step 4: Integrate the TISE across $[-\varepsilon,\varepsilon]$ and take $\varepsilon\to 0$. The full TISE is $-\frac{\hbar^2}{2m}\psi'' - \alpha\delta(x)\psi = E\psi$. Integrating:

$$-\frac{\hbar^2}{2m}\left[\psi'(0^+)-\psi'(0^-)\right]-\alpha\psi(0)=0$$

(the $E\psi$ term vanishes since $E\psi$ is bounded and the interval shrinks to zero).

Step 5: Compute the derivative jump.

$$\psi'(0^+)=-\beta C,\quad \psi'(0^-)=+\beta C.$$

$$\psi'(0^+)-\psi'(0^-)=-2\beta C.$$

Substituting ($\psi(0)=C$):

$$-\frac{\hbar^2}{2m}(-2\beta C)-\alpha C=0 \;\Rightarrow\; \frac{\hbar^2\beta}{m}=\alpha \;\Rightarrow\; \boxed{\beta=\frac{m\alpha}{\hbar^2}.}$$

One unique positive $\beta \Rightarrow$ **exactly one bound state**. (An antisymmetric ansatz requires $\psi(0)=0$, so the jump condition gives $C=0$ – no state.)

Q103 – Bound-State Energy of the Delta Potential

Question 103

Using the result of Q102, find the bound-state energy of $V(x) = -\alpha\delta(x)$.

Step 1: Recall β from Q102.

$$\beta = \frac{m\alpha}{\hbar^2}.$$

Step 2: Relate β to the energy. By definition, $\beta = \sqrt{2m|E|}/\hbar$, so:

$$\beta^2 = \frac{2m|E|}{\hbar^2} \;\Rightarrow\; |E| = \frac{\hbar^2\beta^2}{2m}.$$

Step 3: Substitute β.

$$|E| = \frac{\hbar^2}{2m}\cdot\left(\frac{m\alpha}{\hbar^2}\right)^2 = \frac{\hbar^2}{2m}\cdot\frac{m^2\alpha^2}{\hbar^4} = \frac{m\alpha^2}{2\hbar^2}.$$

Q103 – Bound-State Energy of the Delta Potential

Step 4: State the result. Since the state is bound, $E < 0$:

$$\boxed{E = -\frac{m\alpha^2}{2\hbar^2}.}$$

Step 5: Physical properties.

- Stronger $\alpha \Rightarrow$ deeper (more negative) bound-state energy: $|E| \propto \alpha^2$.
- The decay length $1/\beta = \hbar^2/m\alpha$ decreases as α increases (wavefunction becomes more localised).

Normalised wavefunction. From $\int_{-\infty}^{\infty} |C|^2 e^{-2\beta|x|}\, dx = |C|^2/\beta = 1$:

$$C = \sqrt{\beta} = \sqrt{\frac{m\alpha}{\hbar^2}}, \quad \psi(x) = \sqrt{\beta}\, e^{-\beta|x|}.$$

Question 104

Set up the TISE for $V(x) = \frac{1}{2}m\omega^2 x^2$ and explain qualitatively why the energy spectrum must be discrete.

The TISE for the SHO:

$$-\frac{\hbar^2}{2m}\frac{d^2\psi}{dx^2} + \frac{1}{2}m\omega^2 x^2 \psi = E\psi.$$

This is the quantum analogue of the classical mass-on-a-spring system, and it approximates *any* smooth potential near a stable minimum (by Taylor expansion about the minimum).

Why a discrete spectrum is expected.

For *any* finite energy E, the parabolic potential $V(x) = \frac{1}{2}m\omega^2 x^2 \to \infty$ as $|x| \to \infty$. There always exist two classical turning points $\pm x_0 = \pm\sqrt{2E/m\omega^2}$, beyond which the region is classically forbidden.

This confines the particle on *both* sides for any finite E – exactly as in the infinite well (Q87) or the delta well (Q102–Q103).

The requirement that $\psi(x) \to 0$ as $|x| \to \infty$ (normalisability, Q49) can only be satisfied for *special discrete values* of E. Generic values of E lead to solutions that diverge (as $e^{x^2/2}$) at $\pm\infty$ and are therefore unphysical.

Analogy. The situation is exactly analogous to the infinite square well, except that the "walls" are not hard ($V = \infty$) but softly parabolic – the normalisability condition still selects a discrete (but now infinite) ladder of allowed energies (Q105).

Standard form. Introducing the dimensionless variable $\xi = x\sqrt{m\omega/\hbar}$, the TISE becomes $\psi''(\xi) = (\xi^2 - \lambda)\psi(\xi)$ with $\lambda = 2E/\hbar\omega$. The only values of λ for which ψ is normalisable are $\lambda = 2n + 1$ $(n = 0, 1, 2, \ldots)$, giving $E_n = (n + \frac{1}{2})\hbar\omega$ (Q105).

Q105 – SHO Energy Levels: Ladder Operators

Question 105

State the energy eigenvalues of the quantum harmonic oscillator and outline how they are obtained via the ladder-operator method.

Result.

$$\boxed{E_n = \left(n + \frac{1}{2}\right)\hbar\omega, \quad n = 0, 1, 2, 3, \ldots}$$

Equally spaced levels separated by $\hbar\omega$; ground-state zero-point energy $E_0 = \frac{1}{2}\hbar\omega$.

Outline of the ladder-operator method.
Step 1: Define raising and lowering operators.

$$\hat{a} = \sqrt{\frac{m\omega}{2\hbar}}\left(\hat{x} + \frac{i\hat{p}}{m\omega}\right), \quad \hat{a}^\dagger = \sqrt{\frac{m\omega}{2\hbar}}\left(\hat{x} - \frac{i\hat{p}}{m\omega}\right).$$

Step 2: Rewrite the Hamiltonian. Using $[\hat{x}, \hat{p}] = i\hbar$ (Q76), one can show:

$$\hat{H} = \frac{\hat{p}^2}{2m} + \frac{1}{2}m\omega^2\hat{x}^2 = \hbar\omega\left(\hat{a}^\dagger\hat{a} + \frac{1}{2}\right), \quad [\hat{a}, \hat{a}^\dagger] = 1.$$

Q105 – SHO Energy Levels: Ladder Operators

Step 3: Action on energy eigenstates.
The number operator $\hat{N} = \hat{a}^\dagger \hat{a}$ has non-negative eigenvalues. If $|n\rangle$ is an eigenstate of $\hat{N}$ with eigenvalue n, then:

$$\hat{a}^\dagger |n\rangle = \sqrt{n+1}\,|n+1\rangle \quad \text{(raises by 1)},$$

$$\hat{a}|n\rangle = \sqrt{n}\,|n-1\rangle \quad \text{(lowers by 1)}.$$

Step 4: Ground state from $\hat{a}|0\rangle = 0$.
Since $\hat{N} \geq 0$, there must be a lowest state $|0\rangle$ satisfying $\hat{a}|0\rangle = 0$. From $\hat{H}|0\rangle = \hbar\omega(\hat{N} + \frac{1}{2})|0\rangle = \frac{1}{2}\hbar\omega|0\rangle$:

$$E_0 = \frac{1}{2}\hbar\omega.$$

Repeatedly applying $\hat{a}^\dagger$ generates the entire ladder $|0\rangle, |1\rangle, |2\rangle, \ldots$ with $E_n = (n + \frac{1}{2})\hbar\omega$ – obtained *purely algebraically* without solving the differential equation.

Q106 – Verification of the SHO Ground State

Question 106

Verify by direct substitution that $\psi_0(x) = \left(\frac{m\omega}{\pi\hbar}\right)^{1/4}\exp\left(-\frac{m\omega x^2}{2\hbar}\right)$ solves the SHO TISE and find E_0.

Let $\beta = m\omega/\hbar$ and $N = (m\omega/\pi\hbar)^{1/4}$, so $\psi_0 = Ne^{-\beta x^2/2}$.

Step 1: First derivative.

$$\psi_0' = N(-\beta x)e^{-\beta x^2/2} = -\beta x\psi_0.$$

Step 2: Second derivative.

$$\psi_0'' = -\beta\psi_0 + (-\beta x)(-\beta x\psi_0) = -\beta\psi_0 + \beta^2x^2\psi_0 = (\beta^2x^2 - \beta)\psi_0.$$

Step 3: Substitute into the SHO TISE.

$$-\frac{\hbar^2}{2m}(\beta^2 x^2 - \beta)\psi_0 + \frac{1}{2}m\omega^2 x^2 \psi_0 = E_0\psi_0.$$

Step 4: Collect the x^2 terms.

$$-\frac{\hbar^2\beta^2}{2m} + \frac{m\omega^2}{2} = -\frac{\hbar^2}{2m}\cdot\frac{m^2\omega^2}{\hbar^2} + \frac{m\omega^2}{2} = -\frac{m\omega^2}{2} + \frac{m\omega^2}{2} = 0.$$

The x^2 terms **cancel exactly**. ✓

Step 5: Collect the constant terms.

$$-\frac{\hbar^2}{2m}(-\beta) = \frac{\hbar^2\beta}{2m} = \frac{\hbar^2}{2m}\cdot\frac{m\omega}{\hbar} = \frac{\hbar\omega}{2}.$$

Step 6: Read off the energy eigenvalue.

$$E_0\psi_0 = \frac{\hbar\omega}{2}\psi_0 \;\Rightarrow\; \boxed{E_0 = \frac{\hbar\omega}{2}.} \quad \checkmark$$

This confirms ψ_0 is an eigenfunction with eigenvalue $E_0 = \frac{1}{2}\hbar\omega$, the ground state of Q105.

Question 107

Explain the physical origin and significance of the zero-point energy $E_0 = \frac{1}{2}\hbar\omega$, and contrast it with the classical prediction.

Classical prediction. A classical harmonic oscillator at rest at $x = 0$ with $p = 0$ has energy $E = 0$. This is the achievable minimum – in principle, by cooling the system to remove all kinetic and potential energy.

Quantum result. $E_0 = \frac{1}{2}\hbar\omega > 0$ – even in the ground state, the oscillator retains irreducible energy. The system can *never* be at rest.

Connection to the uncertainty principle. Setting $x = 0$ and $p = 0$ exactly would require $\Delta x = \Delta p = 0$, violating $\Delta x\,\Delta p \geq \hbar/2$ (Q40, Q41).

Minimisation argument. Let $\Delta x = \sigma$. The expected energy is:

$$\langle E\rangle = \frac{\langle p^2\rangle}{2m} + \frac{1}{2}m\omega^2\langle x^2\rangle \geq \frac{(\Delta p)^2}{2m} + \frac{1}{2}m\omega^2(\Delta x)^2.$$

Using $\Delta p \geq \hbar/2\sigma$:

$$\langle E\rangle \geq \frac{\hbar^2}{8m\sigma^2} + \frac{1}{2}m\omega^2\sigma^2 \equiv f(\sigma).$$

Minimise $f(\sigma)$ with respect to σ:

$$\frac{df}{d\sigma} = -\frac{\hbar^2}{4m\sigma^3} + m\omega^2\sigma = 0 \;\Rightarrow\; \sigma^4 = \frac{\hbar^2}{4m^2\omega^2} \;\Rightarrow\; \sigma_{\min} = \sqrt{\frac{\hbar}{2m\omega}}.$$

Substituting back:

$$E_{\min} = \frac{\hbar^2}{8m\cdot\hbar/2m\omega} + \frac{1}{2}m\omega^2\cdot\frac{\hbar}{2m\omega} = \frac{\hbar\omega}{4} + \frac{\hbar\omega}{4} = \frac{\hbar\omega}{2} = E_0. \quad \checkmark$$

Physical significance. Zero-point energy is observable in: residual vibrational energy of molecules at $T \to 0$; the Casimir effect; and the stability of liquid He which remains liquid at atmospheric pressure even at $T \to 0$.

Q108 – Zero-Point Energy of a Diatomic Molecule

Question 108

Estimate the zero-point energy of a diatomic molecule modelled as a harmonic oscillator with $\omega = 3.0 \times 10^{13}$ rad s^{-1}. Express the answer in meV.

Step 1: Apply $E_0 = \frac{1}{2}\hbar\omega$.

$$E_0 = \frac{1}{2}(1.055 \times 10^{-34})(3.0 \times 10^{13}) = 1.58 \times 10^{-21} \text{ J}$$

Step 2: Convert.

$$\boxed{E_0 = \frac{1.58 \times 10^{-21}}{1.6 \times 10^{-19}} \text{ eV} \approx 9.88 \text{ meV}}$$

Q109 – Expectation Values for a QHO Superposition of $|0\rangle$ and $|2\rangle$

Question 109

An electron in the QHO is described by $|\psi\rangle = \frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|2\rangle$. Find $\langle E\rangle$, $\langle x\rangle$, and $\langle x^2\rangle$.

Step 1: Mean energy. $P_0 = P_2 = \frac{1}{2}$:

$$\langle E\rangle = \frac{1}{2}\hbar\omega\left(\frac{1}{2}\right) + \frac{1}{2}\hbar\omega\left(\frac{5}{2}\right) = \frac{3}{2}\hbar\omega$$

Step 2: Mean position. $|0\rangle$ and $|2\rangle$ differ by two quanta, so all relevant matrix elements of $\hat{x}$ vanish:

$$\langle x\rangle = 0$$

Step 3: Mean square position. Using $\hat{x}^2 = \frac{\hbar}{2m\omega}(2\hat{N} + 1 + \hat{a}_+^2 + \hat{a}_-^2)$, the diagonal pieces give $\frac{1}{2}(1) + \frac{1}{2}(5) = 3$ and the cross term contributes $\sqrt{2}$:

$$\boxed{\langle x^2 \rangle = \frac{\hbar}{2m\omega}\left(3 + \sqrt{2}\right) \approx \frac{\hbar}{2m\omega}(4.414)}$$

Question 110

For the state $|\psi\rangle = \frac{1}{\sqrt{5}}|0\rangle + \frac{2}{\sqrt{5}}|1\rangle$ (QHO states), find $\langle\hat{x}\rangle$, using $\hat{x} = \sqrt{\hbar/(2m\omega)}(\hat{a}_+ + \hat{a}_-)$.

Step 1: Apply the ladder operators and project back onto $\langle\psi|$:

$$\langle\psi|(\hat{a}_+ + \hat{a}_-)|\psi\rangle = \frac{1}{\sqrt{5}} \cdot \frac{2}{\sqrt{5}} + \frac{2}{\sqrt{5}} \cdot \frac{1}{\sqrt{5}} = \frac{4}{5}$$

Step 2: Result.

$$\boxed{\langle\hat{x}\rangle = \frac{4}{5}\sqrt{\frac{\hbar}{2m\omega}}}$$

Question 111

Outline how the TISE for a central potential $V(r)$ is separated into radial and angular parts using spherical polar coordinates.

Step 1: Write the 3D TISE.

$$-\frac{\hbar^2}{2m}\nabla^2\psi + V(r)\psi = E\psi,$$

where in spherical polars:

$$\nabla^2 = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial}{\partial r}\right) + \frac{1}{r^2\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial}{\partial\theta}\right) + \frac{1}{r^2\sin^2\theta}\frac{\partial^2}{\partial\varphi^2}.$$

Step 2: Separable ansatz. Since V depends only on r, try $\psi(r,\theta,\varphi) = R(r)Y(\theta,\varphi)$. Substituting and multiplying by $-2mr^2/\hbar^2 RY$, one obtains:

$$\underbrace{\frac{1}{R}\frac{d}{dr}\left(r^2\frac{dR}{dr}\right) - \frac{2mr^2}{\hbar^2}[V(r)-E]}_{\text{function of } r \text{ only}} = \underbrace{-\frac{1}{Y}\hat{\Lambda}^2 Y}_{\text{function of } \theta,\varphi \text{ only}} = \ell(\ell+1),$$

where $\hat{\Lambda}^2$ is the angular part of ∇^2 and $\ell(\ell+1)$ is the common separation constant.

Step 3: Angular equation.

$$\hat{\Lambda}^2 Y(\theta,\varphi) = -\ell(\ell+1)Y(\theta,\varphi),$$

i.e. $\hat{L}^2 Y = \ell(\ell+1)\hbar^2 Y$ (eigenvalue equation for orbital angular momentum squared, Q117).
Solutions are the spherical harmonics $Y_\ell^m(\theta,\varphi)$, with $\ell = 0, 1, 2, \ldots$ and $m = -\ell, \ldots, +\ell$.

Step 4: Radial equation.

$$-\frac{\hbar^2}{2m}\frac{1}{r^2}\frac{d}{dr}\left(r^2\frac{dR}{dr}\right) + \left[V(r) + \frac{\hbar^2\ell(\ell+1)}{2mr^2}\right]R = ER.$$

The term $\hbar^2\ell(\ell+1)/2mr^2$ is the effective “centrifugal” potential.
For hydrogen ($V = -e^2/4\pi\varepsilon_0 r$), requiring $R(r) \to 0$ as $r \to \infty$ quantises E to give
$E_n = -13.6$ eV$/n^2$ (Bohr levels).

Q112 – Four Quantum Numbers

Question 112

List the four quantum numbers n, ℓ, m_ℓ, m_s that specify the state of an electron in a hydrogen-like atom, stating the allowed values and physical meaning of each.

1. **Principal quantum number n.**

$$n = 1, 2, 3, \dots$$

Determines the energy: $E_n = -13.6\ \text{eV}/n^2$ (for hydrogen). Also sets the approximate size of the electron's probability distribution ("shell").

2. **Orbital (azimuthal) quantum number ℓ.**

$$\ell = 0, 1, 2, \dots, (n-1).$$

Determines the magnitude of orbital angular momentum: $L = \sqrt{\ell(\ell+1)}\,\hbar$ (Q117).
Determines the shape of the angular distribution (subshell: $s, p, d, f, \dots$ for $\ell = 0, 1, 2, 3, \dots$).

3. Magnetic quantum number m_ℓ.

$$m_\ell = -\ell, -\ell+1, \ldots, 0, \ldots, \ell-1, +\ell \quad (2\ell+1 \text{ values}).$$

Determines the z-component of orbital angular momentum: $L_z = m_\ell \hbar$ (Q117). Orbitals with the same n, ℓ but different m_ℓ are degenerate in zero field but split in an external magnetic field (Zeeman effect).

4. Spin magnetic quantum number m_s.

$$m_s = +\tfrac{1}{2} \quad \text{or} \quad -\tfrac{1}{2}.$$

Determines the z-component of the electron's intrinsic spin angular momentum: $S_z = m_s \hbar$. Spin is a purely quantum-mechanical property with no classical analogue.

Summary. The set (n, ℓ, m_ℓ, m_s) uniquely labels every distinct quantum state of the electron. Together with the Pauli exclusion principle, this explains atomic structure and the periodic table.

Question 113

Calculate the degeneracy (number of distinct quantum states) of the hydrogen energy level E_n, including electron spin.

Step 1: Count ℓ values for a given n. From Q112: $\ell = 0, 1, 2, \ldots, (n-1)$, giving n distinct values.
Step 2: Count m_ℓ values for each ℓ. For each ℓ: $m_\ell = -\ell, \ldots, +\ell$, giving $2\ell + 1$ values.
Step 3: Sum over ℓ.

$$\sum_{\ell=0}^{n-1}(2\ell+1) = 2\sum_{\ell=0}^{n-1}\ell + \sum_{\ell=0}^{n-1}1 = 2\cdot\frac{(n-1)n}{2} + n = n(n-1) + n = n^2.$$

Step 4: Include spin degeneracy. Each spatial state (n, ℓ, m_ℓ) accommodates $m_s = \pm\frac{1}{2}$, a factor of 2:

$$\boxed{g_n = 2n^2.}$$

Checks.

- $n = 1$: $g_1 = 2$ (the two spin states of the $1s$ orbital – fits the K-shell with 2 electrons).
- $n = 2$: $g_2 = 8$ (the L-shell: $2s^2 2p^6$ – 8 electrons).
- $n = 3$: $g_3 = 18$ (the M-shell – 18 electrons including $3d$).

The $2n^2$ rule underlies the structure of the periodic table (subject to the Pauli exclusion principle, which forbids two electrons from occupying the same set of four quantum numbers).

Question 114

The hydrogen atom energy eigenvalues are $E_n = -13.6/n^2$ eV. (a) Find the ionisation energy from the ground state. (b) Find the wavelength of the H_α line ($n = 3 \to 2$) in nm.

Step 1: (a) Ionisation energy.

$$E_{\text{ion}} = 0 - E_1 = \boxed{13.6 \text{ eV}}$$

Step 2: (b) Transition energy.

$$\Delta E = E_3 - E_2 = 13.6\left(\frac{1}{4} - \frac{1}{9}\right) = 13.6 \times \frac{5}{36} = 1.889 \text{ eV}$$

Step 3: Wavelength.

$$\boxed{\lambda = \frac{1240 \text{ eV nm}}{1.889 \text{ eV}} \approx 656.7 \text{ nm (red)}}$$

Question 115

For the hydrogen ground state $\psi_{100}(r) = (1/\sqrt{\pi})(1/a_0)^{3/2}e^{-r/a_0}$, find the most probable radius (the maximum of $r^2|\psi|^2$).

Step 1: Set $dP/dr = 0$ for $P(r) = (4/a_0^3)r^2e^{-2r/a_0}$:

$$2re^{-2r/a_0}\left(1 - \frac{r}{a_0}\right) = 0$$

Step 2: Non-trivial root.

$$\boxed{r = a_0 = 0.529\ \text{Å}}$$

The most probable radius equals the Bohr radius — a striking link to Bohr's model.

Q116 – Definite Observables for a Hydrogen Eigenstate

Question 116

For the state $|n = 2, \ell = 1, m_\ell = 1\rangle$ of hydrogen, what are the definite values of $\hat{L}^2$, $\hat{L}_z$, and E?

Step 1: Energy. $E_2 = -13.6/4 = -3.4$ eV. **Step 2:** $\hat{L}^2$**.** $\langle \hat{L}^2 \rangle = \ell(\ell + 1)\hbar^2 = 2\hbar^2$. **Step 3:** $\hat{L}_z$**.**

$$\boxed{\langle \hat{L}_z \rangle = m_\ell \hbar = +\hbar}$$

All three are definite (zero uncertainty), since the state is a simultaneous eigenstate of the mutually commuting operators $\hat{H}$, $\hat{L}^2$, $\hat{L}_z$.

Q117 – Angular Momentum Operator $\hat{L}_z$

Question 117

Starting from the classical definition $\mathbf{L} = \mathbf{r} \times \mathbf{p}$, derive $\hat{L}_z$ in spherical coordinates, and state the eigenvalues of $\hat{L}^2$ and $\hat{L}_z$.

Step 1: Classical z-component.

$$L_z = xp_y - yp_x.$$

Step 2: Promote to operators. Using $\hat{p}_x = -i\hbar\partial_x$ and $\hat{p}_y = -i\hbar\partial_y$:

$$\hat{L}_z = -i\hbar\left(x\frac{\partial}{\partial y} - y\frac{\partial}{\partial x}\right).$$

Step 3: Transform to spherical coordinates. Using $x = r\sin\theta\cos\varphi$, $y = r\sin\theta\sin\varphi$ and the chain rule:

$$x\frac{\partial}{\partial y} - y\frac{\partial}{\partial x} = \frac{\partial}{\partial\varphi}.$$

Q117 – Angular Momentum Operator $\hat{L}_z$

$$\boxed{\hat{L}_z = -i\hbar\frac{\partial}{\partial\varphi}.}$$

Q117 – Angular Momentum Operator $\hat{L}_z$

Step 4: Eigenfunctions and eigenvalues of $\hat{L}_z$. The eigenfunctions of $-i\hbar\partial_\varphi$ are $e^{im\varphi}$:

$$\hat{L}_z\, e^{im\varphi} = -i\hbar(im)e^{im\varphi} = m\hbar\, e^{im\varphi}.$$

Single-valuedness of the wavefunction under $\varphi \to \varphi + 2\pi$ requires $e^{im\cdot 2\pi} = 1$, so m must be an integer:

$$L_z = m_\ell \hbar, \quad m_\ell = 0, \pm 1, \pm 2, \ldots$$

Step 5: Eigenvalues of $\hat{L}^2$ (stated). The full angular momentum squared operator (the angular part of $-\hbar^2\nabla^2$) has eigenvalues:

$$L^2 = \ell(\ell+1)\hbar^2, \quad \ell = 0, 1, 2, \ldots, \ |m_\ell| \leq \ell.$$

The eigenfunctions of both $\hat{L}^2$ and $\hat{L}_z$ simultaneously are the spherical harmonics $Y_\ell^{m_\ell}(\theta, \varphi)$ (Q111).

Q118 – Angular Momentum for $\ell = 2$

Question 118

For an electron with $\ell = 2$, find (a) $|\mathbf{L}|$ and (b) all possible values of L_z.

(a) Magnitude of orbital angular momentum. Using $L = \sqrt{\ell(\ell+1)}\,\hbar$ (Q117) with $\ell = 2$:

$$L = \sqrt{2(2+1)}\,\hbar = \sqrt{6}\,\hbar \approx 2.449\,\hbar.$$

Numerically: $L \approx 2.449 \times 1.055 \times 10^{-34} \approx 2.58 \times 10^{-34}$ J s.

(b) Possible values of L_z. For $\ell = 2$: $m_\ell = -2, -1, 0, +1, +2$ (five values, $2\ell + 1 = 5$):

$$L_z \in \{-2\hbar,\ -\hbar,\ 0,\ +\hbar,\ +2\hbar\}.$$

Key observation. $|L_z|_{\max} = 2\hbar < L = \sqrt{6}\,\hbar \approx 2.449\,\hbar$. The angular momentum vector can *never* point exactly along the z-axis in quantum mechanics. If it did, $L_x = L_y = 0$ exactly, violating the angular momentum uncertainty relation $\Delta L_x\,\Delta L_y \geq (\hbar/2)|\langle L_z\rangle|$ (Q139).

Question 119

The orbital angular momentum quantum number is $\ell = 2$. State (a) the possible values of m_ℓ, (b) $\langle \hat{L}^2 \rangle$ in terms of $\hbar$, and (c) the maximum value of $\hat{L}_z$.

Step 1: (a) $m_\ell = -2, -1, 0, +1, +2$ (five values). **Step 2: (b)** $\langle \hat{L}^2 \rangle = \ell(\ell+1)\hbar^2 = 6\hbar^2$.
Step 3: (c)

$$\boxed{(L_z)_{\max} = 2\hbar}$$

Note $\sqrt{\langle \hat{L}^2 \rangle} = \sqrt{6}\,\hbar \approx 2.449\hbar$ exceeds the maximum z-component: $\vec{L}$ can never fully align with any axis.

Question 120

$V = 0$ for $0 < x, y, z < L$; $V = \infty$ outside. Find the energy levels and the degeneracy of the third-lowest distinct level.

Step 1: Separate variables. Since $V(x, y, z) = V_x(x) + V_y(y) + V_z(z)$ with each term identical to the 1D well (Q87):

$$\psi_{n_x n_y n_z} = \sqrt{\frac{8}{L^3}} \sin\left(\frac{n_x \pi x}{L}\right) \sin\left(\frac{n_y \pi y}{L}\right) \sin\left(\frac{n_z \pi z}{L}\right),$$

with $n_x, n_y, n_z = 1, 2, 3, \ldots$ independently.

Step 2: Energy levels.

$$E_{n_x n_y n_z} = \frac{\pi^2 \hbar^2}{2mL^2}(n_x^2 + n_y^2 + n_z^2) = E_1(n_x^2 + n_y^2 + n_z^2), \quad E_1 = \frac{\pi^2 \hbar^2}{2mL^2}.$$

Step 3: Tabulate the lowest distinct levels.

(n_x, n_y, n_z) permutations	$n_x^2 + n_y^2 + n_z^2$	Degeneracy
$(1,1,1)$ only	3	1 (ground state)
$(2,1,1),(1,2,1),(1,1,2)$	6	3 (2nd level)
$(2,2,1),(2,1,2),(1,2,2)$	9	3 (3rd level)
$(3,1,1),(1,3,1),(1,1,3)$	11	3
$(2,2,2)$ only	12	1

Step 4: Answer. The third-lowest distinct energy level is

$$E = 9E_1 = \frac{9\pi^2\hbar^2}{2mL^2},$$

and it is **3-fold degenerate** – the three combinations $(2,2,1),(2,1,2),(1,2,2)$ all give the same energy.
This degeneracy arises from the cubic symmetry (permuting x, y, z), and would be lifted if the box sides were unequal.

Q121 – Operators in Dirac Notation

Question 121

Explain how an operator is defined and represented in Dirac (bra–ket) notation.

Bras and kets. A quantum state is represented by a **ket** $|\psi\rangle$, an abstract vector in Hilbert space (Q123). Its **dual** is the **bra** $\langle\psi|$. The **inner product** of two states is written $\langle\varphi|\psi\rangle$ (a complex number).

Operator definition. An operator $\hat{A}$ maps kets to kets:

$$\hat{A}|\psi\rangle = |\psi'\rangle.$$

This is the abstract, representation-independent analogue of the differential operators used in the position representation (e.g. $\hat{p} = -i\hbar\, d/dx$ acting on $\psi(x)$, Q81).

Matrix elements. The action of $\hat{A}$ between two states is captured by its **matrix element**:

$$\langle\varphi|\hat{A}|\psi\rangle \equiv \langle\varphi|\hat{A}\psi\rangle \in \mathbb{C}.$$

Matrix representation. Choosing a complete orthonormal basis $\{|n\rangle\}$ (satisfying $\langle m|n\rangle = \delta_{mn}$):

$$A_{mn} = \langle m|\hat{A}|n\rangle.$$

This A_{mn} is the matrix representation of $\hat{A}$ in that basis (Q132).

Eigenvalue problem. The abstract eigenvalue equation is $\hat{A}|a_n\rangle = a_n|a_n\rangle$. In the position basis, this becomes the familiar differential equation $\hat{A}\psi_n(x) = a_n\psi_n(x)$ (Q69).

Advantage. The Dirac notation is basis-independent: the same equation $\hat{H}|\psi\rangle = E|\psi\rangle$ represents the TISE regardless of whether we work in position space ($\psi(x)$), momentum space ($\phi(p)$), or any other basis.

Question 122

Define a linear vector space in the context relevant to quantum mechanics, listing the defining axioms.

A **linear (vector) space** $\mathcal{V}$ over $\mathbb{C}$ is a set of elements ("vectors"/kets $|\psi\rangle$) with two operations – addition and scalar multiplication – satisfying the following axioms for all $|\psi\rangle, |\varphi\rangle, |\chi\rangle \in \mathcal{V}$ and $a, b \in \mathbb{C}$:

	Axiom	Statement
1	Closure	$|\psi\rangle + |\varphi\rangle \in \mathcal{V}$; $a|\psi\rangle \in \mathcal{V}$
2	Commutativity	$|\psi\rangle + |\varphi\rangle = |\varphi\rangle + |\psi\rangle$
3	Associativity	$(|\psi\rangle + |\varphi\rangle) + |\chi\rangle = |\psi\rangle + (|\varphi\rangle + |\chi\rangle)$
4	Zero vector	$|\psi\rangle + |0\rangle = |\psi\rangle$
5	Additive inverse	$|\psi\rangle + (-|\psi\rangle) = |0\rangle$
6	Distributivity (vector)	$a(|\psi\rangle + |\varphi\rangle) = a|\psi\rangle + a|\varphi\rangle$
7	Distributivity (scalar)	$(a+b)|\psi\rangle = a|\psi\rangle + b|\psi\rangle$
8	Scalar associativity	$a(b|\psi\rangle) = (ab)|\psi\rangle$
9	Identity scalar	$1 \cdot |\psi\rangle = |\psi\rangle$

These axioms precisely capture the **superposition principle** (Q22): any linear combination $a|\psi\rangle + b|\varphi\rangle$ of allowed states is itself an allowed state.

Question 123

Define a Hilbert space, distinguishing it from a general linear vector space.

A **Hilbert space** $\mathcal{H}$ is a linear vector space (Q122) equipped with two additional structures:

1. Inner product. A map $(|\varphi\rangle, |\psi\rangle) \mapsto \langle\varphi|\psi\rangle \in \mathbb{C}$ satisfying:

- *Conjugate symmetry:* $\langle\varphi|\psi\rangle = \langle\psi|\varphi\rangle^*$.
- *Positive-definiteness:* $\langle\psi|\psi\rangle \geq 0$, with equality iff $|\psi\rangle = 0$.
- *Linearity* in the ket: $\langle\varphi|(a|\psi_1\rangle + b|\psi_2\rangle) = a\langle\varphi|\psi_1\rangle + b\langle\varphi|\psi_2\rangle$.
- *Antilinearity* in the bra: $(\langle\varphi_1|a^* + \langle\varphi_2|b^*)|\psi\rangle = a^*\langle\varphi_1|\psi\rangle + b^*\langle\varphi_2|\psi\rangle$.

This gives the norm $\|\psi\| = \sqrt{\langle\psi|\psi\rangle}$ and a notion of orthogonality ($\langle\varphi|\psi\rangle = 0$) and normalisation ($\langle\psi|\psi\rangle = 1$, Q50).

2. Completeness. Every Cauchy sequence of vectors in $\mathcal{H}$ converges to a vector that is also in $\mathcal{H}$. (This ensures that limits of well-behaved approximating sequences remain valid states.)

Distinction from a general vector space. A bare linear vector space has only algebraic structure (addition and scalar multiplication; Q122). A Hilbert space additionally has:

- **Geometry** (inner product $\Rightarrow$ lengths, angles, orthogonality).
- **Topology** (completeness $\Rightarrow$ convergence of sequences, infinite sums, unbounded operators, expansion in infinite bases).

Both are essential for quantum mechanics. Wavefunctions live in the infinite-dimensional Hilbert space $L^2(\mathbb{R})$ of square-integrable functions, where $\langle\varphi|\psi\rangle = \int_{-\infty}^{\infty} \varphi^* \psi \, dx$.

Question 124

Define $\langle\varphi|\psi\rangle$ in Dirac notation, give its position-representation form, and list its key mathematical properties.

Position-representation formula.

$$\langle\varphi|\psi\rangle \equiv \int_{-\infty}^{\infty} \varphi^*(x)\psi(x)\, dx.$$

This generalises the normalisation integral ($\langle\psi|\psi\rangle = \int |\psi|^2 dx = 1$, Q50) and the expectation-value formula ($\langle A\rangle = \langle\psi|\hat{A}|\psi\rangle$, Q65).

Key properties:

1. Conjugate symmetry: $\langle\varphi|\psi\rangle = \langle\psi|\varphi\rangle^*$. (Swapping bra and ket gives the complex conjugate.)

2. Linearity in the ket (second slot): $\langle\varphi|a\psi_1 + b\psi_2\rangle = a\langle\varphi|\psi_1\rangle + b\langle\varphi|\psi_2\rangle$.

Q124 – Inner Product: Definition and Properties

3. **Antilinearity in the bra (first slot):** $\langle a\varphi_1 + b\varphi_2|\psi\rangle = a^*\langle\varphi_1|\psi\rangle + b^*\langle\varphi_2|\psi\rangle$.
4. **Positive-definiteness:** $\langle\psi|\psi\rangle \geq 0$, with equality if and only if $|\psi\rangle = 0$.
5. **Orthogonality:** $|\varphi\rangle$ and $|\psi\rangle$ are orthogonal iff $\langle\varphi|\psi\rangle = 0$.
6. **Cauchy–Schwarz inequality:** $|\langle\varphi|\psi\rangle|^2 \leq \langle\varphi|\varphi\rangle \cdot \langle\psi|\psi\rangle$, which plays a key role in the proof of the uncertainty principle (Q136).

Q125 – Outer Product and Projection Operators

Question 125

Define the outer product $|\psi\rangle\langle\varphi|$ and show $\hat{P}_n = |n\rangle\langle n|$ is a projection operator; state the completeness relation.

Definition. Whereas $\langle\varphi|\psi\rangle$ (bra times ket) is a *number*, $|\psi\rangle\langle\varphi|$ (ket times bra) is an **operator**. Acting on $|\chi\rangle$:

$$(|\psi\rangle\langle\varphi|)|\chi\rangle = |\psi\rangle\langle\varphi|\chi\rangle = \langle\varphi|\chi\rangle|\psi\rangle.$$

It produces the state $|\psi\rangle$ scaled by the number $\langle\varphi|\chi\rangle$.

Projection operator. Let $|n\rangle$ be normalised ($\langle n|n\rangle = 1$). Define $\hat{P}_n = |n\rangle\langle n|$. Acting on $|\chi\rangle$:

$$\hat{P}_n|\chi\rangle = |n\rangle \underbrace{\langle n|\chi\rangle}_{c_n} = c_n|n\rangle.$$

It *projects* $|\chi\rangle$ onto the direction $|n\rangle$ – exactly like projecting a 3D vector onto a coordinate axis.

Idempotency.

$$\hat{P}_n^2 = |n\rangle\langle n|n\rangle\langle n| = |n\rangle \cdot 1 \cdot \langle n| = \hat{P}_n. \quad \checkmark$$

A projector satisfies $\hat{P}^2 = \hat{P}$ – this is the defining algebraic property of a projection.

Completeness (resolution of identity). If $\{|n\rangle\}$ forms a complete orthonormal basis, summing all projectors gives the **identity operator**:

$$\sum_n |n\rangle\langle n| = \hat{I}.$$

Use. Inserting $\hat{I} = \sum_n |n\rangle\langle n|$ into matrix elements:

$$|\chi\rangle = \hat{I}|\chi\rangle = \sum_n |n\rangle\langle n|\chi\rangle = \sum_n c_n|n\rangle \quad (c_n = \langle n|\chi\rangle).$$

This is the expansion of $|\chi\rangle$ in the basis $\{|n\rangle\}$.

Q126 – Inner Product Calculation

Question 126

Given $|\varphi\rangle = \frac{1}{\sqrt{2}}(|1\rangle + i|2\rangle)$ and $|\psi\rangle = \frac{1}{\sqrt{5}}(2|1\rangle - |2\rangle)$, where $\{|1\rangle, |2\rangle\}$ is orthonormal, compute $\langle\varphi|\psi\rangle$.

Step 1: Write the bra $\langle\varphi|$. Taking the conjugate and swapping ket to bra:

$$\langle\varphi| = \frac{1}{\sqrt{2}}(\langle 1| - i\langle 2|).$$

(Complex conjugate of i is $-i$.)

Step 2: Expand the inner product.

$$\langle\varphi|\psi\rangle = \frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{5}}(\langle 1| - i\langle 2|)(2|1\rangle - |2\rangle) = \frac{1}{\sqrt{10}}\left[2\langle 1|1\rangle - \langle 1|2\rangle - 2i\langle 2|1\rangle + i\langle 2|2\rangle\right].$$

Step 3: Apply orthonormality: $\langle i|j\rangle = \delta_{ij}$.

$$\langle 1|1\rangle = 1, \quad \langle 2|2\rangle = 1, \quad \langle 1|2\rangle = \langle 2|1\rangle = 0.$$

$$\langle \varphi|\psi\rangle = \frac{1}{\sqrt{10}}\left[2(1) - (0) - 2i(0) + i(1)\right] = \frac{2+i}{\sqrt{10}}.$$

Step 4: Compute the magnitude.

$$|\langle \varphi|\psi\rangle| = \frac{|2+i|}{\sqrt{10}} = \frac{\sqrt{4+1}}{\sqrt{10}} = \frac{\sqrt{5}}{\sqrt{10}} = \frac{1}{\sqrt{2}} \approx 0.707.$$

Step 5: Verify the conjugate symmetry.

$$\langle \psi|\varphi\rangle = \langle \varphi|\psi\rangle^* = \frac{2-i}{\sqrt{10}}. \quad \checkmark$$

Q127 – Linear Operators: $\hat{x}$ and $\hat{p}$ are Linear

Question 127

Define a linear operator; verify $\hat{x}$ and $\hat{p}$ are both linear.

Definition: $\hat{A}$ is linear if $\hat{A}(c_1 |\psi_1\rangle + c_2 |\psi_2\rangle) = c_1 \hat{A} |\psi_1\rangle + c_2 \hat{A} |\psi_2\rangle$.

$\hat{x}$ **(multiplication by** x**):**

$$\hat{x}[c_1\psi_1 + c_2\psi_2] = c_1 x\psi_1 + c_2 x\psi_2 = c_1\hat{x}\psi_1 + c_2\hat{x}\psi_2 \quad \checkmark$$

$\hat{p} = -i\hbar\, d/dx$ **(differentiation):**

$$\hat{p}[c_1\psi_1 + c_2\psi_2] = -i\hbar(c_1\psi_1' + c_2\psi_2') = c_1\hat{p}\psi_1 + c_2\hat{p}\psi_2 \quad \checkmark$$

Linearity of all quantum operators guarantees that the superposition principle is preserved under physical operations and time evolution.

Q128 – Hermitian Operators and Observables

Question 128

Define a Hermitian operator and explain why observables must be Hermitian.

Definition: $\hat{A} = \hat{A}^\dagger$, i.e. $\langle\varphi|\,\hat{A}\,|\psi\rangle = \left\langle \hat{A}\varphi \middle| \psi \right\rangle$ for all states.

Why observables must be Hermitian:

1. **Real eigenvalues** (cf. Q82, Q129) — measured values must be real numbers; the same inner-product argument used there for anti-Hermitian and unitary operators shows $a = a^*$ whenever $\hat{A} = \hat{A}^\dagger$.
2. **Orthogonal eigenstates** (Q130) — mutually exclusive outcomes require orthogonality for a consistent probabilistic interpretation.
3. **Completeness** — eigenstates form a basis; the Born rule needs to cover all possible outcomes.

Examples: $\hat{x}$, $\hat{p}$, $\hat{H}$ (for real V). Non-Hermitian: ladder operators $\hat{a}$, $\hat{a}^\dagger$ (each other's adjoints; not self-adjoint).

Q129 – Eigenvalues of a Unitary Operator Have Unit Modulus (Dirac Notation)

Question 129

A unitary operator satisfies $\hat{U}^\dagger \hat{U} = \hat{\mathbb{1}}$. Prove, in Dirac notation, that its eigenvalues all have modulus 1.

Step 1: Set up. Let $\hat{U}|\psi\rangle = u|\psi\rangle$ with $|\psi\rangle \neq 0$.

Step 2: Use the defining property $\hat{U}^\dagger \hat{U} = \hat{\mathbb{1}}$.

$$\langle\psi|\hat{U}^\dagger \hat{U}|\psi\rangle = \langle\psi|\psi\rangle.$$

Step 3: Evaluate the LHS using the eigenvalue equation.
$\hat{U}|\psi\rangle = u|\psi\rangle \;\Rightarrow\; \langle\psi|\hat{U}^\dagger = (\hat{U}|\psi\rangle)^\dagger = u^*\langle\psi|$, so

$$\langle\psi|\hat{U}^\dagger \hat{U}|\psi\rangle = u^*\langle\psi|\big(u|\psi\rangle\big) = |u|^2\langle\psi|\psi\rangle.$$

Step 4: Compare.

$$|u|^2\langle\psi|\psi\rangle = \langle\psi|\psi\rangle \;\Rightarrow\; |u|^2 = 1 \;\Rightarrow\; \boxed{|u| = 1.}$$

So every eigenvalue of a unitary operator can be written $u = e^{i\theta}$ for real θ – exactly why unitary

Question 130

Suppose $\hat{A}$ is Hermitian and ψ_1, ψ_2 are two linearly independent eigenfunctions sharing the *same* eigenvalue a (i.e. a is degenerate), with $\langle\psi_1|\psi_2\rangle = c \neq 0$ in general. Construct an eigenfunction ϕ_2 that is orthogonal to ψ_1 and verify it is still an eigenfunction of $\hat{A}$ with eigenvalue a.

Step 1: Why the original proof fails here. The standard proof of orthogonality (Q130 original) concludes $(a_1 - a_2)\langle\psi_1|\psi_2\rangle = 0$, so orthogonality follows only when $a_1 \neq a_2$. If $a_1 = a_2 = a$ (degenerate), this gives $0 = 0$ – no information, and $\langle\psi_1|\psi_2\rangle$ can indeed be nonzero.

Step 2: Construct an orthogonal combination (Gram–Schmidt). Define

$$\phi_2 \equiv \psi_2 - \frac{\langle \psi_1 | \psi_2 \rangle}{\langle \psi_1 | \psi_1 \rangle}\, \psi_1 = \psi_2 - \frac{c}{\langle \psi_1 | \psi_1 \rangle}\, \psi_1 .$$

Step 3: Verify orthogonality.

$$\langle \psi_1 | \phi_2 \rangle = \langle \psi_1 | \psi_2 \rangle - \frac{c}{\langle \psi_1 | \psi_1 \rangle} \langle \psi_1 | \psi_1 \rangle = c - c = 0.$$

$$\boxed{\langle \psi_1 | \phi_2 \rangle = 0.}$$

Step 4: Verify ϕ_2 is still an eigenfunction with eigenvalue a. Since $\hat{A}$ is linear and $\hat{A}\psi_1 = a\psi_1$, $\hat{A}\psi_2 = a\psi_2$:

$$\hat{A}\phi_2 = \hat{A}\psi_2 - \frac{c}{\langle \psi_1 | \psi_1 \rangle} \hat{A}\psi_1 = a\psi_2 - \frac{c}{\langle \psi_1 | \psi_1 \rangle} a\psi_1 = a\phi_2 .$$

So $\{\psi_1, \phi_2\}$ is an orthogonal basis for the degenerate eigenspace – this is why one can *always* choose a fully orthogonal (in fact orthonormal, after normalising) set of eigenfunctions for a Hermitian operator, even in the presence of degeneracy.

Q131 – $\hat{x}$, $\hat{p}$, $\hat{H}$ in Dirac Notation

Question 131

Write $\hat{x}$, $\hat{p}$, $\hat{H}$ abstractly and show how they act when projected onto the position basis $|x\rangle$.

Operator	Abstract definition	Position basis $\langle x| \hat{O} |\psi\rangle$
$\hat{x}$	$\hat{x}\,|x\rangle = x\,|x\rangle$	$x\psi(x)$
$\hat{p}$	Generator of translations	$-i\hbar\, d\psi/dx$
$\hat{H}$	$\hat{p}^2/2m + V(\hat{x})$	$-\dfrac{\hbar^2}{2m}\psi'' + V(x)\psi$

The familiar differential-equation form of QM is simply the **position-basis representation** of the more general, basis-independent Dirac formalism.
$\langle x| \hat{H} |\psi\rangle = E\psi(x)$ when $|\psi\rangle$ is an energy eigenstate — the TISE (Q73).

Q132 – Eigenvalues and Eigenvectors of a 2×2 Matrix

Question 132

Find eigenvalues and normalised eigenvectors of $\hat{A} = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}$ in basis $\{|1\rangle, |2\rangle\}$.

Step 1: Characteristic equation: $(1-\lambda)^2 - 4 = 0 \Rightarrow \lambda = 1 \mp 2$:

$$\lambda_1 = -1, \quad \lambda_2 = 3$$

Step 2: $\lambda_1 = -1$: $(A + I)v = 0 \Rightarrow v_1 = -v_2$:

$$|a_1\rangle = \frac{1}{\sqrt{2}}(|1\rangle - |2\rangle)$$

Step 3: $\lambda_2 = 3$: $(A - 3I)v = 0 \Rightarrow v_1 = v_2$:

$$|a_2\rangle = \frac{1}{\sqrt{2}}(|1\rangle + |2\rangle)$$

Step 4: Check: $\langle a_1|a_2|a_1|a_2\rangle = \frac{1}{2}(1 - 0 + 0 - 1) = 0$ ✓ (orthogonal, as required by Q130)

Question 133

Suppose $\hat{A}$ and $\hat{B}$ share a *complete* set of simultaneous eigenstates $\{|n\rangle\}$, with $\hat{A}|n\rangle = a_n|n\rangle$, $\hat{B}|n\rangle = b_n|n\rangle$. Prove the converse statement: $[\hat{A}, \hat{B}] = 0$.

Step 1: Expand an arbitrary state in the common eigenbasis. Since $\{|n\rangle\}$ is complete, any state can be written $|\psi\rangle = \sum_n c_n|n\rangle$.

Step 2: Apply $\hat{A}\hat{B}$.

$$\hat{A}\hat{B}|\psi\rangle = \hat{A}\hat{B}\sum_n c_n|n\rangle = \hat{A}\sum_n c_n b_n|n\rangle = \sum_n c_n b_n a_n|n\rangle.$$

Q133 – Converse: A Common Eigenbasis Implies Commutativity

Step 3: Apply $\hat{B}\hat{A}$.

$$\hat{B}\hat{A}|\psi\rangle = \hat{B}\sum_n c_n a_n |n\rangle = \sum_n c_n a_n b_n |n\rangle.$$

Step 4: Compare. Since a_n and b_n are just numbers (they commute trivially), $a_n b_n = b_n a_n$ term by term, so

$$\hat{A}\hat{B}|\psi\rangle = \hat{B}\hat{A}|\psi\rangle \quad \text{for } \textit{every } |\psi\rangle.$$

$$\boxed{\Rightarrow\ \hat{A}\hat{B} = \hat{B}\hat{A} \iff [\hat{A}, \hat{B}] = 0.}$$

Combined with the original Q133, this completes the full "if and only if" statement: two observables have a complete common eigenbasis *if and only if* they commute.

Question 134

Prove (a) antisymmetry, (b) linearity, (c) Leibniz rule for commutators.

(a) Antisymmetry:

$$-[\hat{B},\hat{A}] = -(\hat{B}\hat{A} - \hat{A}\hat{B}) = \hat{A}\hat{B} - \hat{B}\hat{A} = [\hat{A},\hat{B}] \quad \checkmark$$

(b) Linearity:

$$[\hat{A},\hat{B}+\hat{C}] = \hat{A}(\hat{B}+\hat{C}) - (\hat{B}+\hat{C})\hat{A} = [\hat{A},\hat{B}] + [\hat{A},\hat{C}] \quad \checkmark$$

(c) Leibniz (product) rule:

$$[\hat{A},\hat{B}\hat{C}] = \hat{A}\hat{B}\hat{C} - \hat{B}\hat{C}\hat{A}$$

Add/subtract $\hat{B}\hat{A}\hat{C}$:

$$= (\hat{A}\hat{B} - \hat{B}\hat{A})\hat{C} + \hat{B}(\hat{A}\hat{C} - \hat{C}\hat{A}) = [\hat{A},\hat{B}]\hat{C} + \hat{B}[\hat{A},\hat{C}] \quad \checkmark$$

Used in deriving $[\hat{p}^2,\hat{x}] = -2i\hbar\hat{p}$ (Ehrenfest, Q71) and ladder-operator algebra (Q105).

Question 135

Derive $[\hat{x},\hat{p}] = i\hbar$ using the definition of $\hat{p}$ as the generator of spatial translations.

Step 1: $\hat{T}(a)\left|x\right\rangle = \left|x+a\right\rangle$; $\hat{T}(a) \approx \hat{I} - ia\hat{p}/\hbar$ (infinitesimal).
Step 2: $\hat{x}\hat{T}(a)\left|x\right\rangle = (x+a)\hat{T}(a)\left|x\right\rangle$; $\hat{T}(a)\hat{x}\left|x\right\rangle = x\hat{T}(a)\left|x\right\rangle$
Step 3: Subtract: $[\hat{x},\hat{T}(a)]\left|x\right\rangle = a\hat{T}(a)\left|x\right\rangle$, so $[\hat{x},\hat{T}(a)] = a\hat{T}(a)$
Step 4: Expand for infinitesimal a, keep $O(a)$:

$$\left[\hat{x}, \hat{I} - \frac{ia\hat{p}}{\hbar}\right] = a\hat{I} \quad \Rightarrow \quad -\frac{ia}{\hbar}[\hat{x},\hat{p}] = a\hat{I}$$

Step 5:

$$\boxed{[\hat{x},\hat{p}] = i\hbar}$$

Derived from momentum as the generator of translations — a perspective that generalises naturally to other conjugate pairs.

Q136 – General (Robertson–Schrödinger) Uncertainty Relation

Question 136

Derive $\Delta A\,\Delta B \geq \frac{1}{2}|\langle[\hat{A},\hat{B}]\rangle|$.

Step 1: Define $\Delta\hat{A} = \hat{A} - \langle A\rangle$. Consider $|f\rangle = (\Delta\hat{A} + i\lambda\Delta\hat{B})\,|\psi\rangle$, $\lambda \in \mathbb{R}$.
Step 2: $\langle f|f|f|f\rangle \geq 0$:

$$(\Delta A)^2 + \lambda^2(\Delta B)^2 + i\lambda\langle[\hat{A},\hat{B}]\rangle \geq 0 \quad \forall\lambda$$

Step 3: This quadratic in λ is non-negative iff discriminant ≤ 0:

$$\left(i\langle[\hat{A},\hat{B}]\rangle\right)^2 \leq 4(\Delta A)^2(\Delta B)^2 \quad \Rightarrow \quad |\langle[\hat{A},\hat{B}]\rangle|^2 \leq 4(\Delta A)^2(\Delta B)^2$$

Step 4: Take square root:

$$\boxed{\Delta A\,\Delta B \geq \frac{1}{2}|\langle[\hat{A},\hat{B}]\rangle|}$$

Step 5: For $\hat{A} = \hat{x}$, $\hat{B} = \hat{p}$: $|\langle[\hat{x},\hat{p}]\rangle| = \hbar$, giving $\Delta x\,\Delta p \geq \hbar/2$.

Q137 – Uncertainty Product from a Given Commutator Expectation

Question 137

Two Hermitian operators satisfy $[\hat{A}, \hat{B}] = i\hat{C}$ (with $\hat{C}$ Hermitian). Use the Robertson relation to find the uncertainty product for a state in which $\langle \hat{C} \rangle = 2\hbar$.

Step 1: Robertson relation.

$$\Delta A \, \Delta B \geq \frac{1}{2} \left| \langle [\hat{A}, \hat{B}] \rangle \right| = \frac{1}{2} |\langle \hat{C} \rangle|$$

Step 2: Substitute.

$$\boxed{\Delta A \, \Delta B \geq \frac{1}{2}(2\hbar) = \hbar}$$

The minimum uncertainty product is $\hbar$.

Q138 – Minimum Kinetic Energy from General Uncertainty Relation

Question 138

$\Delta x = 2.0 \times 10^{-11}$ m. Find minimum Δp and minimum K for an electron.

Step 1:

$$\Delta p_{\text{min}} = \frac{\hbar}{2\Delta x} = \frac{1.055 \times 10^{-34}}{4.0 \times 10^{-11}} = 2.64 \times 10^{-24}\ \text{kg m/s}$$

Step 2: With $\langle p \rangle = 0$: $\langle p^2 \rangle = (\Delta p)^2$, so

$$K_{\text{min}} = \frac{(\Delta p_{\text{min}})^2}{2m_e} = \frac{(2.64 \times 10^{-24})^2}{2 \times 9.11 \times 10^{-31}} = 3.82 \times 10^{-18}\ \text{J}$$

Step 3:

$$K_{\text{min}} \approx \mathbf{23.9}\ \text{eV}$$

This is the confinement energy estimate of Q41, now derived rigorously from the Robertson–Schrödinger relation.

Question 139

For a spin-$\frac{1}{2}$ particle in the eigenstate $|\uparrow\rangle$ of $\hat{S}_z$ (eigenvalue $+\hbar/2$), compute $\Delta S_x \, \Delta S_y$ explicitly using the Pauli matrices, and check it against the general bound $\Delta S_x \Delta S_y \geq \frac{\hbar}{2}|\langle \hat{S}_z \rangle|$.

Step 1: Write the spin operators.

$$\hat{S}_x = \frac{\hbar}{2}\sigma_x, \quad \hat{S}_y = \frac{\hbar}{2}\sigma_y, \quad \hat{S}_z = \frac{\hbar}{2}\sigma_z, \qquad \sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \, \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \, \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

State: $|\uparrow\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$.

Step 2: Compute $\langle S_x \rangle$ and $\langle S_y \rangle$.

$$\sigma_x|\uparrow\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix} = |\downarrow\rangle, \quad \langle \uparrow |\sigma_x| \uparrow\rangle = \langle \uparrow | \downarrow\rangle = 0.$$

Similarly $\langle S_y \rangle = 0$ (same reasoning with σ_y).

Step 3: Compute $\langle S_x^2\rangle$ and $\langle S_y^2\rangle$. Since $\sigma_x^2 = \sigma_y^2 = \mathbb{1}$ (Pauli matrices square to the identity):

$$\langle S_x^2\rangle = \left(\frac{\hbar}{2}\right)^2 \langle\uparrow|\mathbb{1}|\uparrow\rangle = \frac{\hbar^2}{4}, \qquad \langle S_y^2\rangle = \frac{\hbar^2}{4}.$$

Step 4: Compute the uncertainties.

$$\Delta S_x = \sqrt{\langle S_x^2\rangle - \langle S_x\rangle^2} = \frac{\hbar}{2}, \qquad \Delta S_y = \frac{\hbar}{2}.$$

$$\Delta S_x\, \Delta S_y = \frac{\hbar^2}{4}.$$

Step 5: Compare to the general bound. $\langle S_z\rangle = (\hbar/2)\langle\uparrow|\sigma_z|\uparrow\rangle = \hbar/2$, so the bound is $\frac{\hbar}{2}\cdot\frac{\hbar}{2} = \hbar^2/4$.

$$\boxed{\Delta S_x \Delta S_y = \frac{\hbar^2}{4} = \frac{\hbar}{2}|\langle S_z\rangle| \quad \text{– the bound is exactly } \textit{saturated}.}$$

This makes sense: $|\uparrow\rangle$ is a minimum-uncertainty state for the S_x–S_y pair, analogous to how Gaussian wave packets saturate $\Delta x \Delta p = \hbar/2$.

Q140 – Expectation Value and Standard Deviation in a Three-Outcome State

Question 140

Let $|\psi\rangle = c_1|a_1\rangle + c_2|a_2\rangle + c_3|a_3\rangle$ with $|c_1|^2 = 0.2$, $|c_2|^2 = 0.5$, $|c_3|^2 = 0.3$, and eigenvalues $a_1 = 1$, $a_2 = 3$, $a_3 = 5$. Find $\langle\hat{A}\rangle$, $\langle\hat{A}^2\rangle$, and ΔA.

Step 1: Mean. $\langle\hat{A}\rangle = \sum_i |c_i|^2 a_i$:

$$\langle\hat{A}\rangle = 0.2(1) + 0.5(3) + 0.3(5) = 3.2$$

Step 2: Mean of squares. $\langle\hat{A}^2\rangle = \sum_i |c_i|^2 a_i^2$:

$$\langle\hat{A}^2\rangle = 0.2(1) + 0.5(9) + 0.3(25) = 12.2$$

Step 3: Standard deviation.

Q140 – Expectation Value and Standard Deviation in a Three-Outcome State

$$\boxed{\Delta A = \sqrt{12.2 - (3.2)^2} = \sqrt{1.96} = 1.4}$$

Q141 – Density Matrix of a Pure Qubit State

Question 141

Express the density matrix $\hat{\rho} = |\psi\rangle\langle\psi|$ for the pure state $|\psi\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$ as a 2×2 matrix, and verify that $\mathrm{Tr}(\hat{\rho}) = 1$ and $\hat{\rho}^2 = \hat{\rho}$.

Step 1: Form the outer product.

$$\hat{\rho} = \frac{1}{2}\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$$

Step 2: Trace.

$$\mathrm{Tr}(\hat{\rho}) = \tfrac{1}{2}(1+1) = 1 \checkmark$$

Step 3: Idempotency.

Q141 – Density Matrix of a Pure Qubit State

$$\hat{\rho}^2 = \frac{1}{4}\begin{pmatrix} 2 & 2 \\ 2 & 2 \end{pmatrix} = \frac{1}{2}\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} = \hat{\rho} \checkmark$$

$\boxed{\hat{\rho}^2 = \hat{\rho} \text{ confirms a pure state.}}$

Question 142

A system has observable $\hat{A}$ with eigenvalues $a_1 = 0$, $a_2 = \hbar\omega$ and eigenstates $|1\rangle$, $|2\rangle$. The state is $|\psi\rangle = \frac{\sqrt{3}}{2}|1\rangle + \frac{1}{2}|2\rangle$. (a) Find $\langle\hat{A}\rangle$. (b) After a measurement yields $a_2 = \hbar\omega$, what is the new state?

Step 1: (a) Probabilities.

$$P_1 = \left(\frac{\sqrt{3}}{2}\right)^2 = \frac{3}{4}, \qquad P_2 = \left(\frac{1}{2}\right)^2 = \frac{1}{4}$$

Step 2: Expectation value.

$$\boxed{\langle\hat{A}\rangle = \tfrac{3}{4}(0) + \tfrac{1}{4}(\hbar\omega) = \frac{\hbar\omega}{4}}$$

Step 3: (b) Collapse. Measurement of a_2 projects the state onto the corresponding eigenstate:

$$\boxed{|\psi_{\text{after}}\rangle = |2\rangle}$$

Question 143

In a three-state system with orthonormal basis $\{|1\rangle, |2\rangle, |3\rangle\}$ and energy eigenvalues $E_1 = 1$ eV, $E_2 = 3$ eV, $E_3 = 6$ eV, a state at $t = 0$ is $|\psi(0)\rangle = \frac{1}{\sqrt{6}}|1\rangle + \frac{1}{\sqrt{2}}|2\rangle + \frac{1}{\sqrt{3}}|3\rangle$. Find $|\psi(t)\rangle$ and $\langle E\rangle$.

Step 1: Time evolution. Each eigenstate acquires a phase $e^{-iE_n t/\hbar}$:

$$|\psi(t)\rangle = \frac{1}{\sqrt{6}}e^{-iE_1 t/\hbar}|1\rangle + \frac{1}{\sqrt{2}}e^{-iE_2 t/\hbar}|2\rangle + \frac{1}{\sqrt{3}}e^{-iE_3 t/\hbar}|3\rangle$$

Step 2: Normalisation check. $1/6 + 1/2 + 1/3 = 1$. ✓ **Step 3: Energy expectation value.**

Q143 – Time Evolution and Mean Energy of a Three-State System

$$\langle E \rangle = \frac{1}{6}(1) + \frac{1}{2}(3) + \frac{1}{3}(6) = \frac{1}{6} + \frac{3}{2} + 2$$

$$\boxed{\langle E \rangle = \frac{22}{6} \approx 3.67 \text{ eV}}$$

Q144 – Number Operator, Measurement Probability, and Collapse for a QHO Superposition

Question 144

For the QHO state $|\psi\rangle = \frac{1}{\sqrt{3}}|0\rangle + \frac{1}{\sqrt{3}}|1\rangle + \frac{1}{\sqrt{3}}|2\rangle$, find: (a) $\langle \hat{N} \rangle$ where $\hat{N} = \hat{a}_+\hat{a}_-$; (b) the probability of measuring energy $E_0 = \frac{1}{2}\hbar\omega$; (c) the state immediately after such a measurement.

Step 1: (a) Mean occupation number.

$$\langle \hat{N} \rangle = \frac{1}{3}(0 + 1 + 2)$$

$$\boxed{\langle \hat{N} \rangle = 1}$$

Step 2: (b) Probability of E_0 (corresponds to $n = 0$):

Q144 – Number Operator, Measurement Probability, and Collapse for a QHO Superposition

$$\boxed{P_0 = \left|\frac{1}{\sqrt{3}}\right|^2 = \frac{1}{3}}$$

Step 3: (c) Post-measurement state.

$$\boxed{|\psi_{\text{after}}\rangle = |0\rangle}$$

A subsequent energy measurement now yields E_0 with certainty.

Summary: Five Units at a Glance

Unit	Topic	Key Results
1	Origin of QM	Planck, photoelectric, Compton, de Broglie
2	Formulation	Postulates, uncertainty, wave packets
3	Schrödinger Eq. I	Stationary states, operators, continuity eq.
4	Schrödinger Eq. II	Wells, barriers, tunnelling, SHO, H-atom
5	Dirac Notation	Hilbert space, Hermitian ops, commutators

THANK YOU!!!